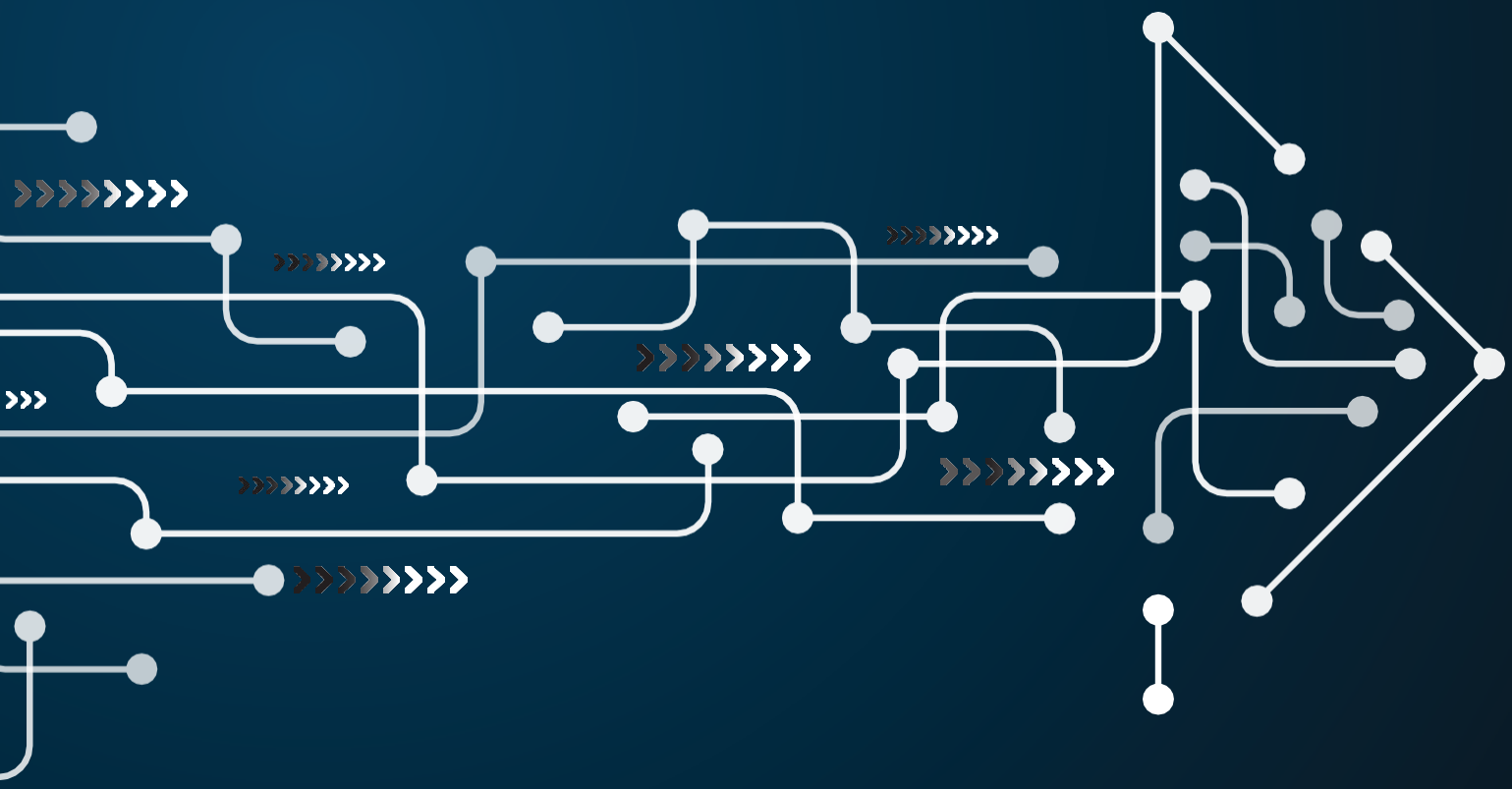
2026

Accredited Faster Payments Professional Handbook

AFPP™

Accredited Faster Payments Professional



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Nacha thanks Lipis & Advisors, the authors of this publication.

Please submit errors, corrections or omissions to education@nacha.org under a subject line "AFPP Handbook correction" with the page number and paragraph number included in the email.

Exam items will be based on information correct as of December 31, 2025.

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Your Payments Association and Nacha are available to assist you in the selection of seminars, training sessions, research documents, and publications that will help you prepare for the AFPP examination.

Follow the links below to find prep courses and other education offerings.

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EPCOR
NEACH
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Nacha's TPI On Campus is a great venue to kick off your AFPP training preparation, which can be further supplemented by Payments Associations training opportunities for AFPP candidates. Many of the PAs are teaming up to offer AFPP Prep Courses and more courses are being added. The offerings listed here represent only a portion of the education courses.

AFPP Exam Blueprint

The AFPP Exam Blueprint

In 2026 the AFPP Exam will consist of up to 120 multiple choice questions, **100 of which are scored**. The AFPP Exam Blueprint illustrates the number of questions per section. For example, in 2026 there will be 18 Fundamentals of Faster Payments questions.

Fundamentals of Faster Payments	18%
Faster Payments Operations and Process Flows	28%
Faster Payments Risk Management	19%
Technology Considerations for Enabling Faster Payments	15%
Faster Payments Governance Framework	20%

Chapter One

Fundamentals of Faster Payments

Knowledge of types, applications, and impact of Faster Payments rails

SECTION BLUEPRINT COMPOSITION: 18%

Types, features, & applications of US Faster Payments network rails

The ACH Network

The US ACH Network, whose rules are set by Nacha, enables Direct Deposit and Direct Payments and reaches all accounts housed by US financial institutions. The ACH Network is composed of FedACH, operated by the Federal Reserve, and EPN, which is owned and operated by The Clearing House (TCH). The two networks are fully interoperable.

The ACH Network enables the distribution and settlement of electronic debits and credits among financial institutions as well as non-monetary entries with payment-related information, such as transmission of notifications of change). Participants in the ACH system exchange information via a nationwide telecommunications network. The ACH Network is open for submission 23 ¼ hours a day and settles payments four times a day (Monday-Friday) via the Federal Reserve's settlement system, the National Settlement Service (NSS). In addition to standard to ACH credits and ACH debits, Nacha Same Day ACH processes credits and debits of up to $1 million.

ACH Network operation

The ACH Network is a batch-processing store-and-forward system, with clearing happening in batch format. FIs create files, sometimes referred to batches, of ACH payment messages and submit the batch file to one of the two ACH Operators for processing during a pre-determined time period. ACH payments typically take between one and two days to process, depending on the time and day of the week that a batch file is submitted. Same Day ACH transactions are also possible.

ACH Network players

Any given ACH transaction involves five different players, though there may be some overlap. ACH transfers involve an originator, originating and depository institutions, and an ACH Operator that processes the payment.

- **Originator:** The end user, either a company or individual, that has authorized an ODFI to initiate a debit, credit, or non-monetary entry to the receiver's account
- **ODFI:** The Originating Depository Financial Institution receives payment instructions from the Originator or a Third-party sender and transmits the ACH entries directly or indirectly to the ACH Operator

- **RDFI:** The Receiving Depository Financial Institution receives ACH entries from the operator and posts entries to the accounts of its receivers.
- **Receiver:** A company or individual with an account authorized to receive payments via the RDFI
- **ACH Operator:** The operator of a network used to clear payments between participating financial institutions. Operators can be a private organization or the Federal Reserve Bank. The Electronic Payments Network (EPN) is the only private ACH Network in the US and The Clearing House is the only private operator of EPN.

Transaction entries are coded by the originating institution as either credits or debits and are submitted to the ACH Network via two primary channels: directly by the ODFI or indirectly by a third-party sender that acts as an intermediary for the submission of payment entries.

ACH Network applications

The ACH Network was originally used to process recurring payments; today it processes a multitude of one-off standard and Same Day transfers. ACH batch entries can be either a credit or debit.

- **ACH Credits:** an originator transfers funds to the receiver's account from the ODFI to the RDFI.
- **ACH Debits:** Funds are collected from the receiver and transferred to originator's account. The originator of the debit order must have a preauthorized authorization enabling them to initiate the debit entry from the receiver's account.

Batch processing makes the ACH Network an attractive option for processing payroll, bills, mortgages, or other recurring payments. ACH payments are also frequently used to process ad-hoc one-time payments between businesses, government disbursements, and P2P transfers.

The Clearing House's RTP Network

The RTP Network enables Participants to initiate credit transfers, receive final and irrevocable settlement for credit transfers, and make available to Receivers funds associated with such credit transfers in real-time, twenty-four (24) hours a day, seven (7) days a week, fifty-two (52) weeks a year.

The RTP Network supports a variety of different use cases and is open to depository institutions in the US as per the RTP rules. The RTP Network is utilized by consumers and businesses. Financial institutions connect directly to the network or through a third-party service providers while non-bank payment service providers connect via a participating financial institution.

Technology providers play several roles in the RTP ecosystem as third-party service providers, channel providers (typically for smaller FIs), gateway providers (for the licensing of applications that FIs use in their data centers to connect to RTP), Value Added Services (VAS), and as a testing provider for FIs to ensure their messaging is in compliance with RTP specifications.

The RTP rules require fraud and risk controls; participating FIs and service providers are required to implement strong authentication, fraud detection and prevention, fraud reporting, and consumer protection policies. All transactions are digitally signed and encrypted.

RTP Network operation

All RTP messages are based on the ISO 20022 messaging standard. The RTP Network facilitates payment and non-payment messages including but not limited to Credit Transfers, Requests for Payment (RfPs), Request for Return of Funds, and Requests-for-Information (RfI).

RTP Network End-user access to the RTP Network is provided by banks, credit unions, and other payment firms, which build payment products upon RTP network basic services.

Account and routing numbers are used to route payments, although other overlay services, such as Zelle, allow for alias-based payment initiation.

Payments made via the RTP Network are settled and posted instantly, are irrevocable, and are limited to $10,000,000 in value

Financial institutions can connect to the Real-Time Payments (RTP) network either:

1. Directly – by becoming a participant of the RTP network themselves, which involves meeting technical and operational requirements set by The Clearing House (TCH).
2. Indirectly – through a third-party service provider (often called a "gateway" or "agent"), which acts on behalf of the institution to send and receive:
    - Payment messages (e.g., credit transfers)
    - Responses (e.g., acknowledgments or errors)
    - Non-payment messages (e.g., requests for information or status updates)

These third-party providers can include core banking platforms, payment processors, or banking-as-a-service (BaaS) platforms. They help smaller institutions or those not ready for direct integration to still participate in RTP services.

RTP Network players

- **The Clearing House:** Operates the RTP and EPN networks and facilitates the movement of payments and payment-related messages between participants. The Clearing House also provides directory and token-related services and anti-fraud alerts.
- RTP Participants: Global banks, regional banks, community banks, and credit unions. FIs may directly or indirectly access the RTP Network, Technology Providers, also referred to as third-party service providers, add to the service by offering Value Added Services (VAS) with real-time capabilities for retailers, small businesses, and corporate customers, such as payment automation services, payment insights and analytics, and treasury/cash management services.
- **Funding Agents:** A Funding Agent is a financial institution that enters into an arrangement with one or more Participants to provide funding for the Participant's Prefunded Requirement and may also provide services to monitor the Participant's liquidity position, add Supplemental Funds to the Prefunded Balance Account as needed and initiate Disbursements of funds from the Prefunded Balance Account when appropriate.
- **Technology providers:** These entities play several roles in the system as Third-Party Service providers, channel providers (to assist smaller FIs in signup and servicing), gateway providers (which use licensed applications for FIs to connect to the platform), Value Added Services (VAS) , and testing providers, which ensure FI messaging capabilities meet RTP specifications.

RTP Network applications

The RTP Network is a credit push network with the payer determining when a credit transfer is initiated even in the case where the payer is responding to an RfP. The network is used for account-to-account (A2A) payments such as payroll, insurance disbursements, merchant disbursements, P2P transactions, and a variety of other use cases.

The system possesses rich data capabilities; all credit transfer and request-for-payment messages can include links to external documents such as remittance data, invoices, bills, deeds, etc. Request-for-payment (RfP) bill pay is also possible via the RTP Network, with the payer determining when the credit transfer is initiated in response to the RfP message.

The Federal Reserve's FedNow Service

FedNow network operation

The FedNow service, created and operated by the Federal Reserve, allows financial institutions to send and receive credit transfers on a 24/7/365 basis with instant clearing, settlement, and posting. The FedNow service is available to all financial institutions eligible for a Master Account with the Federal Reserve, with certain third-party service providers able to perform functions on behalf of a financial institution. These Service Providers provide services for smaller FIs, enabling them to easily connect to the FedNow network to provide payment initiation services or other forms of technological support. FedNow uses the FedLine electronic communication system that connects FIs to all of the Federal Reserve's payment and information infrastructure.

FedNow moves and settles money in central bank-held funds between FI-controlled master accounts. This stands in contrast to the RTP Network, where funds do not move between central bank-held accounts. Instead, ownership of funds within the TCH-controlled account (held at the Reserve Bank of New York) shifts with each transaction.

Payments made via FedNow are settled and posted instantly and are irrevocable.

The transaction value limit, as of November 2025, is $10,000,000, , though FIs can lower this limit for individual customers as they see fit.

FedNow uses the ISO 20022 standard, enabling rich data exchange. Non-payment messaging capabilities include request-for-return (in the event of a misdirected payment), request-for-payment status, information requests (for more information on an instant payment), confirmation-of-posting, account balance requests, and account activity reports.

FedNow allows FIs to use several risk management and participation capabilities such as accept-without-posting and information/return requests. Information security for the service is enabled via digital signatures between FI participants, data encryption, strong authentication and authorization features, and mutual authentication between participants. Fraud mitigation strategies include transaction limits at the network and participant levels, AATs and participant-defined negative lists (suspicious entity blacklist).

Payment messages between FedNow and the RTP Network are not interoperable – in contrast to the ACH Networks EPN and FedACH - limiting transfers between participants on the same network.

FedNow network players

- **The Federal Reserve:** Provides the FedNow interface via FedLine Advantage, the electronic platform that enables FIs to access Federal Reserve payment services, account information, and secure communication.
- **FedNow participants:** Financial institutions that possess a FedNow Participant profile and have established accounting information for payment settlement via a Master Account or another participant.
- **Service providers:** These entities can be FIs with participant profiles or technological service providers with authorized connection profiles. Service providers act as agents for a FedNow participant to: 1) execute or receive payment messages on their behalf, 2) operate or manage connections used to send or receive messages, 3) obtain information or select security/processing procedures for the participant.

- **Correspondents:** Maintain a Master Account with the Federal Reserve or maintain a FedNow settlement account on behalf of another participant.

FedNow platform applications

FedNow enables A2A payments such as consumer-to-business bill pay and P2P; RfP functionalities are also supported. The Federal Reserve is encouraging industry innovation to align with the development of FedNow use cases, including B2B, B2C, C2B, and government payments. These innovations should accommodate various participant and service provider needs such as e-invoicing, on-demand payments, non-recurring disbursements, immediate payroll, e-commerce payments, and transactions involving government entities.

Debit card networks

Debit cards facilitate ATM cash withdrawals, push-to-pay payments, merchant payments, bank deposits, and fund inquiries. The major infrastructure and systems facilitating debit card-based Faster Payments are composed of a set of global payment networks and systems, and include financial institutions and third- party service providers. The card networks act in a rule- and standard-setting capacity as well as connectors between various card schemes and payment networks to facilitate transactions between and among card holders, businesses, and in some circumstances, government entities. The parties involved in facilitating transactions include card issuers (typically FIs), customers, processors, merchants, gateway partners, and acquiring FIs. Within the US the most used methods of debit card-based Faster Payments are Mastercard Send and Visa Direct. These systems facilitate credit transfers of up to $50,000 for P2P transfers and $125,000 for business-to-consumer disbursements on a 24/7/365 basis to card networks across the U.S. Visa Direct also allows users to send payments to accounts linked to the debit cards in over 175 countries. Push-to-debit payments are only used to top-off accounts/ cards. Faster Payments carried out via these networks are immediate (posting typically within seconds, at maximum within 30 minutes) and irrevocable, with all exceptions related to transaction errors, disputes, and reversals being subject to agreement with the card issuing institutions.

Debit card network operation:

Debit card-based push-to-card Faster Payments are facilitated and interoperable across proprietary networks (between Visa, Mastercard, and other players), allowing credit transfers to be sent via a variety of card- and account- based payment rails. These rails offer customers the ability to reach billions of cards and accounts globally.

Settlement methods vary among push-to-card networks. MastercardSend transactions are pre-funded, ensuring funds availability within seconds due to posting agreements between participating FIs. Mastercard manages transactions through internal ledgers and batch netting, alerting FIs to top off minimum balances for system maintenance. This pre-funding model eliminates settlement risk for participating FIs. In contrast, Visa settles Visa Direct transactions once daily, transferring funds from sender to receiver FIs at the end of the business day, which implies a trade-off between efficient use of liquidity and a build-up of settlement risk. Payment routing relies on card details or account/routing numbers, with all payment messages utilizing the ISO 8583 standard.

Neither of the major card networks possesses a unified internal directory. Visa Direct has no single integrated directory in the United States and instead relies on partners to use their proprietary directories. Mastercard Send partners utilize their own directories to support P2P transfers and other use cases, allowing for the tokenization of account credentials for senders and receivers.

The card companies mandate certain security features, with common standards in place for KYC, sanctions screening, consumer disclosures, transaction limits, and other fraud mitigation practices. Partner institutions are also vetted by the major card networks and must comply with anti-fraud and consumer protection standards.

Debit card network participants:

- **Card scheme**: Includes Mastercard and Visa, among other major players. These networks provide connection to the card rails of multiple networks used to facilitate transactions, set rates and fees, and act as arbiters between different parties in cases of disputes.
- **Acquiring banks:** Establishes and maintains receiver accounts, allowing users to accept payments from the card networks.
- **Issuing banks:** FIs that issue the debit cards associated with the card networks via their relationship with the card associations.
- **Payment processors:** companies that process debit card transactions, connecting users, FIs, and card networks to facilitate transactions. These entities handle the processing and authorization of transactions. ebit card network applications:

Debit card-based faster payment networks allow users to send payments for various use cases including P2P, B2C, and G2C. These networks typically leverage existing relationships with financial institutions to facilitate transactions. Services provided by Visa and Mastercard enable insurance disbursements, refunds, brokerage account transfers, domestic P2P transfers, tax refunds, social security payouts, cash pick-ups, and a variety of other applications and use cases.

Conclusion

Any Faster Payment Professional should be familiar with the basics of the payments rails that operate within the US market. Knowing the functions of these payment rails and the participants involved is essential to have a comprehensive understanding of the broader fast payments landscape.

TABLE 1. BASIC CHARACTERISTICS OF FASTER PAYMENT RAILS

	SAME DAY ACH	***RTP NETWORK***	***FEDNOW***	***DEBIT CARD NETWORKS***
Operator	Nacha owns Rules Federal Reserve or EPN are ACH Operators	The Clearing House	The Federal Reserve	Visa, Mastercard, other card companies
Payment type	Credit/debit	Credit	Credit	Credits (cam allow debits for account top ups)
Messaging standard	Proprietary legacy	ISO 20022	ISO 20022	ISO 8583
Transaction limit	$1,000,000	$10,000,000	$10,000,000 (as of November 2025)	By use case P2P: $10,000 Disbursements: $50,000
Settlement	Deferred Net Settlement 3 times daily Batched funds availability	Real-time Gross Settlement Immediate 24/7/365 Instant funds availability	Real-time Gross Settlement Immediate 24/7/365 Instant funds availability	Deferred Net Settlement Once daily Near immediate funds availability
Payment routing	Account and routing number	Account number and routing transit number	Account number and routing transit number	Card credentials or account number
Fraud and risk controls	Transaction limits and monitoring, mandatory account validation, information sharing	Mandatory SCA, fraud reporting, detection, and consumer protection policies	Transaction limits, data encryption, Strong Customer Authentication (SCA), mutual authentication between participants, suspicious entity blacklist, AATs	AML measures, pre- transaction controls, mandatory risk-management controls, sanctions screening

Practical use cases for Faster Payments

Faster Payments offer solutions across a variety of use cases. The expedited posting speed inherent to Faster Payments is particularly attractive for disbursements where increased speed is a major value-add, such as B2C

payments like payroll and emergency loans. Instant P2P transactions provide a quicker and more convenient way to pay microbusinesses, friends, and family in a variety of circumstances. Faster Payments can also be used in C2B transactions where speed or confirmation of payment are key to reduce late fees or improving the customer experience, such as with bill payments, electronic invoicing, or e-commerce transactions. Businesses can utilize Faster Payments for B2B transactions, optimizing cash flow, while government entities, consumers, and business can all benefit from quick posting times of social benefits, tax refunds, and instant tax payments.

TABLE 2. FASTER PAYMENTS USE CASES AND STAKEHOLDER BENEFITS

USE CASE TYPE	POTENTIAL USE CASES	STAKEHOLDER BENEFITS
Account-to-Account payments (A2A)	**Me-to-me:** Faster Payments allow individuals to quickly transfer funds between accounts at different institutions **Mobile wallet top-ups:** Instant cashout or funding of mobile wallets **Pre-paid cards:** Reloadable cards can be topped off without a wait **Investment account management:** Funds can be moved from & to an individual's FI to & fromt their brokerage account at a different FI instantly **Corporate cash pooling:** Fund consolidation into a single account for purposes such as payroll	• Improved liquidity management due to speed of transfers • Avoidance of late fees and declined payments due to fast transfers of funds between user accounts
Consumer-to-Business payments (C2B)	**Bill pay:** Bills such as loans and utilities can be paid by individuals and businesses with immediate posting to the biller's account, avoiding late fees and improving cash flow. **E-invoicing:** Using Faster Payments, billers can send requests-for- payment to customers directly for quicker clearing and settlement. **E-commerce:** Consumers can select Faster Payments options when purchasing goods online	• Reduced transaction costs due to displacement of expensive legacy payment methods (checks, wires, cards etc.) • Enhanced transparency and information around payments due to richer data messaging standards • Better data and easier analysis of user habits and trends stemming from richer data standards
Peer-to-Peer payments (P2P)	**Micro-business payments:** Individuals are able to quickly pay micro-businesses for goods and services **Payments to family and friends:** Individuals can pay family and friends quickly and electronically	• Improved cash flow and efficiency for micro- businesses • Improved user experience for sending and receiving parties due to fast posting
Business-to-Business payments (B2B)	**On-demand payments:** Businesses can pay suppliers later due to the increased posting speed, freeing up working capital **E-invoicing:** Businesses can use Faster Payments to send e-invoices via request-for-payment with attached remittance information to be included in the returned payment	• Improved liquidity management and cashflow for businesses • Improved invoice reconciliation and information transparency from richer data messaging standards • Greater predictability around payment processing allows for more efficient resource allocation
Business-to-Consumer payments (B2C)	**One-time payments:** Non-recurring disbursements such as insurance payouts, refunds, etc. can be paid instantly, providing improved customer service **Immediate payroll:** Payroll instructions can be carried out instantly without having to be scheduled in advance, allowing for easier and more immediate access to funds for employees	Improved user experience due to faster and more transparent returns and disbursements
Government payments (G2X)	**Payments to businesses/ individuals:** Welfare benefits, tax refunds, and emergency relief payments can be paid out more quickly to individuals, while also building consumer trust and familiarity with Faster Payments **Payments to the government:** Tax payments can be paid more easily to national, state- level, and municipal government entities more quickly and efficiently:	Quicker fund access for government disbursements, faster processing of government payments

Impact and benefit of Faster Payments for stakeholders

Speed of settlement

The improved speed of settlement of some faster payment systems (specifically the RTP Network and FedNow) comes with a number of different benefits and effects. Fast settlement of funds eases or even eliminates settlement risk, increasing the safety of a payment system, potentially alleviating regulator fears around the buildup of settlement risk. However, not all Faster Payments rails feature faster settlement. While the instant systems (FedNow and the RTP Network) and Same Day ACH have faster settlement than traditional ACH, push-to-card systems do not. Furthermore, faster settlement – especially in instant systems – are associated with increases in costs and liquidity that participant FI's must keep available to process each transaction.

Systems featuring instant settlement not only have higher liquidity needs because they settle on a transaction-by-transaction basis rather than on a net basis, they also operate 24/7/365. This means that participating FIs need to adequately forecast their liquidity needs for times when the NSS is not operating. Some participants may not settle on their own behalf and use a "banker's bank" or settlement agent to settle their payments for them. This, however, also comes with its own set of costs. Regulators often like instant settlement because it eliminates settlement risk altogether (a payment cannot be processed if the ODFI or ODFI's settlement agent lacks the liquidity to settle the payment).

Same Day ACH payments settle during business hours and provide a compromise between instant payments and standard batch payments, which reduce liquidity requirements for financial institutions. The Clearing House's RTPNetwork settles payments in real time via a pre-funded account, completely eliminating settlement risk. The FedNow service offers 24/7 real-time settlement, ensuring no settlement or credit risk build-up. Card networks like Mastercard and Visa differ; Mastercard Send utilizes-pre-funding, reducing settlement risk, while Visa's push-to-card network settles on a net basis once a day, incurring slightly more settlement risk.

Availability of funds

Posting funds can take several days in traditional payment systems, depending on when a payment is initiated; in comparison, faster payment methods allow for instant or Same Day access to funds. Faster availability of funds primarily benefits consumers and merchants because they have access to their financial resources more quickly or can pay their bills at the last minute, greatly improving their liquidity management

Different payment systems offer varying posting speeds. Same Day ACH payments typically post on the day the transaction is initiated or the following day, ensuring faster fund availability compared to traditional ACH payments. The RTP Network and FedNow posts within seconds, providing instant access to funds for consumers and corporations. Push-to-card transactions by Visa and Mastercard require funds to be available within 30 minutes of authorization.

Predictability

Traditional payment systems such as ACH, checks, or wires often involve multiple steps in processing (submission windows, compliance checks, settlement times, posting windows, etc.), making it difficult to predict when a payment will be posted to the beneficiary. In comparison, faster payment systems speed this process up significantly and offer stakeholders a greater degree of predictability due to functionality such as instant confirmation and expedited posting speed. Notification can come in the form of a push-notification to a smart phone or an automated message via the FI's online banking platform. These notifications are sent following the posting of a payment to a beneficiary's account for payments made via the RTP Network and FedNow.

This enables consumers and businesses to better manage their cash flows and decision-making process.

Efficiency

Faster payment systems can be more efficient than traditional payment methods, especially checks and wires. Not only do Faster Payments settle and post more quickly than checks and wires, but they are also much more automated, cutting down on the cost to FIs, as well. Value-added services (VAS), such as confirmation- of-payee and the use of alias-based routing, can also cut down on error rates associated with Faster Payments compared to check and wires.

While Faster Payments can cut down on manual intervention needed to process payments and deal with exceptions handling, they can be less liquidity-efficient, especially systems that settle instantly. Same Day ACH and Visa Direct, however, settle on a net basis, and therefore do not have these concerns. Mastercard Send is pre-funded, which ties up FI liquidity.

Security

The introduction of faster payment systems in the US has not only revolutionized the speed of transactions but has also significantly influenced security dynamics within the payments industry. The speed and irrevocability of faster payment systems presents an appealing opportunity for fraudsters in ways that traditional ACH payments do not. This underscores the essential need for robust security features such as tokenization, strong customer authentication (also known as multi-factor authentication) (SCA/MFA), data encryption, and fraud monitoring mechanisms. Together these features can form a protective barrier for end users, including consumers and businesses. Real-time transaction alerts, transaction and value limits, and enhanced authentication methods allow end users to respond quickly to fraud incidents and provide "smart" friction against fraud. Fraud detection tools at the infrastructure and FI levels further reduce end user vulnerabilities.

FedNow has a series of security features to benefit its users, such as transaction value limits, data encryption and tokenization services, and strong authentication and authorization services for end users. The Clearing House's RTPNetwork enforces stringent security measures, such as strong authentication, transaction value limits, encrypted and tokenized transactions, and robust fraud monitoring. Same Day ACH transfers are rendered more secure by Nacha Operating Rules requiring encryption and tokenization of sensitive account data. Lastly, push-to-card payments offer significant security benefits and maintain transaction integrity through comprehensive controls including recipient eligibility standards, use case validation, predefined value limits, and anti-money laundering measures (such as transaction monitoring, sanctions screening, and KYC measures), with configurable options for enhanced security.

Enriched payment data

The usage of data-rich messaging standards such as ISO 20022 by FedNow and the RTP Network allows for the provision of improved payment data that can benefit a variety of stakeholders. ISO 20022 data provides significantly more information about each message in a structured format. Richer, robust data improves the reconciliation and integration process for business end users, making the matching of payments to specific invoices easier, and increasing the efficiency of e-invoicing and billing. Enriched data also allows financial institutions and PSPs to build more sophisticated fraud prevention models, reducing risks for themselves and their customers. In turn, better structured and more robust data standards reduce friction in security screenings by providing these mechanisms with more information through which they can evaluate transactions. This makes screening for fraud, sanctions, and Anti-Money Laundering (AML) purposes far more efficient. Richer payments data also opens potential opportunities for new product developments and revenue streams for financial institutions and PSPs, through a more thorough understanding of their client's needs and behaviors.

24/7/365 availability

Unlike traditional payment rails, one of the hallmarks of Faster Payments is their increased availability. Most faster payment systems, including FedNow, the RTP Network, Mastercard Send, and Visa Direct, process payments 24/7/365. This also means that FIs need to account for 24/7/365 processing. System maintenance, upgrades, and other issues that required downtime to address issues is made more complicated when a system must be available 24/7/365, though all card transactions currently require 24/7/365 already

Request-for-Payment (RfP)

Request-for-Payment (RfP) is a type of ISO 20022 non-payment message that allows businesses or individuals to initiate a formal request for an instant credit transfer from a recipient. Rather than sending funds, the sender issues a structured message asking the recipient to authorize and complete the payment. This mechanism supports real-time payment workflows, improves cash flow visibility, and enhances customer experience by enabling faster and more secure transactions. Unlike with a direct debit, an RfP still gives the recipient of the RfP (the sender of the payment) control as to a) whether the transaction is accepted or not and b) the timing of said payment. The FedNow and RTP Networks provide RfP functionality, although the two are not interoperable between the networks.

This functionality provides several benefits, including greater convenience and flexibility for consumers in paying their bills – including the ability to avoid late payments – or requesting funds from friends or family. For billers, RfPs enable a sort of account validation when setting up a new business relationship (if the account does not exist, the RfP will not work and if the account does not belong to the intended payee, the RfP is unlikely to be accepted). They also do not require mandates to be set up or held onto because the recipient of the RfP has complete control as to if and when the payment is made. Billers also benefit from straight-through-processing (STP) and enhanced reconciliation due to the RfP ISO 20022 message itself can contain the remittance information that the corresponding payment can incorporate into the ISO 20022 credit transfer message

Request-for-Information (RfI)

The usage of modern payment messaging standards for Faster Payments facilitates the exchange of much more than simple value-related messages for transactions. Systems that utilize ISO 20022 (FedNow and the RTP Network) can send messages that enable financial institutions to seek additional details pertaining to credit transfers or RfP instructions, inquire about the status of information request processing, and promptly respond to information requests. As a result, all parties have access to information needed to resolve inquiries, disputes, and security concerns effectively. The debit card networks also possess enhanced data messaging capabilities, with users able to receive notifications on purchases that may exceed transaction limits.

Difference between instant and Same Day payments.

While both instant and Same Day payments are considered faster payment types, there are numerous differences in terms of speed, use cases, risk, and liquidity management. As its name suggests, instant payments are posted within seconds of initiation and can be initiated 24/7/365. In comparison, Same Day ACH payments post and settle on the same day they are initiated and are subject to the payment systems' and financial institutions' daily cut-off times.

For instance, as of October 2024, Nacha's Same Day ACH's cut-off times are at 10:30 AM, 2:45 PM and 4:45 PM ET, with settlement occurring at 1:00 PM, 5:00 PM, and 6:00 PM, Monday-Friday. Second, Same Day payments are file-based whereas instant payments are message-based and the ISO 20022 messages support much more structured data elements than the fixed file format of ACH. Third, both payment types provide faster funds availability compared to traditional credit and debit transfers and support similar use cases, such as faster payroll payments, emergency payments for payouts and bill pay, and B2B invoicing. The difference is naturally the speed of the availability of funds which, in some cases, needs to be shorter than a business day, for instance to avoid late fees due to payment deadlines. Lastly, from a risk management perspective, instant payments are irrevocable in case of fraudulent activities or if incorrect payment details have been submitted, while Same Day payments can be returned between the cut-off and settlement times if a return request is submitted in time.

TABLE 3. SETTLEMENT CHARCTERISTICS BY FASTER PAYMENT NETWORK

PAYMENT RAILS	SPEED OF POSTING	SETTLEMENT FREQUENCY	TYPE OF FUNDS SETTLED	FORMAT
FedNow	Seconds	Continuous	Central bank funds	ISO 20022
RTP Network	Seconds	Continuous	Prefunded balance account using central bank funds	ISO 20022
Same Day ACH	Not defined	Three times/day	Central bank funds	Nacha
Mastercard Send	Up to 30 minutes	N/A	Commercial funds	ISO 8583
Visa Direct	Up to 30 minutes	Once daily	Commercial funds	ISO 8583

End-user funds availability timing and corresponding interbank settlement methods for each faster payment rail

Settlement differs across Faster Payments schemes in terms of frequency, funding method, and timing of settlement.

Same Day ACH has three processing windows with three corresponding settlement windows, settling participant positions on a net basis.

FedNow settles transactions in real time between participating FIs Federal Reserve master accounts.

In order to achieve immediate funds availability with no risk to the Receiving Participant, RTP Payments are immediately settled between the Payer's FI and the Payee's FI by the RTP System. This is accomplished through the use of an external Prefunded Balance Account held by the Federal Reserve Bank of New York. The Account is jointly owned by all Funding Participants and Funding Agents and TCH is the sole agent of the Account.

Mastercard Send does not have a fixed settlement window and lets participants organize the settlement of transactions among each other.

Visa Direct also settles net positions, but only once a day, and it ensures that funds are available to the recipient in seconds (but up to 30 minutes with certain exceptions).

Figure 1. End-to-end payment processing cycle

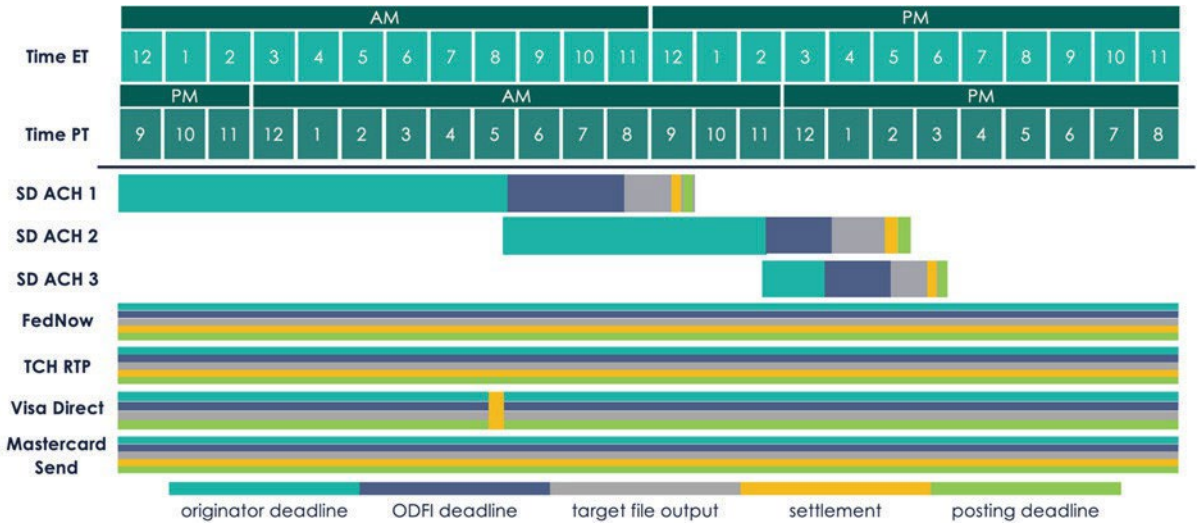


TABLE 4. SETTLEMENT METHODS BY FASTER PAYMENT NETWORK	
PAYMENT RAILS	***SETTLEMENT METHOD***
FedNow	Real-time gross settlement
RTP Network	Real-time gross settlement
Same Day ACH	Multilateral deferred net settlement
Mastercard Send	Multilateral deferred net settlement (pre-funded)
Visa Direct	Multilateral deferred net settlement

Relevance of irrevocability of final funds for each faster payment rail

Irrevocability refers to the point at which a payment becomes final, meaning that neither the originator or the originating entity can recover the funds – even in some cases of human error or fraud. Traditional ACH payments can be recalled by the originator or originating entity up until the payment is submitted for processing (which can sometimes be hours or even more than a day if a payment is originated at night or during a weekend or holiday). ACH payments, for example, have extensive protections regarding fraud, so payments can be requested back from the RDFI or returned by the RDFI once processed through the network. FedNow and the RTP Network have Request for Return of Funds messages which have corresponding rules of when these requests are required for certain cases of fraud or payments send in error.

Instant Payments, on the other hand, often have rules in place that make payments irrevocable once they are submitted. This requires certain protections to be in place to prevent misdirected payments (such as confirmation-of-payee or alias-based payment initiation) as well as sophisticated anti-fraud systems being in place.

TABLE 5. SPEED OF POSTING, TIMING OF SETTLEMENT, & TIME BEFORE PAYMENTS ARE FINAL

PAYMENT RAILS		SPEED OF POSTING	TIMING OF SETTLEMENT	LENGTH OF TIME BEFORE FINAL
FedNow		Seconds	Continuous 24/7/365	Immediate after settlement
RTP Network		Seconds	Continuous 24/7/365	Immediate after settlement
Same Day ACH	**1st**	same day Cut off time is 10:30 AM ET Posting by 1:00 PM ET	Not defined	Return entries can be submitted within 2 or 60 days after settlement, depending on the reason for return
	2nd	same day Cut off time is 2:45 PM ET Posting by 5:00 PM ET	Not defined	
	3rd	same day Cut off time is 4:45 PM ET Posting by 6:00 PM ET	Not defined	
Mastercard Send		Up to 30 minutes	Acquirer or sponsor bank is responsible for timing of settlement	Immediate after posting
Visa Direct		Within 1 minute	Net settlement occurs Mon-Fri 8:00 AM MT	Immediate after posting

Factors impacting the adoption & application of Faster Payment solutions

Many Faster Payment solutions are not currently used every day by consumers and their wide adoption requires much more than just the technical ability to initiate a Faster Payment. The value of new payment solutions to any given end-user depends on the ability to reach a substantial number of other end users as well as many other factors, such as convenience, safety, price, additional services, etc.

Reach, interoperability, & ubiquity

Interoperability as a concept refers to two separate technical systems being able to communicate or exchange and then settle payment messages with one another, much like FedACH and EPN in the ACH space. The Federal Reserve defines it to mean that payment messages are routed or exchanged and settled such that they seamlessly reach the receiver, without the individual or business even being aware of the path taken by the payment. In terms of Faster Payments, FedNow and the RTP Network, are not interoperable with one another. Mastercard Send and Visa Direct are interoperable with one another, but not with the other Faster Payment solutions. Same Day ACH is interoperable as described above, but it is not interoperable with the other Faster Payments solutions as well.

Ubiquity and interoperability are distinct, but similar: while interoperability is concerned with distinct systems ablility to work with one another, ubiquity describes the degree to which a specific service can reach all end users. In this sense, reach and ubiquity measure the same thing: ubiquity is the end point of reach.

ACH is ubiquitous: ACH payments can reach any end user with a bank account in the US. The card networks are also ubiquitous, but FedNow and the RTP Network are not yet until all banks are participants on these networks and any bank can reach any bank account.

Culture and technology

Standard ACH and cards have become everyday parts of people's lives. Familiarity and trust mean that it could be difficult for some end users to move away from these trusted payment methods and try something new, especially if the newer options – namely Faster Payments – do not seem to offer much of an advantage compared to the other payment options.

Value-added services (VAS), such as aliases to ease payment initiation or confirmation-of-payee, which helps prevent misdirected payments and fraud, can go a long way to make Faster Payments either more convenient, more trustworthy, or add some other type of value that provides end users with a reason to use these systems instead of other options. Others, such as consumer-facing services like Zelle or biller-facing technologies like Request-for-Payment (RfP), can help FIs improve their relationship with consumers or offer businesses new ways to get paid. QR codes and NFC technology are other ways to make initiating a payment easier and more secure.

Established corporate processes and goals influence the adoption of Faster Payment systems, as certain businesses or organizations may be less willing to try a new payment option and prefer to stick to systems that they know and trust, such as checks. Changing these habits can be both a challenge and a hindrance to the wide adoption of a more digital payment system.

In some Faster Payment Implementations, the user experience would not expect the user to know or care about the payment type and service that is being used behind the scenes and may just be asked to choose a timing along with any associated fees which may vary with the payment type and service being used behind the scenes.

Security and trust

Unfortunately, Faster Payments has sometimes led to faster fraud. Fraudsters often benefit from the irrevocability that some types of Faster Payments offer, making fraud prevention key to protecting end users rather than relying on rules that allow payments to be reversed once fraud is established. This has forced financial institutions to monitor transactions in real time and for Faster Payments network to integrate certain types of information sharing between participants.It has led to improved user experience by enabling confirmation that the payee and their account are valid and match the intended recipient. Additionally, AI-driven enhancements now support behavioral and anomaly detection for both the payer and payee, as well as the transaction amount. These capabilities help prevent fraud and reduce errors, making the payment process more secure and reliable.

Payer authorization requirements by rail

Payment authorization refers to how the users authorize funds to be debited from their account in order to execute a payment. One example is when a user's account is debited when making a payment, such as a P2P transfer to a family member, or when a business auto debits a customer's account (i.e. bill payment). Both payments required authorization: in the P2P example the user authorizes their bank to make a payment to their family member's bank whereas in the bill payment, the consumer provided the business with authorization (i.e. a mandate) to periodically debit the consumer's account.

Payer authentication (both for individuals and business customers) refers to how participants ensure that the person initiating a payment instruction is allowed to do so. Faster Payments networks either require or recommend specific methods to authenticate users. In addition to those rules enforced by the respective networks, the Federal Financial Institution Examination Council (FFIEC) released guidance to provide financial institutions with examples of effective risk management principles and practices for access and authentication. These principles and practices address business and individual customers, employees, and third parties that access digital banking services and financial institution information systems.

The table below describes payer authentication and authorization requirements per Faster Payments network.

TABLE 6. PAYER AUTHENTICATION AND AUTHORIZATION RULES BY FASTER PAYMENTS NETWORK

FASTER PAYMENTS NETWORK	PAYER AUTHENTICATION	PAYER AUTHORIZATION
FedNow	Participants are advised to use "strong user authentication practices"	Implied during the authentication process
RTP Network	Sending participants must, at a mini- mum, utilize multi-factor authentication to authenticate the identity of customers who transmit payment Instructions to the Sending Participant.	Implied during the authentication process
Same Day ACH	Participants are advised to follow the FFIEC's guidance on authentication methods. For WEB debits, Originators are advised to verify account ownership, though Nacha Operating Rules are neutral with regard to specific methods or technologies that could be used.	Depends on the SEC. Authorization can be oral, written, or even not necessary (see appendix for a detailed view of authorization types by SEC)
Mastercard Send	There are no mandatory authentication methods. Participants are advised to use secure authentication methods such as multi-factor authentication, address verification, and/or other methods. Participants may also use value-added services to increase security such as Mastercard's Account Verification Service (AVS).	Authorization is implied during the authentication process
Visa Direct	There are no mandatory authentication methods. Participants are advised to use secure authentication methods such as multi-factor authentication, address verification, and/or other methods. Participants may also use value-added services to increase security such as Visa's Payment Account Validation: Address Verification.	Originators must provide the Terms and Conditions to the customer who must accept them before using any Visa Direct service. Authorization is implied during the authentication process

Sources: FedNow service risk management capabilities; RTP Risk Management and Fraud Control Requirements and RTP Network Operating Rules; Nacha Supplementing Fraud Detection Standards for WEB Debits; Visa Developer Center, Mastercard Developer Center

The purpose and benefits of confirmation-of-payee

Confirmation-of-payee (CoP) is an initiative in the payments industry designed to help prevent fraudulent authorized push payments and misdirected payments by confirming the receiver's information before approving a payment transaction. CoP enables payment initiators to check the name associated with the recipient's account and compares it to the name provided by the sender. If the names match, the payment proceeds as usual. If there is a mismatch, the bank alerts the person making the payment, who can then check the details and either correct the mistake, cancel payment initiation, or continue with the payment.

This service serves two main purposes and benefits. First, it reduces errors in payments stemming from inputting the wrong account information or account holders with similar names. Second, it helps prevent fraud, especially Authorized Push Payment (APP) fraud, which is where customers are tricked into making payments to a fraudster. Services like CoP increase user confidence and contribute to improving the user experience by preventing consumers from making misdirected transactions or sending money to fraudsters.

Nacha requires any ACH originators to have a fraudulent transaction detection system that validates first-use consumer account numbers when originating debit payments that are either authorized or initiated over an online channel. This has been in place since March 2021.

Other than the commercially available validation services like confirmation-of-payee, ACH prenotification and ACH micro-transaction verifications can serve the same purpose by validating an account number and attributes, such as ownership and address. A prenotification is a non-monetary entry (i.e., no funds move as a result of the transaction) that precedes the first live entry. The prenotification entry looks identical to the subsequent live-dollar entries apart from the dollar amount and transaction code. The Receiving Depository Financial Institution (RDFI) verifies the accuracy of the account data.

Micro-Entries are defined by Nacha as ACH credits of less than $1, and any offsetting debits, used by an originator for the purpose of verifying a receiver's account or an individual's access to an account. Nacha further requires originators of micro-entries to use commercially reasonable fraud detection, including the monitoring of micro-entry forward and return volumes.

Message types

Value messages instruct funds be moved between participants and are cleared and settled through each rail. Non-value messages, on the other hand, are messages that do not generate an accounting entry and typically contain details or the status of a value message, report, or request for information, request for payment, and request for return of funds.

ACH Network payments rely on Standard Entry Class Codes (SECs), indicating the payment message type, encompassing both value and non-value messages. Certain codes are versatile for both ACH credits and debits, while others are exclusive to one or the other. Codes can also be used for consumer or non-consumer transactions. A few common value-related codes in this section. Please consult the appendix for the code/message list in its entirety.

For Mastercard Send, originating entities can use Mastercard's Funding API to request the creation of a Funding Transaction to secure the funds being sent to the payment recipient. Once approved by the issuer and Mastercard, the Payment Transaction can take place.

Originating entities using Visa Direct have the option to initiate their Visa Direct payments either by ISO-formatted messages or as an API call submitted directly to the Visa network using the Funds Transfer APIs. Other APIs that provide value-added services apart from initiating payments are also available.

FedNow message types

Value messages:

- **Customer credit transfer:** A credit transfer is sent by a sending FI to a receiving FI to initiate a funds transfer from a sender's Account to a receiver's Account.
- **Liquidity management transfer (LMT):** Also referred to as Financial Institution Credit Transfer, it is used to transfer funds between the Master Accounts or Correspondent account(s) of two FedNow Participants, or between a FedNow Participant's Master Account and a joint account.
- **Payment Return:** Used to return previously received funds.

Non-value messages

- **Service Rejections:** A message can be rejected if it fails validation. All rejection messages from the FedNow Service, for instance, include an error code to inform the reason for rejection.
- **Accept Without Post (ACWP):** By responding with ACWP, the participant does not make funds immediately available to its customer and have an opportunity to investigate (until midnight the next day) the payment for legal and compliance reasons.

- **Request for Information:** A request for information is used to communicate for more information on an instant payment.
- **Request for Payment:** A request for payment allows a sender FI to initiate a payment request on behalf of itself or its customer.
- **Return for Return of Funds:** A request made by the sending participant to have funds returned from the receiving participant only for special cases such as for errors and fraud.
- **Standalone Remittance Advice:** This is where structured B2B payment remittance details can be sent separate from the payment made through any of the network rails. The same structured remittance advice can also be embedded along with credit transfer and some of the non-value cash management report messages.

RTP message types

Value messages

- **Credit Transfer:** A credit transfer is sent by a sending FI to a receiving FI to initiate a funds transfer from a sender's Account to a receiver's Account. Note that RTP Network uses enhanced credit transfers for payment returns.
- **FI-to-FI Credit Transfer:** A FI-to-FI Credit Transfer is a payment from one FI to FI to increase the Receiving FI's Current Prefunded Position during hours when the Fedwire Funds service is closed.

Non-value messages (not exhaustive)

- **Request for Information (RFI):** A Payment-related Message that a Receiving Participant submits to the RTP System for delivery to Sending Participant to so it may request from a Customer additional information in connection with an RTP Payment or Request for Payment (RFP).
- **Request for Payment (RFP):** A Payment-related Message that a Message Sender submits to the RTP System to for delivery to a Message Receiver to request an RTP Payment from a Customer .of a Message Receiver.
- **Request for Return of Funds (RFRF):** A Payment-related Message that a Sending Participant submits to the RTP System to request return of funds related to an RTP Payment.
- **Remittance Advice:** A Payment-related message that a Sending Participant submits to the RTP System that is associated with, and that provides additional information about, an RTP Payment.
- **Stand-alone Remittance Advice:** A Payment-related message that a Sending Participant submits to the RTP System that is associated with, and that provides additional information about, an RTP Payment.

Same Day ACH message types

Value messages

- **Accounts Receivable Entry (ARC):** Applies to a single direct debit transaction, initiated by a consumer or other entity. Used to convert an eligible source document (i.e. scanned check) received by mail, in person, or at a manned location to collect funds from the other parties' accounts.
- **Customer Initiated Entry (CIE):** Applies to customer-initiated credit transfers through a bill payment service provider, authorizing the transfer of funds from the consumer account to the account of the biller along with the identification of the biller's customer information.

- **Corporate Credit or Debit Entry (CCD):** refers to a recurring or non-recurring credit or debit entry made between two corporate accounts. Frequently used to concentrate funds from outlying accounts, fund payroll, B2B payments, or other disbursement accounts. Can contain a single addendum to record payment-related information.
- **Point of Sale Entry (POS):** Refers to a single consumer-initiated credit or debit entry at a POS terminal using a card or other access device that may have scanned a consumer check at the point of sale.
- **Website Entry (WEB):** Used for website initiated customer authorized debit payments.
- **Telephone Entry (TEL):** Used for over the phone, including IVR customer authorized debit payments.
- **Corporate Trade Exchange (CTX):** Refers to a single B2B initiated payment which can include remittance details typically formatted using EDI 820 formatted messages recorded in addenda records.

Non-value messages

- **ACH payment acknowledgment (ACK):** Non-monetary entry sent to acknowledge a CCD credit entry.
- **Notification of change (COR) or refused notification of change:** A non-monetary entry used to inform participating FIs about changes in a customer's banking information or to indicate the rejection of such changes, used to ensure accurate and updated information for fund transfers.

Mastercard Send message types

Value messages

- **Disbursement Payment:** Refers to a single payment initiated by a business, non-profit, or governmental organization to disburse funds either to businesses or individuals.
- **Person-to-Person Payment:** Refers to a single payment initiated by an individual to another using a "Payment Transfer API."
- **Funding Transaction:** Refers to a single payment initiated by an entity to pull funds from an individual's account.

Non-value messages

- **Account Information:** Retrieves eligibility, card type, card brand, and funds availability information of a card account.
- **Account Verification:** Refers to an account check with a card's issuer, on account status, validity, address, and/or Card Verification Code

Visa Direct message types

Value messages

- **Original Credit Transaction (OCT):** Refers to a single payment initiated by a business, non-profit, or governmental organization to disburse funds either to businesses or individuals.
- **Account Funding Transaction (AFT):** Refers to a single payment initiated by an individual to fund another person's non-merchant account.

Non-value messages

- **Watch List Screening API:** Provides a score based on a check on an individual's information that matches with the OFAC SDN watch list.
- **Reports API:** Provides transaction capabilities including transaction reconciliation data.
- **Query API:** Provides an option to query the processing status of Visa Direct transactions in real-time.
- **Refund API:** Refers to processing merchandise returns, in other words, refund, transactions.
- **Receive Side API:** Facilitates the implementation of mobile push payments, for clients to receive merchant payment, cash-in, and cash-out transactions.

How Faster Payments impact end-user cash flow management

One of the key values of Faster Payments lies in the quick availability of funds for receivers. This impacts end-user cash flow management by posting funds within a few seconds for instant payments, within minutes for push-to-card payments, or in a few hours for Same Day ACH transactions. Faster Payments benefit consumers, merchants, and corporates by offering enhanced visibility into cash flow and enabling easier and better cash and liquidity management. It is easier for individuals and businesses to track and monitor their incoming and outgoing funds when payments are processed more quickly, eliminating problems associated with the gap between payment submission and posting. This immediacy allows for more accurate and up-to-date financial data, improving the ability to make informed cash flow decisions.

On the payer side, Faster Payments grant consumers and businesses the flexibility to complete last-minute or emergency payments, lowering the likelihood of late fees, account overdrafts, and issues with suppliers. Having immediate access to funds means payers are less likely to rely on short-term and often costly financing like payday loans or short-term liquidity infusions. Consumers can also benefit from earning of interest for longer time frames when funds come in sooner and funds going out as late as possible.

Resource-strained small merchants and businesses benefit from more timely receipt of payment, improving cash flow. Merchants and businesses benefit from more working capital, reducing the need for loans. Benefits of straight-through processing, which is aided by request-for-payment and remittance advice structured messages, decreases the need for physical documentation or manual involvement, lowering operational costs.

The impact of Faster Payments on financial institution liquidity management

One challenge all FIs face is liquidity management. The implementation of one or more Faster Payment systems necessitates more and improved liquidity management processes, which could lead to strategic enhancements and operational adjustments.

Batch payment files, such as with Same Day ACH, are processed at specific times throughout the business day. This regularity allows FIs to plan their resources, striking a balance between maintaining enough liquidity to settle net obligations while avoiding the excessive holding of cash that could otherwise be used elsewhere. Real-time payments, on the other hand, are by their nature 24/7/365, which leads to more unpredictable payment behavior and in turn higher liquidity costs and more uncertainty. Push-to-card are a bit in-between: while settlement is not real-time, they are pre-funded, which costs liquidity. Push-to-card obligations settle in commercial funds, however, so this requires a separate pool of liquidity.

Financial institutions already forecast their liquidity needs. The transition to instant payments is still ongoing for many FIs and they will lack data during their early days or weeks of implementation to forecast liquidity requirements in the beginning; this can be a challenge for higher-value payments, such as business-to-business payments. Greater utilization of Faster Payments presents heightened uncertainty, potentially complicating FIs' efforts to forecast liquidity flows and manage them efficiently.

Some faster payment systems, such as FedNow and the RTP Network, settle immediately, necessitating increased automation of processes (sanctions screening, fraud checks, authentication, authorization, etc.). Other processes, such as liquidity forecasting, gain significance because they are essential to the sound operation of these systems. The 24/7/365 nature of instant payments also means that backup plans for emergencies become crucial to ensure uninterrupted service.

Financial institution liquidity management in correspondent/ respondent relationships

Correspondent-respondent relationships in Faster Payments refer to bilateral agreements where one FI (the correspondent) offers settlement services to another FI (the respondent), enabling access to payment infrastructures. The respondent FIs benefit because they can free up liquidity for other purposes and can save on personnel and operations costs because they do not need to monitor their liquidity position as tightly as they would if they settled on their own behalf. Lastly, these relationships can also save on connectivity costs. From the correspondent side, while their liquidity costs go up, they can charge a fee to the respondent for these services, creating a new revenue stream. Correspondents are also then able to spread their own connectivity costs around, creating economies of scale. These arrangements assume several roles when it comes to liquidity management:

- **Provider and receiver of liquidity:** Correspondent FIs typically have more capital and more extensive resources than respondent FIs. Respondent FIs rely on correspondent FIs for settlement services, freeing up their liquidity for other uses.
- **Payment clearing and settlement:** While respondent FIs initiate payments and transactions for their customers, correspondent FIs facilitate the clearing and settlement of payments on behalf of respondent FIs. They provide infrastructure access or even process transactions efficiently and securely for their respondents.
- **Intraday liquidity management:** Intraday liquidity refers to funds that can be accessed during the business day or the opening hours of the payment and settlement system for financial institutions to make payments. Respondent FIs manage their intraday liquidity needs with the help of correspondent FIs, which ensures that they have the necessary funds to fulfill payment obligations throughout the business day.

The introduction of Faster Payments has impacted these roles by requiring FIs to have the liquidity to settle these transactions promptly, in some cases instantly. In the case of pre-funding, liquidity needs are increased compared to Same Day ACH or traditional ACH. These higher costs bring their own potential issues, including:

- **Risk management implications:** Managing risk is a critical aspect of these relationships. For smaller FIs, Faster Payments can have an outsized impact because 24/7 operations force smaller FIs to forecast liquidity needs for non-traditional banking hours (nights, weekends, holidays). This adds risk because payments will fail if sufficient liquidity is not there. For correspondents, they often impose limits on how much liquidity the respondent can use, enabling the correspondent to forecast their own liquidity costs, which can be difficult as the number of respondents they are responsible for increases

Knowledge of the roles, responsibilities, and requirements of Faster Payments stakeholders

Roles, responsibilities, and requirements of faster payment stakeholders

FedNow stakeholder roles and responsibilities

Financial institutions:

The various types of participants in the FedNow network are able to perform different functions. FIs can have multiple roles within the FedNow service, such as offering instant credit transfers to their own customers as well as offering settlement services to other FIs for obligations that other FIs incur on the FedNow network. The table below outlines the various types of participation possible:

TABLE 7. FEDNOW PARTICIPATION TYPES & SERVICE OFFERINGS

FEDNOW STAKEHOLDERS	PARTICIPATION TYPES	CONNECTION SETUP	SERVICE OFFERINGS*
Financial Institution	Financial Institution	Direct	**Customer credit transfers and Request-for-Payment:** • Receive customer credit transfers • Send and receive customer transfers • Receive requests for payment **Financial institution transfers:** • Liquidity management transfers • Settlement services for financial institutions
		Via service provider	
		Direct and service provider	
	Correspondent	Direct	
		Service provider	
		Direct and service provider	
	Service Provider	Direct	
Service Provider	Service Provider	Direct	

The FedNow Service Operating Procedures, through Operating Circular 8, outlines the responsibilities participating financial institutions when they join the system. This includes 24/7/365 reachability on the network, the ability to respond to various message types (see operating procedures for more information on messaging types by participant type), and remaining in compliance with FedNow operating procedures service procedures as well as security and risk management procedures. This includes upholding standards for mutual payment message signing, key pairing, encryption, and authentication laid out as a part of the FedNow operating guidelines. Regardless of participation type, it is a best practice for FedNow participants to maintain and update their own negative lists of suspicious entities and are obliged to reject transactions where one of the parties' routing/account numbers is included on an suspicious entities list. Participants are also expected to manage their master account balances to avoid overnight overdrafts and to stay within intraday overdraft capacity limits.

FedNow differentiates between two types of FI participants, respondent FIs and correspondent FIs:

- **Respondent FIs:** Access the FedNow network via a third party correspondent that maintains a Master Account with the Fed Reserve and settles transactions on the respondent's behalf. Respondent FIs may also rely on correspondents for liquidity management services.
- **Correspondent FIs:** A financial institution that maintains a Federal Reserve Master account on behalf of another financial institution that provides payments and other services for FIs that are participants in the FedNow system. These entities are responsible for conducting transactions or settlement for respondent FIs in compliance with FedNow operating procedures concerning timeliness, posting, and security, as well as for acting in their own capacity as full system participants

Infrastructure operators

Network operator, regulator, and settlement services: The Federal Reserve is the developer and operator of the FedNow faster payment service. The Federal Reserve's responsibilities include system development, operation, oversight, stakeholder engagement, regulatory compliance enforcement, and systemic risk management within the payment service. The Federal Reserve is also the settlement system for the service, with access to payment services, account information, and secure communication provided via FedLine Advantage, the Federal Reserve's electronic interface platform.

Third-party service providers

Service providers in the FedNow network are entities that perform various functions on the participant's behalf and possess an Authorized Connection Profile with the Federal Reserve. This allows them to process and manage messages and payments on behalf of their respondent parties. Third party service providers within the FedNow ecosystem can take the following forms:

- **Payment processors:** These entities connect FIs to the FedNow network and are responsible for processing transactions and providing compliance tools and fraud management services. Processors initiate, transmit, and receive payment messages on behalf of their FI clients while also managing their connection to the system. Additional responsibilities can include selecting security procedures and processing options.
- **Correspondents:** examples include bankers banks and corporate credit unions. These entities are responsible for providing FedNow services that their respondents have not elected to provide, such as the liquidity management and clearing and settlement services described above.
- **Banking operations:** These entities can provide accounting, reconciliation, operational, and fraud and risk management services as part of FedNow. Responsibilities primarily include the support of core banking and accounting operations as well as customer support and the provision of customer interface platforms.

RTP Network stakeholder roles and responsibilities

Financial institutions:

Participant FIs:

Participant financial institutions are responsible for submitting, processing, and receiving transactions on behalf of their customers, including businesses and consumer). Only depository institutions with a Master Account with the Federal Reserve (or a funding arrangement with a FI that has one) are eligible to become participants in the RTP Network. Participant FIs can be divided into two categories:

- **Funding Participants:** A Participant that has become a party to the RTP Prefunded Balance Account Agreement with the Prefunded Balance Account Bank and that requests and receives disbursements from the Prefunded Balance Account, as permitted by the RTP Operating Rules, to its Federal Reserve account and if the Participant is a Sending Participant, prefunds for itself in accordance with the RTP Participation Rules and these RTP Operating Rules.
- **Non-funding Participant:** A Participant that is not a Funding Participant and that has an agreement with either (i) a Funding Manager to act on the Participant's behalf with respect to its funding obligations under the RTP Participation Rules and these RTP Operating Rules or (ii) a Funding Provider to be a Non-funding Group Member..

Participants must ensure that they are compliant with the RTP Network Operating Rules , technical specifications, risk management, and fraud reporting and controls on a 24/7/365 basis. Participants are also required to immediately make available information regarding the status of an RTP payment to both the sending and receiving parties. Security requirements include the mandatory usage of multi-factor authentication and fraud/risk-screening measures before the submission of a customer's payment message to the RTP Network.

Funding Agent A depository institution that has become a party to the Prefunded Balance Account Agreement and is either a Funding Manager or a Funding Provider.facilitate transactions on behalf of FIs that are not direct funding participants within the RTP Network. Funding Agents have two profiles, funding managers, which provide separate funding for each non-funding RTP participant; or funding providers, which provide shared funding for participants in a "non-funding" group. These FIs are responsible for adhering to the same operating procedures and security protocols as all other participant FIs in the RTP Network.

TABLE 8. RTP PARTICIPATION TYPES AND SERVICE OFFERINGS

TCH RTP STAKEHOLDERS	PARTICIPATION TYPES	CONNECTION SETUP	SERVICE OFFERINGS
Depository institution	Receive-only participant	Direct	• Receive credit transfers • Initiate credit transfers • Non-payment messaging services • Liquidity management services
		Third-party access	
	Sending and receiving participant	Direct	
		Third-party access	
Non-depository third-party service providers	Third-party service provider	N/A	May offer additional VAS such as alias directory services and security services such as tokenization

Infrastructure operators:

Network operator: The Clearing House

The Clearing House manages the RTP Network and scheme rulebook as well as security services such as network-level anti-fraud alerts and screenings. The Clearing House is responsible for payment clearing services, compliance monitoring, activity reporting to federal banking regulators, and emergency response in the event of service disruption. The Clearing House also facilitates the settlement of payments within its internal RTP ledger. To this effect, The Clearing House ensures that a sending participant has the minimum pre-funded position necessary to facilitate a transaction before initiating a payment and proceeding with settlement within the ledger.

The Prefunded Settlement Model:

All RTP Payments that are Accepted or Accepted without Posting by the Receiving Participant are immediately and settled in real-time with finality on an individual (rather than net) basis. This form of settlement is real-time gross settlement or RTGS. The RTP System settles each Payment by debiting the Net Position of the Sending Participant and crediting the Net Position of the Receiving Participant in the amount of the Payment. The RTP System maintains a Current Prefunded Position (CPP) for Funding Participants. 2 These Participants send funds to the Prefunded Balance Account via Fedwire to fund their CPP. Once the Funding Participant has provided initial funding to the Prefunded Balance Account, it can begin submitting Payment Messages to RTP. Upon receipt of a Payment Message from the Sending Participant, the RTP System checks to ensure that the Sending Participant has a sufficient CPP to cover the amount of the Payment. As the RTP System increases and decreases the Net Positions of Participants to settl

Same Day ACH Network stakeholder roles and responsibilities

Financial institutions:

Originating Depository Financial Institution: The ODFI is the FI of the sending party in a transaction and is the institution responsible for submitting entries to the ACH Network in accordance with scheme rules. Not all FIs are required to initiate ACH entries and participate as ODFIs, and by consequence, not all RDFIs are ODFIs. When submitting an ACH entry an ODFI is responsible for having an established originating agreement with the sender and complying with Nacha risk management standards. ODFIs must receive authorization from the receiving entity to initiate entries to their account for transactions that are not credit entries between natural persons. Regarding security, ODFIs are prohibited from passing on sensitive information to third parties, such as account/routing numbers, without expressed permission. ODFIs must also implement commercially viable fraud detection systems in-house and conduct identity verification of other FIs/entities in transactions. ODFIs are responsible for accepting and processing rule-compliant return entries for transactions.

Receiving Depository Financial Institution: The RDFI is responsible for receiving payment instructions from the ACH submitted by originators or third-party senders and providing funds availability for receivers. RDFIs must accept entries that comply with Nacha operating rules. All Depository FIs that are a part of Nacha are required to comply with the requirements of an RDFI, although not all RDFIs are also ODFIs. RDFIs are obligated to accept rule-compliant ACH entries and provide information (date and amount) about entries to receivers while adhering to Operating Rules on timely processing and posting rules. RDFIs are responsible for processing and sending rule-compliant returns and notification of change entries for transactions.

For Same Day ACH transactions made within the days first processing window, RDFIs are responsible for ensuring that funds are posted to the receiver's account no later than 1:30 pm (ET), and no later than 5:00 pm (ET)for the second processing window.

TABLE 9. SAME DAY ACH PARTICIPATION TYPES AND SERVICE OFFERINGS

SAME DAY ACH STAKEHOLDERS	PARTICIPATION TYPES	CONNECTION SETUP	SERVICE OFFERINGS
DFIs	ODFI	Direct	• Credit and debit transfer submission and processing services • Receiving of credit and debits • Correspondent settlement services
		Third-party access	
	RDFI	Direct	
		Third-party access	
	Third-party service provider	Direct	
		Third-party access	
Non-FI third-party service providers	Third-party service provider	N/A	• Data processing services

Infrastructure operators:

ACH Operators: The role of ACH Operators in the Same Day ACH scheme is to facilitate the electronic transfer of funds between financial institutions by processing Same Day ACH debit and credit entries. The ACH Operator is the central facility for the clearing and settlement of entries among participating FIs. The operator's responsibilities include compliance with Nacha's operating rules, especially provisions concerning national performance standards, Federal Reserve policies on settlement, processing obligations, interoperability with other operators, and risk management obligations. ACH Operators must have also entered into processing agreements with a minimum of 20 independent participating DFIs. The operator must also calculate the settlement amounts for each banking day for all entries processed by the operator and forward these balances to the Federal Reserve for settlement. The ACH Operator is also responsible for returning or reversing ACH entries that cannot be completed due to technical mistakes or erroneous filing of an entry.

Settlement systems:

Settlement via Same Day ACH in the US is provided via the Federal Reserve as well as The Clearing House's Electronic Payments Network (EPN), the two national ACH operators. All participating FIs must maintain, or have access via a correspondent, to a Federal Reserve Master Account. The Federal Reserve and EPN play a crucial role in the finalization of Same Day ACH transactions. Their primary responsibility is to ensure that the funds transferred through the ACH Network are settled accurately and securely between the participating financial institutions. This involves reconciling the credits and debits associated with Same Day ACH transactions, managing the movement of funds between Master bank accounts, and guaranteeing that the net settlement positions are balanced. In doing so, the ACH operators maintain the integrity and reliability of the ACH Network.

Regulators:

Nacha: provides the operating rules all parties must abide by in conjunction with input from the Federal Reserve and Nacha member institutions. Nacha operates as the authoritative governing body responsible for crafting and upholding the regulatory framework, standards, and operational guidelines for Same Day ACH transactions. These frameworks and operational guidelines come primarily in the form of the Nacha operating rules, which lay out the rules and regulations for the ACH Network. Nacha's responsibilities include the monitoring of data for ACH quality and rule compliance as well as standards enforcement. Nacha is also responsible for enacting data security policies and complying with national privacy laws.

Push-to-Card networks roles and responsibilities

Card networks:

Card networks are central parties within a push-to-card scheme, acting as facilitators, rule-makers, infrastructure providers, and arbiters in disputes. Card networks like Visa and Mastercard are responsible for routing transactions, authorization, and sending payment instructions. These networks also manage settlement between sending and receiving FIs, by determining net settlement position between parties at the end of the settlement window (in the case of Visa) or by maintaining a pre-funded account for the scheme's maintenance, as in the case of Mastercard. Settlement is provided by the Net Settlement Service of the card networks, which all participating financial institutions are required to join. Other security practices include the setting of transaction limits, which differ by network and use case. The card networks are also responsible for scheme rule formulation and compliance enforcement.

TABLE 10. CARD NETWORK PARTICIPATION TYPES AND SERVICE OFFERINGS

CARD NETWORK STAKEHOLDERS	PARTICIPATION TYPE	CONNECTION SETUP	SERVICE OFFERINGS
Financial institutions	Originating Institution	Direct	• Correspondent settlement services • Push-to-card credit transfers • Transaction monitoring
	Sponsor	Direct	
		No connection (Monitoring a Transaction Initiator)	
Receiving institution	Issuer	No official connection	• Receiving of push-to-card payments • Settlement involvement
Third parties	Transaction Initiator	Direct (with sponsor)	• Transaction initiation services

Financial institutions:

Receiving FIs:

Within the push-to-card schemes the receiving FI is the receiver's bank. In a push-to-card scheme, receiving FIs are responsible for enrolling and onboarding consumers (and small merchants) and their card accounts into the broader card system and verifying the end user's eligibility. These FIs are responsible for ensuring that funds sent from the sender are deposited into the receiver account/added to their card balance. Receiving FIs also are involved in aspects of settlement. In the Mastercard scheme this requires receiving FIs ensure their position within Mastercard's pre-funded account is maintained at the minimal required levels for participation whereas in the Visa scheme Visa collects the funds from the sending FI for delivery to the receiving FI.

Sending FIs:

As the FI of the sending end-user, sending FIs have a critical role in push-to-card schemes. Sending FIs are responsible for authorizing transactions, making payments to the receiving bank, debiting the cardholder's account, and collecting usage fees from cardholders. Sending FIs are also responsible for ensuring compliance with regulatory standards and scheme-specific operational rules and procedures. This includes meeting authorization timelines, security protocols, and settling within the timelines outlined by card networks as a part of their broader operating protocols.

Eligibility requirements of Faster Payments stakeholders

Participant eligibility requirements

To be eligible, all instant payments participants are obligated to satisfy the requirements set forth in the applicable participant rules. These include compliance with applicable schedules, Information security standards and requirements, technical specifications and risk management and fraud control requirements.

FedNow eligibility requirements

Eligibility requirements are based on the organization's type, role, and connection set up. The following table shows some common combinations of participant type and connection set ups along with the applicable profile requirements.

TABLE 11. ELIGIBILITY REQUIREMENTS FOR FEDNOW

FEDNOW STAKEHOLDERS	ACTING AS A..	CONNECTION SETUP	ELIGIBILITY REQURIEMENTS
Financial institution	Financial institution	Direct	Participation and Authorized Connection Profiles
		Service provider	Participation Profile
		Direct and service provider	Participation and Authorized Connection Profiles
	Correspondent	Direct	Participation and Authorized Connection Profiles
		Service provider	Participation Profile
		Direct and service provider	Participation and Authorized Connection Profiles
	Service provider	Direct	Authorized Connection Profile
Service provider	Service provider	Direct	Authorized Connection Profile

All financial institutions, regardless of their sizes, that are eligible for Federal Reserve Financial Services are able to use the FedNow service, which depends on whether or not a financial institution has access to a Federal Reserve Master Account. Following an application process from depository institutions the Federal Reserve grants access to Master Account, and those that do not wish to have their own or fall short of the central bank's expectations can use a correspondent FI to settle. Financial institutions that are connected in any type of setup must follow the Participation Profile requirements while those that act as Service Providers to other depository institutions must follow the Authorized Connection profile requirements.

TABLE 12. ELIGIBILITY REQUIREMENTS FOR THE RTP NETWORK

TCH RTP STAKEHOLDERS	ACTING AS A...	CONNECTION SETUP	ELIGIBILITY REQUIREMENTS
Depository institution	Receive-only participant	Direct	• Business located in the USA • Supervised and regulated by US regulatory authorities • Has a Federal Reserve Master Account or arrangement with a Funding Aagent • Can directly (or through third-party service provider) operate and manage RTP activity on a continuous basis, 24/7/365 • Approved by TCH as a Participant
		Third-party access	
	Sending and receiving participant	Direct	
		Third-party access	
	Third-party service provider	Direct	
Non-depository third-party service providers	Third-party service provider	N/A	• These players lack a formal role in the RTP environment and therefore lack eligibility requirements other than on behalf of an eligible participant

Same Day ACH eligibility requirements

TABLE 13. ELIGIBILITY REQUIREMENTS FOR SAME DAY ACH (ACH)

SAME DAY ACH STAKEHOLDERS	ACTING AS A...	CONNECTION SETUP	ELIGIBILITY REQUIREMENTS
DFIs	ODFI	Direct	• Agreement to Terms and General Participant Information • Reserve bank Operating Circular 4 • FedACH Participation Agreement • Compliance with General Rules (exceptions apply upon suspected illegality) • Conduct an annual audit of compliance with General Rules • Retain proof of completion of the annual audit • Conduct risk assessment of ACH activities and implement risk management program accordingly
		Third-party access	
	RDFI	Direct	
		Third-party access	
	Third-party service provider	Direct	
		Third-party access	
Non-FI third-party service providers	Third-party service provider	N/A	These players lack a formal role in the FedACH environment and therefore lack eligibility requirements other than on behalf of an eligible participant

Mastercard Send eligibility requirements

TABLE 14. ELIGIBILITY REQUIREMENTS FOR MASTERCARD SEND

MASTERCARD SEND STAKEHOLDERS	ACTING AS A...	CONNECTION SETUP	ELIGIBILITY REQUIREMENTS
Financial institutions	Originating Institution	Direct	• Adhere to Mastercard Send Program Standards and Rules • Ensure all connected Third Parties adhere, while using the platform and services.
	Sponsor (for third party)	Direct	
		No connection (Monitoring a Transaction Initiator)	
Receiving institution	Issuer	No official connection	• No need to register to participate in MoneySend Payment Transactions Program • Notify Mastercard when they are ready to accept payment transactions • Adhere to Program Standards.
Third parties	Transaction Initiator	Direct (with sponsor)	• Adhere to Mastercard Send Program Standards and Rules
		Through Originating Institution	

Visa Direct eligibility requirements

TABLE 15. ELIGIBILITY REQUIREMENTS FOR VISA DIRECT

VISA DIRECT STAKEHOLDERS	ACTING AS A...	CONNECTION SETUP	ELIGIBILITY REQURIEMENTS
Financial Institution	Enabler (Acquirer/ Processor)	Direct	• Card Capture and Verification Functionalities • Customer Support requirements • Financial and Back Office Management functionalities • Make connection to Visa via ISO or API connection, either through
		Service Provider	
	Service Provider	Through Acquiring Sponsor	
	Acquiring Sponsor	Direct	
Service provider	Gateway provider	Through Acquiring Sponsor	
Receiving entity	Issuer	No official connection	Unknown

Relationships between Faster Payments stakeholders

Regulators

Regulators are responsible for putting in place and maintaining regulatory frameworks needed to ensure the security and robustness of payment schemes while safeguarding end user interests and ensuring competition among participants. Regulators create frameworks to ensure that infrastructure operators conduct their business in a sound manner and that financial institutions manage their operations in a way the limits the risk of financial runs and collapse. Regulators might make specific requirements to different payment types, use cases or systems, or they can implement industry-wide regulatory packages that cover numerous payment types and stakeholders, such as the Electronic Funds transfer Act. Regulators often issue public consultations that end users and financial institutions can respond to, expressing their opinions on relevant matters.

Infrastructure operators

The main responsibility of infrastructure operators is to facilitate transactions between participating entities by ensuring the safety and security of clearing and settlement activities. Operators are responsible for running payment systems, enforce regulatory compliance and managing systemic risk within their payments environment. Infrastructure operators must receive license(s) to operate payment systems and often rely on (inter)national best

practices to ensure they are in compliance with both national regulations and market practices. Infrastructure operators are also subject to regulators' rulemaking.

Participants

System participants are financial institutions that originate or receive transactions. Their responsibility is to transmit/receive payments initiated by or to their customers. Participants must comply with regulatory frameworks, rules, and standards set up by regulators and infrastructure operators to participate in payment systems. This includes practices to limit the buildup of risk within the payment system. Their relationship with end users depends on their offerings and ability to handle payments.

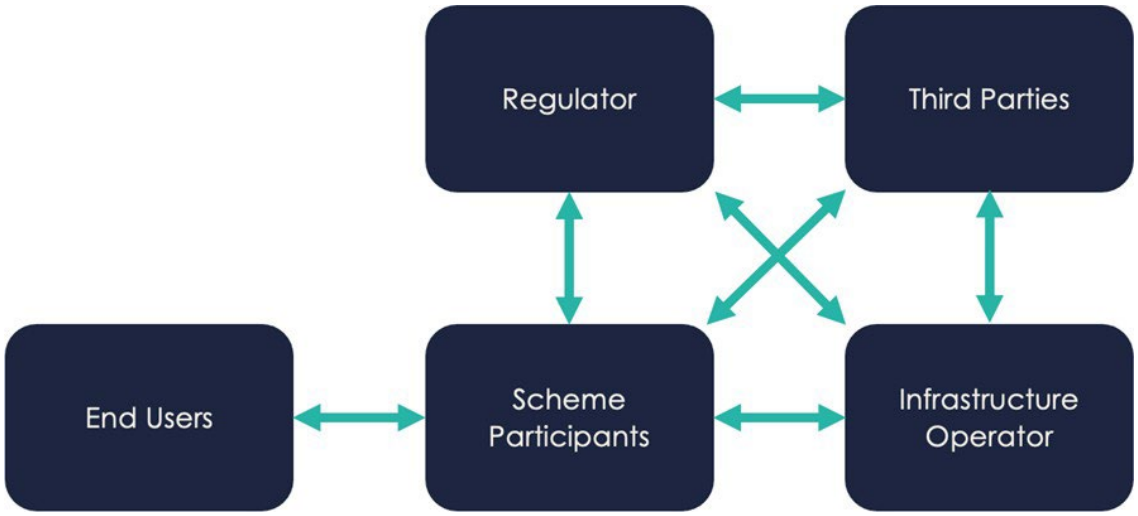
End users

Payment schemes serve end users, ensuring that they can originate and receive transactions. End users utilize payment systems to meet their obligations and receive payment for goods or services they provide. End users may also provide feedback to other stakeholders such as their financial institution, infrastructure operators, and even regulators. End users can be individuals, businesses, non-profits, and other organizations such as government.

Third parties

Third parties provide services to different stakeholders to either ensure, secure or assist with their participation in the payment ecosystem. Their responsibilities range from providing gateways for financial institutions to connecting to payment systems, facilitating payments processing on behalf of financial institutions, facilitating testing environments for participants, or creating overlay and value -added software products. Third parties, depending on the exact services they offer, must keep up with market demands and the needs of financial institutions. Moreover, they must comply with rules set by the regulator or infrastructure operators to connect to payment systems and offer services.

Figure 2. Relationships between Faster Payments stakeholders



Credits and debits in Faster Payments

A credit or "push" payment describes payment methods where the transaction is initiated by the payer; examples include peer-to-peer and funds disbursements. Debit or "pull" payments are where the beneficiary initiates the transaction (typically a business or biller) from the sender's account. The receiver is authorized to initiate this transaction based on a contractual agreement between the two parties, often referred to as a mandate. This authorizes the receiver to pull funds from the originator's account without asking for consent prior to every transaction, though the transaction must conform to the original agreement. Typical examples of debit payments include consumer bill payments.

Among the Faster Payments schemes covered in this handbook only Same Day ACH offers debits. Both instant payment systems and push-to-card schemes are limited to credit payments.

Types of messaging capabilities for each Faster Payments rail

Considering messaging capabilities, the two key characteristics payment schemes have are data standard messaging formats they use, and the type of messages they support. Data standards are important to keep messages consistent and structured, and to provide interoperability within and between payment systems. Regarding the type of messages, one can differentiate between value and non-value messages. Value messages represent funds transactions, while non-value messages ensure information exchange between scheme participants, that can include account information, transaction status, and information or payment requests.

Different payment schemes allow different messaging rights to participants that differ in their status (e.g., receiving only participant, or sending and receiving participant). This section will give a comparison of the messaging capabilities between Faster Payment rails. While details of the messaging types per participation rail can be found on section "1004.2, Message types required to be supported depend on participation types".

TABLE 16. TYPES OF MESSAGING CAPABILITIES FOR EACH FASTER PAYMENTS RAIL

PAYMENT RAILS	DATA STANDARDS AND MESSAGE FORMATS	TYPES OF VALUE MESSAGES	TYPES OF NON-VALUE MESSAGES
FedNow	ISO 20022	• Customer Credit Transfer • FI to FI Credit Transfers (Liquidity Management Transfers) • Return of Funds	• Request for Payment (RfP) • Information Request • Return Request • Payment/Request Status • Stand Alone Remittance Advice
RTP Network	ISO 20022	• Instant Credit Transfer • FI-to-FI Credit Transfer (to adjust prefunded position when Fedwire service is closed)	• Request for Payment (RfP) • Request for Information • Payment Acknowledgement • Stand Alone Remittance Advice
Same Day ACH	Nacha	• Credit Entry • Debit Entry	• Payment Acknowledgement • Automated Accounting Advice • Notification of Change/Refusal • Death Notification • Automated Enrollment • Truncated/ Destroyed Check Notification
Mastercard Send	ISO 8583	• Disbursement Payment • Person-to-Person Payment • Funding Transaction	• Account Information • Account Verification
Visa Direct	ISO 8583	• Original Credit Transaction • Account Funding Transaction	• Watch List Screening • Reports • Transaction Query • Account Look-up • Payment Account Validation • General Attributes

Transaction value limits for each Faster Payments rail

There are considerable differences in the amount of money that can be sent over the various Faster Payments rails, which have tended to increase over time. Same Day ACH has gradually increased its value limit since going live and now allows transactions up to $1 million. In a similar vein, the RTP Network started off at $25,000 and increased to $100,000, $1,000,000, and then $10 million whereas FedNow went live in July 2023 with an upper limit of $500,000 which was increased to $1,000,000 and will be $10,000,000 as of November 2025. FedNow participants are assigned a default value limit of $100,000, which they can decide to increase. The card networks have a much lower value limit given the nature of the use cases they target (i.e.,, typically P2P and B2C). Visa Direct sets limits on the value per transaction as well as maximum daily, weekly, and monthly value limits, and a maximum number of transactions. Master Send limits vary by use case.

Increases in the transaction value limits have enabled Faster Payments to expand into more business-centric use cases, which tend to require higher-value limits. The increase in value limits will also necessitate careful planning by Faster Payments participants to accommodate higher value flows at various times during the day, affecting financial institutions' credit risk assessments, liquidity costs, and their decisions regarding whether to offer higher dollar limits to payment initiators. Overall, increases to transaction value limits are poised to facilitate greater adoption and expand the utility of Faster Payments.

It is worth noting that network participants can and do impose their own limits – which are lower than the operator-imposed limits, – depending on a number of factors including the type of account initiating the payments (corporate versus small merchant versus consumer), the risk profile of the originator, etc. which may be lower than the limits allowed by the Faster Payment networks.

TABLE 17. DOLLAR LIMITS BY FASTER PAYMENT NETWORK

	DOLLAR LIMITS
FedNow	$10,000,000 (as of November 2025) (Default limit set at $100,000)
RTP Network	$10,000,000
Same Day ACH	$1,000,000
Mastercard Send	P2P: $10,000 A2A: $25,000 B2C: $50,000
Visa Direct	$50,000 per OCT transaction $100,000 per/day (150 txn limit) $250,000/week (250 txn limit) $500,000/month (750 txn limit)

Network service level agreements applicable to Faster Payments

Faster Payments networks define minimum standards that participants and the system itself must abide by – known as Service Level Agreements (SLAs). SLAs usually relate to the system's operations and the quality and/or speed of the services provided. These are typically defined in each Faster Payments operational and technical rules/guidelines.

Common SLAs include the maximum amount of time after which a payment message must be submitted to the Faster Payments network after the sender submits the payment instruction to the financial institution, the maximum response time to payment messages, and posting times. Other SLAs that Faster Payments networks typically define relate to system availability as well as participant availability. The table below describes some common SLA's for Faster Payments Networks.

TABLE 18. EXAMPLES OF SERVICE LEVEL AGREEMENTS

SERVICE LEVEL AGREEMENT (EXAMPLES)	DEFINITION
System availability and operating hours	Defines when the system is available for participants to send and receive messages. Example: "The network is a 24x7x365 service with no expectations of planned downtime."
Participant availability	Defines when participants must be reachable by the system. Typically expressed as a percentage of the total time of a certain time period. Example: "Participants must satisfy at least a standard of 99.5% continuous connectivity to the system, measured on a monthly basis."
Downtime for planned maintenance	Defines the maximum time, usually in hours, that participants can remain unavailable due to planned maintenance. Some networks also determine when maintenance should be carried out. Example: "Participants' planned downtime should not exceed two sequential hours or 24 hours total per quarter, and any planned downtime should be scheduled during hours with expected minimal transaction volume."
Maximum response times / Time-out	Defines the amount of time (usually seconds) after which the payment message is cancelled/rejected if there is no response from the relevant participant or system.
Posting speed	Defines the maximum amount of time within which the receiving institution must post funds to the receiver's account. Example: "Participants should make funds immediately available, as soon as practicable, but no longer than a few seconds, to its customers on a 24- hour basis each calendar day, including weekends and holidays."
Throughput time	Determines the total time a transaction can take to be processed.

Source: Nacha, based on operational guidelines and rules from the Faster Payments networks
More specifics to each of these type of SLAs are covered in Chapter 2

Functions of each participant type

Types of participation by faster payment type

Just as with traditional payment systems, faster payment networks enable financial institutions to participate in the system in various ways. Each participation type affects how FIs interact with the network, including send and/or receive capabilities, how the participant settles its obligations arising from its activity within the network, and the type of connection it has to the network.

Send and/or receive: the participant's ability to send and/or receive payments. In some networks FIs may choose to only receive payments instead of enabling their clients to send and receive.

Settlement: how the participant settles payments, either directly or via a sponsoring (funding/correspondent) agent

Connection: how the participant connects to the Faster Payments network (e.g., directly, through another participant, or a third-party provider).

The following sub-sections detail how each network identifies participant types.

FedNow participant types

Customer credit transfers and requests for payment: FIs are able to select from several options to enable send and/or receive credit transfers and Request-for-Payments (RfPs).

- **Receive credit transfers (receive only):** A participant that is able to receive payments but is not able to initiate payments except to return payments. Receive-only participants have the ability to send RfPs but are not eligible to receive RfPs because they cannot initiate customer credit transfers as a response to a payment request.
- **Send and receive credit transfers:** A participant that is able to send and receive payments as well as return payments. This participant is only able to send RfPs.
- **Receive RfPs:** This participation type allows FIs to receive RfPs. An FI enabled to send and/or receive credit transfers may send an RfP, but only those enabled to send payments are eligible to receive an RfP.

Liquidity management: FIs can participate in FedNow for the sole purpose of supporting liquidity management transfers and funding or defunding participant accounts.

- Send and receive FI Liquidity Management Transfers
- Receive-only FI Liquidity Management Transfers

Settlement only: FIs may participate in FedNow to only provide settlement services, acting as settlement agents for other FIs.

RTP participant types

Funding Participant: A Participant that has become a party to the RTP Prefunded Balance Account Agreement with the Prefunded Balance Account Bank and that requests and receives disbursements from the Prefunded Balance Account, as permitted by the RTP Operating Rules, to its Federal Reserve account and if the Participant is a Sending Participant, prefunds for itself in accordance with the RTP Participation Rules and these RTP Operating Rules. **Non-funding Participant:** A Participant that is not a Funding Participant and that has an agreement with either (i) a Funding Manager to act on the Participant's behalf with respect to its funding obligations under the RTP Participation Rules and these RTP Operating Rules or (ii) a Funding Provider to be a Non-funding Group Member. **Receiving Participant:** The Participant that holds the Receiver's Account and that receives a Payment Message.

Sending Participant: The Participant that holds the Sender's Account and initiates an RTP Payment.

Any participant must be (i) either a Funding Participant or a Non-funding Participant, and (ii) a Receiving Participant. A participant also may be a Sending Participant. This means that a participant can be a "Receiving only participant", or a "Sending and Receiving participant".

Same Day ACH participant types

Originating Depository Financial Institution (ODFI): receives payment instructions from originators (e.g. senders) and forwards the entries to the ACH Operator.

Receiving Depository Financial Institution (RDFI): receives ACH entries from the ACH Operator and posts them to the receiver's account.

An FI may participate in the ACH Network as an RDFI without also being an ODFI; however, if an FI chooses to originate ACH entries, it must also agree to act as an RDFI.

Mastercard Send financial institution participant types

Financial institutions decide whether to participate in Mastercard Send as well as the programs they want to participate in. Mastercard Send comprises three programs and Originating Institutions register their

organization and implementation partners for the program(s) relevant to their intended funds transfer solution. The three available programs are shown below.

TABLE 19. MASTERCARD SEND AVAILABLE PROGRAMS

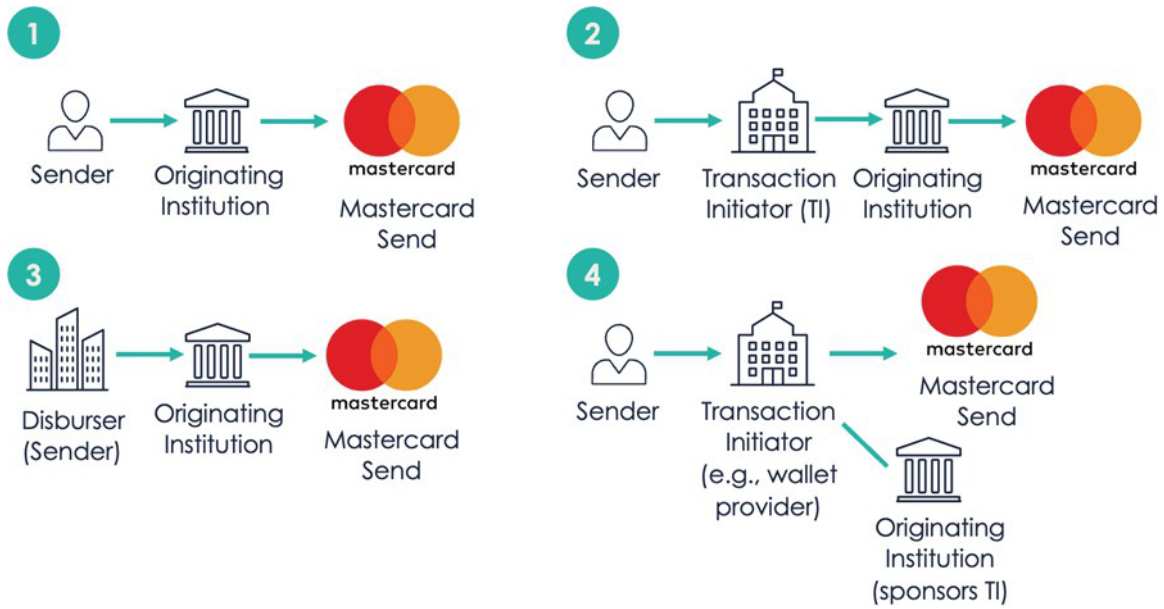
MASTERCARD SEND PROGRAM(S)	
MoneySend Payment & Funding Transactions	Governs Payment and Funding Transactions to transfer funds from/to Mastercard
Other Network Transactions	Governs various types of other network Payment and Funding Transactions that transfer funds from/to non-Mastercard accounts. Mastercard Send will route the transaction to the appropriate network.
Gaming and Gambling Payments	Governs Payment Transactions that transfer gaming and gambling payouts to Mastercard accounts.

Source: Mastercard Send, Developers portal, Documentation, Registration and Eligibility: https://developer .mastercard .com/mastercard-send/documentation/send-eligibility/

Mastercard Send allows several participation models. Participants may work alone or with licensed partners to facilitate different use cases. Some example models are shown below, which introduce some of the main Program Participant types/roles.

1. The Originating Institution (OI) (a financial institution) provides funds transfer services to its customers (senders).
2. The OI facilitates funds transfer requests from its Transaction Initiator (TI) partner.
3. The OI facilitates disbursements from a large disburser (sender).
4. The OI sponsors a TI, e.g. a digital wallet provider. The TI has a direct connection to Mastercard Send and the OI sets limits and monitors the TI's activity.

Figure 3. Examples of program participant types – Mastercard Send



Source: Mastercard Send, Developers portal, Documentation, Registration and Eligibility: https://developer .mastercard .com/mastercard-send/documentation/send-eligibility/

Mastercard issuers (Receiving Institutions) do not need to register to participate in the MoneySend Payment Transactions Program, but they do need to notify Mastercard when they are ready to accept those Payment Transactions and they must adhere to the program standards.

Visa Direct participant types

As with Mastercard Send, financial institutions decide whether to participate in Visa Direct. Originators must be a licensed Visa acquirer or be sponsored by one. Third-party service providers who offer push payment services to end users (consumers or businesses) can be sponsored by a Visa licensed acquirer to offer push payment capabilities to Visa card accounts.

Summary and conclusions

Faster Payments networks typically allow FIs to participate in the network in several different ways to allow for flexibility while satisfying each participant's needs. Some networks also enable institutions to provide services to participants (such as liquidity provision or settlement services) without participating in the network by sending or receiving payment messages. This provides participants with several options to carry out their activities and fulfill their obligations as network participants.

TABLE 20. PARTICIPANT TYPES SUMMARY

	CAPABILITIES	SETTLEMENT	CONNECTION
FedNow	• Receive-only (CTs) • Send and receive CTs • Receive RfPs • Send and receive FI liquidity management transfers • Receive-only FI liquidity management transfers • Settlement only	• On their own behalf • Through a settlement agent (correspondent)	• Directly • Through a service provider
RTP Network	• Sending and Receiving participant • Receiving participant	• Funding participant Non-funding participant through a Funding Agent arrangement	• Directly • Through a third-party service provider
Same Day ACH	• Originating and Receiving FI • Receiving only FI	• On their own behalf • Through a settlement agent	• Directly • Through a third-party service provider
Mastercard Send	• Participant • Non-participant	• On their own behalf • Through a sponsor	• Directly • Through a third-party processor
Visa Direct	Non-participant Through a sponsor	• On their own behalf • Through a sponsor	• Directly • Through a third-party processor

Source: Nacha, based on operational guidelines and rules from the Faster Payments networks

Liquidity management

- **Send and receive FI LMT:** this participant can send and receive FI credit transfers (pacs.009)
- **Receive only FI LMT:** this participant can receive FI credit transfers (pacs.009) **Settlement only:** this participant cannot send nor receive FI credit transfers (pacs.009) or customer payment messages (pacs.008). The participant can exchange other system messages (e.g., camt.060 – Account Reporting Request, camt.052 – Account Balance Report, admi.002 – Message Reject) and others.

Message types used by the various network participant types

FIs have to support specific messaging types – both value- and non-value – depending on the type of participation model and faster payment network, while other message types may be optional or even restricted to certain participant types.

This section illustrates the message types that are needed for each participation type per Faster Payments network.

FedNow message types by participant type

TABLE 21. DETAILED MESSAGE TYPES BY PARTICIPANT – FEDNOW (FINANCIAL INSTITUTION CREDIT TRANSFERS)						
FI CREDIT TRANSFERS MESSAGES (ISO 20022)	***SETTLEMENT ONLY***		***LMT RECEIVE ONLY***		***LMT SEND/RECEIVE***	
	SEND	***RECEIVE***	***SEND***	***RECEIVE***	***SEND***	***RECEIVE***
pacs.009 – Financial Institution Credit Transfer	Prohibited	Prohibited	Prohibited	Mandatory	Mandatory	Mandatory
pacs.002 – Payment Status Report	Prohibited	Prohibited	N/A	N/A	N/A	Mandatory
pacs.028 – Payment Status Request	Prohibited	Prohibited	Prohibited	N/A	Optional	N/A
ACCOUNT REPORTING MESSAGES (ISO 20022)	***SETTLEMENT***		***LMT RECEIVE ONLY***		***LMT SEND/RECEIVE***	
	SEND	***RECEIVE***	***SEND***	***RECEIVE***	***SEND***	***RECEIVE***
camt.060 – Account Reporting Request	Optional	N/A	Optional	N/A	Optional	N/A
camt.052 – Account Balance Report	N/A	Optional	N/A	Optional	N/A	Optional
camt.052 – Account Activity Totals Report	N/A	Optional	N/A	Optional	N/A	Optional
camt.052 – Account Activity Details Report	N/A	Optional	N/A	Optional	N/A	Optional
camt.054 – Account Debit/Credit Notification	N/A	Optional	N/A	Optional	N/A	Optional
SYSTEM MESSAGES (ISO 20022	***SETTLEMENT***		***LMT RECEIVE ONLY***		***LMT SEND/RECEIVE***	
	SEND	***RECEIVE***	***SEND***	***RECEIVE***	***SEND***	***RECEIVE***
admi.002 – Message Reject	Mandatory	Mandatory	Mandatory	Mandatory	Mandatory	Mandatory
admi.004 – FedNow Broadcast	N/A	Mandatory	N/A	Mandatory	N/A	Mandatory
admi.004 – Participant Broadcast	Prohibited	N/A	Mandatory	N/A	Mandatory	N/A
admi.011 – FedNow System Response	N/A	N/A	N/A	Mandatory	N/A	Mandatory
admi.006 – Retrieval Request	Prohibited	N/A	Optional	N/A	Optional	N/A
admi.998 – FedNow Participant File	N/A	Mandatory	N/A	Mandatory	N/A	Mandatory

Source: FedNow, Readiness Guide, jumpstarting the onboarding process, Participation Types: https://explore .fednow .org/resources/readiness-guide-participation-types .pdf

Customer credit transfers and RfPs

- **Receive credit transfers (receive only):** this participant is not able to send payments (ISO message pacs.008) but can return messages (ISO message pacs.004). This participant can send RfPs

(ISO message pain.013), which allows the participant to receive funds from the sending institution. This participant can also send and receive FI credit transfers (ISO message pacs.009) to meet liquidity needs.

- **Send and receive credit transfers:** this participant is able to send and receive payments (pacs.008) and return customer payment messages (pacs.004) as well as send RfPs. The participant may also choose to receive RfPs (pain.013). This participant can send and receive FI credit transfers (pacs.009).
- **Receive Requests-for-Payment (RfPs):** this participant can receive RfPs (pain.013) and can send and receive payments (pacs.008).

The table below summarizes message types by participant. For more information see "Appendix 1: FedNow – message types required by type of participant".

TABLE 22. MESSAGE TYPES BY PARTICIPANT – FEDNOW

MESSAGE TYPE (ISO 20022)	CREDIT TRANSFER RECEIVE ONLY		CREDIT TRANSFER SEND/ RECEIVE		SEND/RECEIVE WITH RECEIVE RFP	
	SEND	RECEIVE	SEND	RECEIVE	SEND	RECEIVE
pacs.008 – Customer Credit Transfer	Prohibited	Mandatory	Mandatory	Mandatory	Mandatory	Mandatory
pacs.002 - Payment Status Report	Mandatory	Mandatory	Mandatory	Mandatory	Mandatory	Mandatory
camt.056 – Return Request	Optional	Mandatory	Optional	Mandatory	Mandatory	Mandatory
pacs.004 – Payment Return	Optional	Optional	Mandatory	Mandatory	Mandatory	Mandatory
pain.013 – Request for Payment (RfP)	Optional	Prohibited	Optional	Prohibited	Optional	Mandatory
Pain.014 – Request for Payment Status	N/A	Conditional	N/A	Conditional	Mandatory	Conditional

Source: FedNow, Readiness Guide, jumpstarting the onboarding process, Participation Types: https://explore .fednow .org/resources/readiness-guide-participation-types .pdf

RTP message types by participant type

Participants enabled to send and receive customer credit transfers must be able to exchange pacs.008 messages while receive-only participants must be able to receive these messages despite not being enabled to send them. Receive-only participants cannot initiate customer credit transfers. Both receiving-only and sending and receiving participants must exchange payment status report (pacs.002) messages. If the participant supports RfP, other message types in addition to those mentioned above should be utilized, namely pain.013 and pain.014.

TABLE 23. EXAMPLE OF MESSAGE TYPES BY PARTICIPANT – RTP

MESSAGE TYPE (ISO 20022)	RECEIVING-ONLY PARTICIPANT		SENDING AND RECEIVING PARTICIPANT	
	SEND	RECEIVE	SEND	RECEIVE
pacs.008 – Customer Credit Transfer	Prohibited	Mandatory	Mandatory	Mandatory
pacs.002 – Payment Status Report	Mandatory	Mandatory	Mandatory	Mandatory

Source: Nacha, assumed based on RTP's operational guidelines and rules

There are other message types that participants should support when it comes to system management. For example, messages to sign-on from the system (admn.001) and to sign-off (admn.003) from the system in case of planned downtime, liquidity transfer messages, and echo response messages (admn.006), to name a few.

Same Day ACH message types by participant type

It is mandatory for all ACH payments to include an SEC. It is the responsibility of the Originating Depository Financial Institution to include the correct SEC in the ACH payment. SECs are attached to the ACH payment request as they are a part of the Nacha file format.

Nacha currently allows several SEC on ACH payments. Note that some codes can be used either for ACH credits and debits while others can only be used for one or the other. Similarly, certain codes can be used for one-time as well as recurring payments while others can only be used for one-time payments. Moreover, some SECs are only used for non-monetary transactions.

Certain SECs can only be used for consumer or retail transactions, while others are reserved for business and government transactions. Certain codes, such as WEB (for online authorizations) and TEL (for phone authorizations), describe where the transaction was authorized.

Refer to the appendix for a detailed view of available SECs.

Mastercard Send message types by participant type

Mastercard Send enables payments and funding via a transaction type called "MoneySend". To use Mastercard Send and send "MoneySend" messages, institutions must become a Mastercard Send program participant, which involves registering for the required program(s) and services and meeting program eligibility requirements. Participants will not be able to use the service until they are approved as a Mastercard Send Program Participant and have been onboarded and set up for the service.

Mastercard Send funds transfers may be composed of one or both of these steps:

- Payment Transaction:

  Originators make a "Payment Transfer API" (POST request) for each payment transfer. Mastercard Send validates the request and sends a Payment Transaction request message to the Receiving Institution. The Receiving Institution must respond by approving (Approved = 201 response with "status": "APPROVED") or declining (Declined = 402 error response with "ReasonCode": "DECLINE") the transaction.

  The Payment Transfer API can also be used by the originating institution to retrieve other transfer details (GET request) such as the status of the transaction.

- Funding Transaction:

  The Mastercard Send Funding service enables a participant to secure funds from a consumer's sending account that is then transferred to a Recipient. Funding a transaction is done through the "Funding API", which creates a Funding Transfer (POST request). The Receiving Institution must respond by approving (Approved = 201 response with "status": "APPROVED") or declining (Declined = 402 error response with "ReasonCode": "DECLINE") the transaction.

  The Funding API can also be used by the originating institution to retrieve other transfer details (GET request) such as the status of the transaction.

  If the overall funds transfer could not be completed successfully, Originators should use a "Funding Reversal API" (POST request) to reverse the Funding Transaction.

- Disbursement Transaction:

  Originators make a "Disbursement API" (POST request) for each disbursement. Mastercard Send validates the request and sends a Payment Transaction request message to the Receiving Institution. The

Receiving Institution must respond by approving (Approved = 201 response with "status": "APPROVED") or declining (Declined = 402 error response with "ReasonCode": "DECLINE") the transaction.

The Disbursement API can also be used by the originating institution to retrieve other transfer details (GET request) such as the status of the transaction.

TABLE 24. SUMMARY OF MASTERCARD SEND MESSAGE TYPES

MESSAGE TYPE	ORIGINATING INSTITUTION	RECEIVING INSTITUTION
Payment Transfer API (POST request)	Mandatory	Mandatory (must approve or decline)
Payment Transfer API (GET request)	Optional	Conditional
Funding Transfer (POST request)	Mandatory	Mandatory (must approve or decline)
Funding API (GET request)	Optional	Conditional
Funding Reversal API (POST request)	Optional	Mandatory (must approve or decline)
Disbursement API (POST request)	Mandatory	Mandatory (must approve or decline)
Disbursement API (GET request)	Optional	Conditional

Besides the message types mentioned above, there are other additional message types (i.e.,, additional services) to enhance the funds transfer flow. Some examples include the "Account Information Service", which retrieves information about card accounts, including sending and receiving eligibility; and the "Account Verification Service (AVS)", which checks card accounts with issuers (e.g., checks that the account is open and valid).

Visa Direct message types by participant type

Visa Direct participants must ensure that all stages of transaction processing, including transaction submission, financial settlement, dispute management, and reporting are in place before launching Visa Direct. Only originators who can correctly process every aspect of the value chain are allowed to go live.

Visa Direct provides originators with a service to push payments directly to Visa card accounts. This push payment capability is delivered through the use of the Original Credit Transaction (OCT). In addition to the OCT, Visa Direct also supports the Account Funding Transaction (AFT), which enables funds to be pulled from an eligible Visa account.

- **Original Credit Transaction (OCT):** Visa Direct originators can submit OCTs either as ISO-formatted messages through a Visa network endpoint connection or directly into the Visa network as an API call, the "Funds Transfer APIs".
- **Account Funding Transaction (AFT):** As with the OCT, Visa Direct originators can submit AFT transactions as either an ISO message or directly through an API call, "the Funds Transfer APIs".

Should a push payment be declined, the Funds Transfer API can also be used to return the funds to the sender's funding source.

Visa Direct provides other APIs for optional use such as:

- Mobile Push Payment API
- Visa Alias Directory Service
- Reports API
- Query API
- Refund API
- Receive Side API
- Adjustment API

Chapter Two

Faster Payments Operations and Process Flows

SECTION BLUEPRINT COMPOSITION: 28%

Knowledge of financial institutions' procedures for payment processing via Faster Payments rails

Participants in Faster Payments systems are required to follow certain procedures before originating and/or receiving messages. For example, there are requirements related to testing, fraud, compliance issues, ensuring participant availability, validating the receiving account, and message validation, to name a few.

Participants must ensure they have a steady network connection and the ability to respond to system messages as well as have sufficient liquidity to settle payments. Both FedNow and RTP require participants to sign-on to the system in order to confirm that they are fully connected and active participants prior to receiving transactions.

The rest of this section covers the requirements of the various Faster Payments networks that participants must comply with as well as the required message validations prior to message submission to the network.

Requirements for financial institutions before receiving originating transactions

FedNow® requirements to receive/originate transactions

Participants must establish a participant profile according to their business needs and risk tolerance before sending or receiving payments. Each FedNow participant and service provider is required to follow the FedNow Services. operating procedures and technical specifications, including, but not limited to, ensuring continuous system availability and making funds immediately available.

Participants are required to notify the Federal Reserve if they are unable to send/receive messages to or from the system and must establish monitoring and alerting capabilities to resolve any issues. Participants are expected to have an information security program for physical and logical system components and are expected to conduct an annual assessment of compliance with the Federal Reserve's FedLine security requirements.

All messages exchanged through FedNow require a digital signature generated by the participants. Participants are required to use public/private key pairs to support digital message signing. Each FedNow participant shall adopt and maintain:

- Procedures for screening sanction lists against their customer base

- A compliance program
- A Bank Secrecy Act (BSA)/anti-money laundering (AML) compliance program

TABLE 25. SUMMARY OF FEDNOW REQUIREMENTS PRIOR TO RECEIVING/ORIGINATING TRANSACTIONS

SUMMARY OF FEDNOW PARTICIPANT REQUIREMENTS
• Establish a participant profile
• Follow FedNow's operating procedures and technical specifications
• Ensure continuous availability and the ability to make funds immediately available to users
• Designate a settlement account, either a Master Account or a settlement account maintained by a correspondent
• Ensure appropriate monitoring and alerting capabilities are in place to resolve any issues that may arise
• Have information security programs in place, as well as compliance and AML programs

RTP® requirements to receive/originate transactions

Each participant must comply with the RTP Operating Rules and technical specifications, which must be tested and audited. Participants must also specify a settlement account and ensure adequate funds to settle obligations. Other requirements include the implementation of an OFAC compliance program.

Prior to submitting a payment message to the RTP Network, sending participants must:

- Have satisfied prefunding requirements
- Ensure that the payment message complies with the applicable RTP operating rules and technical specifications
- Provide the sender with the name of the receiver associated with the receiving account for payments originating from consumer accounts.

Participants must first transmit a sign-on request (ISO 20022 message ID: admn.001) message to be reachable on the RTP Network.

Same Day ACH requirements prior to receiving/originating transactions

Participants must be able to initiate ACH files in the format established in the Nacha Operating Rules. The Nacha Operating Rules include a general requirement that ACH Participants comply with U.S. law, including but not limited to compliance with sanctions programs. Other requirements include that participants conduct risk assessments of their ACH activities and implement risk management programs as well as have security policies and procedures related to the initiation, processing, and storage of information in place.

Other requirements include:

- Verification of originator or third-party sender identity
- Enter into an origination agreement with each originator for which the ODFI will originate entries

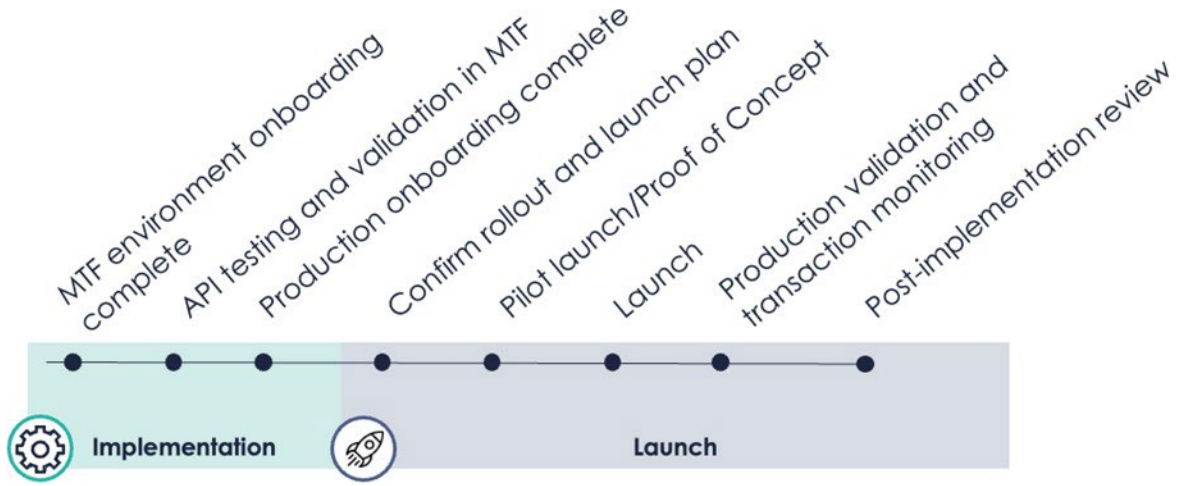
Mastercard Send requirements prior to receiving/originating transactions

Becoming a Mastercard Send participant involves registering for the required programs and services, and meeting program eligibility requirements as well as adhering to the service's standards and rules (including agreements, licenses, and establishing a settlement account).

Originators will not be able to use the APIs in the Mastercard Test Facility (MTF) or production environments until they are approved and onboarded. Once registered and onboarded, originators are given keys and a unique partner reference ID.

The table below summarizes some of the steps required prior to launching Mastercard Send.

Figure 4. Mastercard Send – simplified implementation overview



Visa Direct requirements prior to receiving/originating transactions

An originating acquirer (either itself, or through its merchants or service providers that originate an Original Credit Transaction), must:

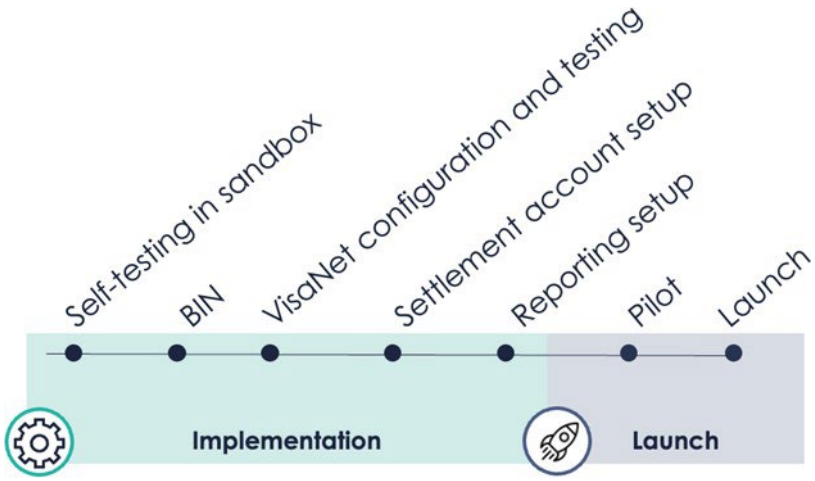
- Validate sender data and comply with applicable anti-money laundering laws and regulations and anti-terrorist financing standards
- Provide proper disclosure to the sender regarding the collection of sender data
- Notify Visa before it or its Merchant or service provider start to process any Original Credit Transactions.

Visa Direct requires the originator (who can also be merchants, P2P payments providers, etc.) to ensure that all stages of transaction processing, including transaction submission, financial settlement, dispute management, and reporting, are in place before launching Visa Direct.

Originators are expected to perform Know Your Customer (KYC), fraud, and eligibility checks on the sender and/or recipient. Visa also recommends that institutions screen transactions for high-risk and/or sanctioned transactions and set velocity limits. Participants must also ensure compliance with the Payment Card Industry (PCI) Data Security Standards and applicable law(s).

The table below summarizes the implementation steps that originators must follow prior to launching Visa Direct.

Figure 5 Visa Direct – simplified implementation overview



Source: Visa Direct Funds Disbursement Program Considerations for Enablers and Processors Push to Card, January 2020

Steps involved in Faster Payments transactions

The payments process flow consists of multiple messages. These steps may vary according to how Faster Payments networks process transactions. The information below summarizes these steps in five stages: the initiation stage, validation stage, response/processing stage, settlement stage, and posting and notification stage.

In addition to the above stages, Faster Payments networks also define the processes involved in instances when the payment fails as well as the process involved in payment returns.

FedNow Steps involved in Faster Payments transactions

1. Initiation stage:
   - User initiates a payment
2. Validation stage:
   - Sender FI sends the Customer Credit Transfer (CCT) (pacs.008) to FedNow
   - FedNow receives CCT (pacs.008) message from the sender FI and runs through all applicable validations.
     - If the message fails one of the validation checks, FedNow either:
       - Rejects the message immediately and sends a message reject (admi.002) with the appropriate reason code to the sender FI, or
       - Continues processing through validations to collect any additional errors. The FedNow Service then sends a payment status message (pacs.002) with the consolidated appropriate error reason code(s).
   - The message passes all validations and is sent to the Receiver FI.
3. Response stage:
   - The receiver FI begins to validate the CCT (pacs.008) message from FedNow.
   - The receiver FI reviews the message and responds with either:
     - Accept

- Accept without post
- Reject

4. Settlement stage:
   - FedNow processes the pacs.002 message from the receiver FI and runs it through all applicable validation checks.
   - FedNow checks that the payment timeout clock has not expired for the CCT (pacs.008).
   - FedNow confirms settlement relationships, assigns a settlement date, and settles the transaction by recording it.
5. Posting and notification stage:
   - FedNow sends a payment status report (pacs.002) within seconds as confirmation to the sender FI as acknowledgement, and to the receiver FI as advice of credit.
   - Upon receipt of advice of credit (pacs.002), the receiver FI makes the funds available to the recipient immediately and notifies the recipient of funds availability.
   - FedNow delivers the confirmation of posting payment status report (pacs.002) with the code "ACCC" ("accepted") to the Sender FI regarding the availability of funds if received by the FI.

TABLE 26. SIMPLIFIED STAGES IN A FASTER PAYMENTS TRANSACTION – FEDNOW

	INITIATION STAGE	VALIDATION STAGE	RESPONSE STAGE	SETTLEMENT STAGE	POSTING & NOTIFICATION STAGE
Sender	Initiates a payment message				
Sending FI	Authenticates the sender and receives the payment instruction	Sender FI sends pacs.008			Upon receipt of pacs.002, notifies sender (optional)
Faster Payments system		FedNow receives pacs.008 and runs validations, then sends message to receiver FI		Processes pacs.002 and runs validations. If timeout not expired, FedNow settles the payment	Sends pacs.002 as confirmation to sender and receiver FIs
Receiver FI		Receives pacs.008 and validates	Responds with pacs.002 and either Accept, Accept without post, or Reject		Makes funds available and notifies the recipient

RTP Steps involved in Faster Payments transactions

1. Initiation stage: the sending FI authenticates the sender and receives a payment instruction from the payer via a supported channel.
2. Validation stage:
   - The sender's FI performs account verification and payment approval, such as:
     i. Check the sender's account balances
     ii. Earmark funds for the debit
     iii. Perform fraud checks
   - The sender's FI creates a credit transfer message (pacs.008) and sends it to the RTP Network.

- RTP performs a series of formatting, process, and business rule validations.
- The message passes validations and is routed to the receiver FI.
- RTP awaits a response for a period defined as the "time-out period".

3. Response stage:
   - The receiver FI receives the message from RTP and conducts its own validations to ensure its correctness and security.
   - The receiver FI sends RTP a "message status report" (pacs.002) indicating that the credit transfer has either been accepted, rejected, or accepted without posting.
4. Settlement stage:
   - RTP System validates that the response was received within the time-out period.
   - Following, RTP settles the transaction.
   - RTP then passes the response along to the sending FI which then informs the sender of the payment's status.
5. Posting and notification stage:
   - RTP provides a message status report called a "message receiver confirmation" message to the receiver FI, notifying that the transaction has been settled.
   - The receiver FI makes the funds immediately available to the receiver.

TABLE 27. SIMPLIFIED STAGES IN A FASTER PAYMENTS TRANSACTION – RTP NETWORK

	INITIATION STAGE	VALIDATION STAGE	RESPONSE STAGE	SETTLEMENT STAGE	POSTING & NOTIFICATION STAGE
Sender	Initiates a payment message				
Sending FI	Authenticates the sender and receives the payment instruction via a supported Channel Application	Performs account and payment verification and approval. Creates a pacs.008 and sends to RTP			Informs the sender
Faster Payments system		Performs a series of formatting, process, and business rule validations. Message is routed to the receiver's FI		Validates the response and (assuming no time-out), settles the payment	Confirms to the sending and receiving FI pacs.002
Receiver FI		Receives the message and conducts its own validations	Sends RTP a message status report pacs.002 indicating either Accepted, Accepted Without Posting, or rejected		Makes funds available and notifies the recipient

Same Day ACH steps involved in Faster Payments transactions

1. Initiation stage: the originator initiates a payment instruction.
2. Validation stage: the ODFI next validates the payment instruction.
3. Processing stage:
   - The ODFI debits the sender's account.
   - The ODFI then sends their ACH file to an ACH Operator.
4. Settlement stage:
   - The ACH Operator forwards aggregated batches to the RDFI
   - The ACH Operator calculates and instructs interbank net settlements through Fed Master Accounts (four times daily).
5. Posting and notification stage:
   - The RDFI updates the receiver's account accordingly (i.e., posts funds) unless conditions require a return.
   - For updated receiver accounts, the receiver sees actual or pending fund balances reflected in their account.

TABLE 28. SIMPLIFIED STAGES IN A FASTER PAYMENTS TRANSACTION – SAME DAY ACH

	INITIATION STAGE	*VALIDATION STAGE*	*PROCESSING STAGE*	*SETTLEMENT STAGE*	*POSTING & NOTIFICATION STAGE*
Originator	Initiates a sending ACH instruction				
Sending FI /ODFI	Authenticates the sender and receives the ACH instruction	Validates the instruction and debits the sender's account	Sends their ACH files to the ACH operator		
EPN / FedACH			Forwards aggregated batches to receiver FI	Calculates and instructs interbank net settlement through Fed Master Accounts	
Receiver FI /RDFI				Validates the receiver's account and updates them accordingly	Notifies receiver

Visa Direct steps involved in Faster Payments transactions

1. Initiation stage: the sender initiates a payout or P2P money transfer through a payment solution.
2. Validation stage: the sending FI validates the sender data and submits the payment to the network.
3. Processing stage: the network routes the payment using debit card or international bank routing credentials.
4. Posting and notification stage:
   - The network forwards the money transfer to the receiver FI.
   - The receiver FI receives the payment, posts the funds to the receiver's account, and notifies the receiver.
5. Settlement stage: payment transactions are settled once per day.

TABLE 29. SIMPLIFIED STAGES IN A FASTER PAYMENTS TRANSACTION – VISA DIRECT

	INITIATION STAGE	VALIDATION STAGE	PROCESSING STAGE	POSTING & NOTIFICATION STAGE	SETTLEMENT STAGE
Sender	User initiates payment message				
Sending FI		Performs account & payment verification and approval. Submits payment			
Visa Direct			Network routes payment using debit card or international bank routing credentials	Forwards money transfer to receiver FI	Settles payment transactions once daily
Receiver FI				Receives payment, posts funds to receiver's account & notifies receiver	

Mastercard Send steps involved in Faster Payments transactions

1. Initiation stage: the sender initiates a payment transaction.
2. Validation stage:
   - The sending FI authenticates and screens the sender.
   - The sending FI makes a payment transfer API POST request to Mastercard Send
   - Mastercard Send validates the request, including:
     i. API request checks
     ii. Transaction controls
     iii. Limit validations
   - When the request is validated, Mastercard Send transmits the payment transaction request to the receiving FI via the relevant network for approval.
3. Response stage:
   - The receiving FI approves or declines the payment transaction request and then sends a response back to Mastercard Send.
   - Mastercard Send then transmits the API response to the sending FI.
4. Posting and notification stage: the receiving FI credits the receiver's account and notifies the receiver.
5. Settlement stage: Mastercard Send instructs settlement.

TABLE 30. SIMPLIFIED STAGES IN A FASTER PAYMENTS TRANSACTION – MASTERCARD SEND

	INITIATION STAGE	VALIDATION STAGE	RESPONSE STAGE	POSTING & NOTIFICATION STAGE	SETTLEMENT STAGE
Sender	User initiates payment message				
Sending FI	Requests payment transfer by making payment transfer API Post request to Mastercard Send				
Mastercard Send		Validates request with API request checks, transaction controls, limit validations etc. Message is transmitted to the receiver's FI via relevant network,			Instructs settlement
Receiver FI			Approves or declines the payment transaction request and a sends response	Credits the receiver's account and notifies the receiver	

Process requirements for message validation

Before participants submit messages to the rail, they must ensure that messages are consistent with system requirements and message specifications.

Specific requirements for each Faster Payments rail are described below.

FedNow process requirements for message validation

All ISO 20022 messages, including value and non-value messages, are subject to FedNow validations. ISO rules and guidelines, security procedures, and processing standards. System preparation applies to all participants and must be implemented before sending messages, including:

- Ensure accuracy of message content
- Ensure connection to the system, including meeting technical, operational, and security protocols
- Ensure that the participant's profile is appropriate:
  - A participant profile to send and receive customer credit transfers
  - A participant profile for receiving credit transfers only
  - LMT send and receive
  - LMT receive only
- Ensure that message queues (endpoints) are established with the FedNow Service
- Reference FedNow's participant list and Broadcast Messages to validate that the intended receiver FI is active and signed on to the system
- Use of a reliable time server and include the timestamp in the payment message to determine start of the payment timeout clock.

- Adhere to the FedNow Service network limits
- Ensure having an active public key(s) for message signature
- Use a unique message identifier, also referred to as the unique message ID, assigned to each new message
- Ensure that the value message is $0.01 or greater
- Implement a system to support the defined receiver response time rules and guidelines
- Ensure that the XML syntax and FedNow Service ISO 20022 schema rules are followed

Some message types have other items that participants must validate. The table below shows validation items for FedNow participants sending customer credit transfers. See Appendix 4 for validation items for other message types.

TABLE 31. VALIDATIONS FOR CUSTOMER CREDIT TRANSFER – FEDNOW

VALIDATIONS FOR CUSTOMER CREDIT TRANSFER
Valid message signature
Valid message size
XML syntax
Authorized sender
Confirm unique message ID
FedNow Service ISO 20022 schema rules followed
Sending and Receiving FIs are FedNow participants, are active and signed on, and have profiles to send & receive payments or receive only. Sender must have a profile for sending and receiving payments
Payment timeout clock has not been exceeded
Message is not future dated
Message is within Receiver FI reserved response time
Message is within FedNow Service network transaction limit or Sender FI transaction limit
Sender and Receiver FIs have valid settlement relationships
Optional: Negative list check, if enabled

RTP process requirements for message validation

Messages must be submitted according to RTP Network's technical documentation which defines the standard message formats used within the RTP Network. Prior to submitting credit transfers to the RTP Network, the sending institution checks the sender's account balances, earmarking funds for the debit, and performs fraud checks. Participants must validate that the payment does not exceed the maximum transaction limit for each message.

Same Day ACH process requirements for message validation

Prior to submitting ACH entries to the ACH Operator, originating financial institutions must validate:

- That the file is not a duplicate of one that has previously been processed
- The originator information, including qualification to originate
- The field contents, such as codes and numeric requirements. Each entry type has specific conditions that must be met in order for the entry to be considered properly authorized.
- Eligibility for SameDay ACH processing
- Dollar totals at batch and file levels and

- That origination activity does not exceed agreed-upon limits
- That there is a valid agreement/authorization from the end user to initiate this payment

When originating financial institutions consolidate multiple ACH files, they must:

- Ensure that the file contents are valid and will meet the ACH Operator's criteria
- Assign trace numbers to each record which must be unique within a batch

Mastercard Send process requirements for message validation

Mastercard Send participants must send payment transfer messages in the formats requested by Mastercard Send. It is implied that sending participants must validate that messages follow the guidelines and that participants must enforce the applicable transaction limits and conduct value and velocity checks to confirm compliance with those limits.

Aside from payment message validation, Mastercard Send provides participants with optional Value Added Services (VAS) to help them validate the existence and eligibility of the accounts used for funds transfers. Mastercard Send recommends, but does not mandate, participants to use the Account Information Service and Account Verification Service (in that order) to check sender and recipient card accounts prior to making payment transfer API calls.

Account Information Service to check:

- The sending card account is eligible to send funds
- The receiving card account is eligible to receive payment transfers

Account Verification Service (AVS) to check:

- The sending card account is still valid/open and verify the provided sender's details
- The receiving card account is still valid/open and verify the provided recipient's details

Visa Direct process requirements for message validation

Visa provides the technical specifications for exchanging messages through the available APIs within its API Developer Center. The type of information that must be included when using the different APIs available differs slightly due to the different functionalities they deliver, but all generally require the below information. Participants are expected to validate that this information is included and that the message complies with the processing guidelines.

- An acquiring BIN and its country code
- Business Application Identifier (BAI): This is a two-character code that identifies the intended use of a push payment.
- Information about an individual sender or sending entity: These fields identify the sender and sender transaction narrative as it will appear on the cardholder's statement.

Participants also validate that the card is eligible to receive OCTs and that it is a valid account. Visa Direct also makes VAS available to help participants validate the information of the user's account, such as the Account Look-up API, which helps validate key characteristics of a recipient card before initiating the transfer, such as country, card type, block status, etc.

Knowledge of the factors that impact message acceptance and rejection

Numerous parties and processes play a role in the Faster Payments transaction flow. Each party must ensure that its part of the payment flow operates smoothly and adheres to both operational and technical rules.

This section describes the points of failure for messages across the payment flow, the reasons why payment messages fail, the techniques implemented to mitigate payment rejections and failures, and the responsibilities of sending financial institutions when receiving message rejections and failures.

Status responses from receiving financial institutions and their impact on payment completion rates.

Participants exchange a series of messages, including value and non-value messages, as part of the process flow in Faster Payments transactions. When it comes to FedNow and RTP, one particular non-value message exchanged in the response stage is the payment status report (pacs.002).

In the FedNow and RTP Networks, receiving financial institutions respond to the sending institution with a pacs.002 message indicating whether the credit transfer has been accepted, rejected, or accepted without posting. FedNow and RTP also use pacs.002 to notify the receiving FI that the response has been received within the time-out period and the transaction has been settled, allowing the receiving FI to make the payment immediately available to the receiver. See the figure below for a simplified process flow of a credit transfer within RTP showing when are message status reports (pacs.002) used.

Participants can also send a payment status request (pacs.028) to FedNow/RTP or another participant to inquire about the status of a previously sent message. Status report messages and response messages help participants acknowledge every step of the payment flow and allow them to react when one of these steps fails.

Figure 6. Example of a successful credit transfer in RTP

Source: The Clearing House Customer Documentation, Introduction to the RTP Network, October 2020

Similarly, participants in push-to-card networks exchange messages that indicate if the receiving institution approved or declined the transaction. Participants can also exchange messages to obtain the latest status of the payment message.

Potential points of failure in the payment flow

A payment message can fail for any number of reasons, though specific points of failure are linked to the involved parties at each step. For instance, the sending FI and its access channels are the potential point of failure during the initiation stage, though the sending end user needs to ensure that sufficient funds are avail- able in the sending account to initiate the payment. If the end user is initiating the payment via a third-party provider, this is another potential point of failure because the third-party provider needs to have a secure connection to the Faster Payments network.

Once a payment message moves to the validation stage the sending FI must ensure the necessary information is included in the payment message that it is submitted in the correct format. Payment messages can be rejected or fail at different points throughout the Faster Payments flow, which is illustrated in the table below.

TABLE 32. POTENTIAL POINTS OF FAILURE IN THE FASTER PAYMENTS FLOW – FEDNOW & RTP

	Initiation stage	Validation stage	Response stage	Settlement stage	Notification and posting stage
Sender (initiation channels)	X				
Sending FI	X	X			X
System connection	X	X			X
Liquidity provider				X	
Faster Payments network		X	X	X	X
Liquidity provider				X	
System connection			X		X
Receiver FI			X		X

Note that some Faster Payments networks within the scope of this handbook post funds to the receiver's account before funds are settled. This means the above figure will look slightly different for different types of Faster Payments rails.

TABLE 33. SUMMARY OF POTENTIAL FACTORS INVOLVED IN FASTER PAYMENTS FAILURE – FEDNOW & RTP

	INITIATION STAGE	VALIDATION STAGE	RESPONSE STAGE	SETTLEMENT STAGE	POSTING AND NOTIFICATION STAGE
Sender (initiation channels)	Insufficient funds, failure in initiation channels				
Sending FI	Suspected fraud or illicit activities, exceeding value limits, 24/7 availability, and failure in connection to third-party service provider	Inaccuracy of the message contents, inappropriate participant profile, incorrect keys for message signature, lack of a unique message identifier, incorrect XML syntax and ISO 20022 schema, and issues with system connection			Inability to receive payment status report and notify sender
System connection	Lack of 24/7 availability, malfunctioning systems and connection to sending FI	Lack of 24/7 availability, malfunctioning systems and connection to the network			Lack of 24/7 availability, malfunctioning systems and connection to the network and sendingFI
Liquidity provider				Unavailable settlement account with sufficient liquidity	
FedNow RTP Network		Invalid message signature, incorrect XML syntax and ISO 20022 schema, lack of a unique message ID, timeout, exceeding network transaction limits, future dated payment, inappropriate or incorrect participants and profiles	Inability to exchange response messages with sending and receiving institution	Unable to process messages and validation, timeout, unsuccessful transaction settlement	Inability to send payment status reports to sender and receiver.
Liquidity provider				Unavailable settlement account	
System connection		Lack of 24/7 availability, malfunctioning systems and connection to the network	Lack of 24/7 availability, malfunctioning systems and connection to the network		Lack of 24/7 availability, malfunctioning systems and connection to the network and receiving FI
Receiver FI		Inability to open, read and validate the message, and timeout	Inability to send a message response		Inability to receive payment status report and post funds to receiver

The following section describes the specific factors that cause failure in the processing of Faster Payments.

Factors involved in Faster Payments failures

Validation takes place at various stages of the payments message flow. Potential factors contributing to payment message failures include connection issues, system availability, lack of reach, inaccurate information in the payment message, and data formatting issues. Security failures, such as those pertaining to message signing, and settlement issues, such as a lack of liquidity, are also potential points of failure.

Issues around fraud, and money laundering can also cause a payment to fail, even if the payment in question is not actually fraudulent or be related to money laundering.

The figure below summarizes the factors that may cause a message to fail at each stage of the payment flow for an RTP and FedNow payment message.

The factors that can lead to payment message failure are the same with Visa Direct and Mastercard Send, though the failures occur at different times within the payment flow due to differences in how settlement takes place.

The figure below summarizes potential factors involved in a Mastercard Send funds transaction failure.

TABLE 34. SUMMARY OF POTENTIAL FACTORS INVOLVED IN FASTER PAYMENTS FAILURE – MASTERCARD SEND

	INITIATION STAGE	VALIDATION STAGE	RESPONSE STAGE	POSTING AND NOTIFICATION STAGE	SETTLEMENT STAGE (POST POSTING)
Sender (initiation channels)	Failure in initiation channels				
Sending FI	Suspected fraud or illicit activities, lack of user agreement, invalid funding source and failures while securing funds	Invalid sending and receiving account, invalid message formats, exceeding value and velocity, lack of unique ID			Unavailable settlement account
System connection		Malfunctioning systems and connection to network			Malfunctioning systems and connection to the network and sending FI
Liquidity provider					Unavailable settlement account with insufficient liquidity at the time of settlement
Mastercard		Failures in API request checks, transaction controls, and exceeding limits	Timeout, network communication issues		
Liquidity provider					Unavailable settlement account
System connection			Malfunctioning systems and connection to network		Malfunctioning systems and connection to network and receiving FI
Receiver FI			Validation errors, missing required data	Invalid receiving account, and user agreement	

With regards to Same Day ACH payments, suspected fraud or illicit activities, failures in the user authentication and authorization process, duplicated files, missing or incorrect information within the payment message, lack of eligibility for same-day ACH processing, lack of liquidity for payment settlement, and exceeding limits on the dollar totals at the batch and/or file levels are all potential causes of failures.

Methods and techniques for mitigating message rejections and failures

FedNow methods and techniques for mitigating message rejections and failures

Both participants and Faster Payments system operators use tools and processes to minimize message rejections and failures throughout the payment flow. The specific validations that every participant should conduct prior to submitting messages to the network and at every ensuing step are clearly defined (e.g., validating message syntax and the ISO 20022 syntax, validating unique message IDs, etc.) within FedNow's operating guidelines. There are validation requirements and processes for every step of the payment flow and participants (through FedNow) share the status of the payment to provide certainty to the counterparty. FedNow also recommends participants refer to a reliable time server to support accurate tracking of the processing time and avoid unnecessary rejections (i.e.,, due to timeouts).

Participants must be continuously available and ensure appropriate monitoring and alerting capabilities to resolve potential issues. When issues occur, participants are expected to sign off from receiving credit transfer messages. Broadcast messages are used to notify participants of changes to participant availability to receive credit transfers as well as a list of participating routing transit numbers. When it comes to preventing rejections and failures related to settlement, FedNow enables FI-to-FI liquidity management transfers in support of liquidity needs.

RTP methods and techniques for mitigating message rejections and failures

When it comes to RTP, as with FedNow, participants must be continuously available and validate messages before submitting them to the network. Also, payment messages are validated by the relevant party at every step of the payment flow. Message status reports and confirmations are used in RTP as they are in FedNow. RTP also uses system-level messages to control which participants are active on the system.

Participants must provide immediate responses to messages and receiving participants must accept generally all payment messages that are consistent with RTP operating guides. Regarding settlement issues, participants must monitor their prefunded positions and provide supplemental funding when needed to ensure settlement. Participants can receive alerts from the system when their prefunded position falls below a certain value and they are encouraged to have arrangements with each other to transfer liquidity if needed during non-Fedwire hours.

Same Day ACH methods and techniques for mitigating message rejections and failures

Participants in Same Day ACH validate messages prior to submission. The Nacha Operating Rules specify how files, batches, and entries should be structured and the type of information to be included. In the ACH Network the RDFI may create a "notification of change" to notify the ODFI that a posted entry contains invalid or erroneous information and should be changed (e.g., notification that something about a bank account has changed). Participants are advised to implement analyses of transaction errors.

Push-to-card methods and techniques for mitigating message rejections and failures

Both Visa Direct and Mastercard Send provide value-added services to help participants validate the information provided in payment messages prior to payment submission. For example, sending FIs can validate that the card is eligible to receive push-to-card transactions and that it is a valid account.

TABLE 35. EXAMPLES OF METHODS USED TO MITIGATE MESSAGE REJECTIONS AND FAILURES IN FASTER PAYMENTS RAILS
EXAMPLES OF METHODS USED TO MITIGATE MESSAGE REJECTIONS/FAILURES IN FASTER PAYMENTS RAILS
Provide a list of necessary validations prior to message submission
Validating the existence and availability of receiving accounts
Sharing the status of the payment throughout the payments flow
Ensuring continuous availability and ensure proper monitoring and alerting of issues
System-wide messages that communicate participant availability
Flexible options to support liquidity needs
Monitoring settlement accounts
Analyzing transaction errors

Responsibilities of sending financial institutions when receiving message rejections and failures

Faster Payments networks define specific actions that each party has to conduct when messages are rejected by the network or by another participant. This section describes the actions to correct failures or react to failures by Faster Payments rail.

FedNow

The steps below summarize the sending FI's responsibilities at each stage of the payment message (pacs.008) flow.

- If FedNow does not receive the pacs.008 from the sending FI, the sending FI should wait a minimum of 25 seconds to inquire about its status.
  - The sending FI can send a payment status request to the FedNow Service (pacs.028) or search in the FedNow interface Adhoc Query Tool. If FedNow responds "message is not found", the sending FI can resend the message with the same previously used message ID. If FedNow does not respond, the sending FI should check its connection.
- If FedNow does receive the pacs.008 from the sending FI but the message fails a validation (i.e.,, the message is rejected by FedNow), the sending FI needs to review the error code description and correct the error to ensure that subsequent messages meet requirements.
- If a payment is rejected, the FedNow Service sends a message (ISO message pacs.002) to the Sender FI notifying it that the payment was rejected.
- If the Receiver FI has received the contents of a payment message in a request for confirmation but does not respond before the timeout clock expires, the Receiver FI also receives a message notifying that the payment has been rejected.
- If the Sender FI resubmits a payment message that the FedNow Service already received, it needs to include a new unique message identification number. Otherwise, the service rejects the message based on the duplicate identification number.
- Sending FIs should wait a minimum of 25 seconds if it does not receive a pacs.002 status report message before inquiring as to its status.

RTP Network

The RTP Network cancels payment messages and notifies the sending and receiving FIs if the receiving FI fails to respond to a payment message from the RTP Network within the established timeframe. Rejected RTP payments are not settled. Sending FIs should ensure that sending users receive confirmation that their

payment was rejected (or successful) within seconds. When it comes to rejections related to insufficient settlement funds, sending participants must make sure that they have sufficient liquidity before submitting payment messages and leverage the existing processes and tools for liquidity management (e.g., FI-to-FI transfers, prefunded balance warning notifications, etc.).

Particiapnts can send a payment status report (pacs.028) to inquire about the outcome of the transaction if a processing error results in the participant not knowing the outcome of the transaction.

Same Day ACH

The ACH Network supports return/reject entries for specific reasons. Returns/ rejections can be initiated from one of two sources: the ACH Operator and the RDFI.

This section will focus on returns by the ACH Operator, for example, those transactions that never reached the RDFI. In these cases, the ODFI must:

- Forward the return to the originator for action: the originator should be notified that entries were returned by the ACH Operator so that it can initiate a corrected entry or contact the receiver about using an alternate method of payment.
- Reinitiate an entry for the originator if it has been returned because of insufficient or uncollected funds. Sending FIs should be aware of the processing windows offered by ACH Operators for the processing of returns.

Mastercard Send

After sending a transaction (Payment Transfer API), the sending FI must await a response from the receiving FI. In some cases, such as timeouts or network communication issues, a transaction response can have an 'unknown' status. Mastercard Send will continue trying to retrieve the status from the receiver network, though sending FIs are recommended to perform an API lookup (Payment Transfer API GET request) on the relevant transaction after a minute to retrieve more information for a payment transfer, including its latest status. Sending FIs should not resubmit a transaction that has an 'unknown' status because the original transaction may have been processed successfully. In order to resubmit the same transaction, sending FIs must first use a GET request to check the status of the original transaction request to ensure that Mastercard Send has not financially processed the transaction. Only when the original transaction is in 'ERROR' should they resubmit the transaction.

Given that Mastercard Send transactions may consist of one or both of these steps: 1) Funding transaction to secure the funds that will be transferred (e.g., in the case that the originating institution is not the sender's account issuer) and 2) Payment transaction to transfer the funds to the Receiving Institution. When a payment transaction is not completed (i.e.,, declined), the originator must reverse the funding transaction that secured the funds i.e.,, they must return the funds to the funding source. The funding reversal must be submitted within 30 minutes of the funding transfer request.

Visa Direct

As with Mastercard Send, a Visa Direct transaction can consist of pull funds transaction and a subsequent push funds transaction. This means that originators can use the Funds Transfer APIs (or the ISO transaction) to pull funds from a Visa card account and to push funds into a Visa card account. In this context, if the push funds transaction fails, originators must return funds to the sending user (i.e.,, they must reverse the pull transaction).

Knowledge of the exceptions handling processes for Faster Payments rails

Exceptions processes are carried out when messages are rejected, fail, or in other scenarios, such as when a receiving FI accepts a payment but will not post the funds to the receiver until further validations are conducted ('accept without posting' message). Both FedNow and RTP allow participants to accept payment messages without posting. Other examples include exceptions processes that participants carry out to request a return of funds or "notification of change" messages utilized in Same Day ACH to notify the ODFI that a posted entry contains invalid or erroneous information and should be changed.

As new use cases for Faster Payments rails emerge, new types of exceptions processes are developed. Request- to-pay, for example, has required defining processes for canceling requests. The following section describes the exceptions-handling processes in different Faster Payments networks and the tools enabling these processes to be carried out.

Payment messages and exceptions processes

Different payment messages can lead to different types of exceptions. This section describes the main exceptions processes related to each Faster Payments rail.

TABLE 36. MAIN EXCEPTIONS PROCESSES PER FASTER PAYMENTS RAILS (FEDNOW, RTP AND SAME DAY ACH)

MAIN EXCEPTIONS PROCESSES PER FASTER PAYMENTS RAIL	
FedNow	Those related to payment returns (return requests), message rejects (i.e., the receiving FI rejects the payment message), canceling RFP messages, and 'accept without posting' messages.
RTP Network	
Same Day ACH	Reversing files and entries, notifications of change, and returning entries (i.e.,, the receiving FI or the ACH operator rejects the payment).

Tools to enable exception processing

Faster Payments rails provide tools to enable exceptions processing in a reasonable amount of time such as non-value messages, participant and system alerts, system notifications, and participant service support centers. This section summarizes the tools available to participants by Faster Payments rail.

FedNow – Tools to enable exceptions processing

FedNow participants use non-value messages to initiate exceptions processes such as payment return requests or RFP cancellations as well as to receive information regarding the status of a payment.

Messages

- **Payment status (pacs.002):** informs the sender of the status of a previously sent message to another participant or FedNow. The response will indicate whether the credit transfer was Accepted, Accepted without Posting, or Rejected. "Accepted without Posting" responses and "Rejected" responses imply an exception handling process.
- **Payment status request (pacs.028):** request the processing status (from the system or another participant) of a previously sent message.
- **Information request (camt.026):** Receiving FI asks the sending FI to provide further information on a previously sent customer credit transfer to understand the purpose of the payment.
- **Ping message (admi.004):** to check the connection to FedNow

- **Message reject (admi.002):** inform a participant or FedNow that the message has been rejected or that the participant cannot process the message.
- **Additional Payment Information (camt.028): Provide additional information about a transaction, such as customer credit transfers or requests for payment.**
- **Return request (camt.056):** Request a participant to return some or all of a previously settled payment or liquidity transfer.
- **Payment return message (pacs.004):** Return previously received funds.
- **Return request response (camt.029):** Processing status of a previously sent return request (camt.056).
- **RFP cancellation request (camt.055):** Request to cancel a previously sent RFP (pain.013).
- **RFP cancellation request response (camt.029):** Processing status of an RFP cancellation request.

Other tools:

- **FedNow interface Adhoc Query Tool:** participants can run ad-hoc queries for information retrieval via the FedNow interface and filter by time to see messages that were sent to FedNow.
- **FRB Services Support Center:** The support center provides technical and service setup support for using, installing, or navigating the service as well as problem reporting.
- **FedNow Service public key list:** Participants are required to maintain a list of all active FedNow public keys. Participants must initially use the FedNow interface to obtain the FedNow public key. After the participants have the initial list, they can use the FedNow interface or MQ Messaging to maintain it.
- **Activity reports**: includes account activity reports

RTP – Tools to enable exceptions processing

Particpants in the RTP Network have specific messages available to initiate exceptions processes.

Messages:

- **Message status report (pacs.002):** informs the sender of the status of a previously sent message to another participant or the RTP Network. The response will indicate whether the credit transfer was Accepted, Accepted without Posting, or Rejected.
- **Payment status request (pacs.028):** request the processing status from another participant of a previously sent message.
- **Request for information (camt.026):** Receiving FI asks the sending FI to provide further information on a previously sent customer credit transfer to understand the purpose of the payment.
- **Response to request for information (camt.028):** the sending FI sends a response to camt.026 message sent by the receiving FI
- **Request for return of funds (camt.056):** Request a participant to return some or all of a previously settled payment or liquidity transfer. Camt.056 messages are also used to indicate "request for payment expiry (RFPE)": Sent by receiving FI to mark a previously sent RFP as expired so that a payment is not made in response to the RFP. In addition, Camt.056 messages are also used to notify system time-out: Sent by RTP to the receiving FI when such FI does not respond within the system SLAs.

- **Response to request for return of funds (camt.029):** Processing status of a previously sent return request (camt.056)

Other tools:

- **Activity reports:** includes account activity reports, customer support teams and a portal for retrieving information

Same Day ACH – Tools to enable exceptions processing

For Same Day ACH, exceptions handling refers to entries that do not post to the receiver's account, questions or requests for additional information, file/entry reversals, etc.

When an entry does not post, the RDFI must either manually post the entry; manually post the entry and initiate a notification of change; or return (i.e.,, reject) the entry. Payment returns are exchanged through return reason code messages that contain the reason for rejection.

Messages/processes:

- **Return entries:** The ACH Network allows for returning (i.e.,, rejecting payments), the process allows participants not to accept an entry and to return it to the originator through the ACH Network. An ACH return is a communication that indicates why the entry could not be completed as intended. The reason for return is communicated via a return reason code to the participant receiving the return (e.g., insufficient funds, invalid account number, closed account, etc.). These are called return codes and they consist of an R followed by two numeric digits (i.e.,, R01). Originators may reinitiate returned entries only if:
  - the entry was returned for insufficient funds
  - the entry was returned for stopped payment and re-initiation has been authorized by the receiver (e.g., recurring funds collections/debit transfers), or
  - the sending FI / its sender has taken corrective action to remedy the reason for the return In certain cases, the ODFI can request the RDFI to return entries (request returns).
- **Reversing files and reversing entries:** An originator, its ODFI, or an ACH Operator may initiate a reversing entry to reverse all entries of an erroneous or duplicate file or to correct an erroneous or duplicate entry.
- **Notifications of change:** are used to notify the sender of an ACH payment that something about a bank account has changed.

Push-to-card – Tools to enable exceptions processing

Mastercard Send and Visa Directs' APIs are also used to reverse funding transactions when the subsequent push payment fails (usually because the push payment was declined by the receiving issuer or in the event of a timeout). In such case, the originating participant must return the funds to the sending user.

Mastercard Send – Messages / APIs:

- **Funding API (POST request):** Creates a funding transaction to secure (pull) funds from a sending account that will be sent to a receiver. Includes a "Funding Reversal API" that enables participants to reverse a funding transaction (within 30 minutes) if the associated payment transaction is declined or rejected (due to an error).

- **Payment Transfer API (POST request):** This API is used to push funds to a receiving account.
  - When the payment is sent to the network and processed, Mastercard includes the Network Response Codes (ISO Codes) from the receiving network in the payment transfer API response messages. The response code indicates an issuer's reason for approving or declining a transaction. Mastercard send provides participants with a list of error reason codes. This helps participants understand the reason for decline and carry out the appropriate exceptions process.
  - In case of failed payments (or those with 'unknown' status), Mastercard Send allows participants to process transactions with "Idempotency". This means that participants can make the same Payment Transfer API POST request multiple times without worrying about the transaction being processed multiple times. If Mastercard Send detects that a request matches a previous transaction request submitted within the preceding 24-hour period, it will ensure that the transaction is processed only once. If the transaction has already been processed, Mastercard Send will respond with the current status of the transaction. If Mastercard Send did not receive or process the original request (in other words, Mastercard Send has not seen this transaction request before), it will process the transaction and provide a response.
- **Payment Transfer API (GET request):** Used to retrieve more information for a payment transfer, including its latest status. Example uses include getting the network decline code and description when a transaction is declined, getting the latest status when a transaction response was 'Unknown', retrieving transaction history and research purposes.

Visa Direct – Messages / APIs:

- **Funds Transfer API:** The Funds Transfer API can pull funds from the sender's Visa account (in preparation for pushing funds to a recipient's account) in an Account Funding Transaction (AFT). The Funds Transfer API also pushes funds to the recipient's Visa account using an Original Credit Transaction (OCT). If a transaction is declined, the Funds Transfer API can return the funds to the sender's funding source in an Account Funding Transaction Reversal (AFTR).
  - The API response indicates if the transaction was approved or declined. If declined, participants can obtain a reason code for the decline. Visa shares a list of reason codes and descriptions which help participants carry out the appropriate exceptions handling process.
- **Query API (GET request):** This API is used to check whether a PullFunds, PushFunds or ReverseFunds transaction was processed successfully if received by Visa. This API is also used to retrieve the detailed lifecycle or history of a transaction (e.g. original transaction, chargeback, chargeback reversal etc). The Query API allows participants to query in real-time the processing status of Visa Direct (Account Funding and Original Credit) transactions as well as other related transactions that are part of the Visa Direct suite of transactions (reversals, adjustments, chargebacks and re-presentments). Participants can invoke the Query API when there is no response returned from Visa or when the response is returned with errors (e.g., 500, 400, 404, etc.). Additionally, it allows service providers to query the history of transactions and return the entire transaction set related to the original Visa Direct transaction. A transaction set will include approved and settled original Visa Direct transactions, reversals, chargebacks, adjustments, and re-presentments.
- **Refund API:** The Refund API is used to process a merchandise return (refund) transaction to a Visa card. The refund transaction is typically associated with a previous merchant payment. The refund can be for the entire amount of a previous transaction or portions of it. Visa also supports a Merchandise Return Reversal API, which is used to reverse a refund that was previously processed.

Other tools:

- **Visa Resolve Online:** a dispute resolution platform that helps issuers and acquirers manage the dispute life cycle. Users of VROL may retrieve transaction information online, receive financial information related to disputes, exchange information and documentation, electronically, submit pre-filing and case filings electronically, receive financial messages, report fraud, and manage exception file listings.

Methods used by Faster Payments rails to aggregate exception data and identify patterns

Faster Payments rails offer participants specific tools designed to consolidate failed or rejected payment messages. This enables participants to comprehend and analyze the reasons behind these issues, facilitating a thorough examination of the information. By identifying patterns, participants can enhance their processes for better efficiency. These methods include various types of reports (such as Mastercard's daily transaction report) and the ability to query the system (such as the FedNow interface Adhoc Query Tool) to retrieve information on failed messages.

FedNow reports:

- Various activity reports and the FedNow interface Adhoc Query Tool
- Participants can also send a retrieval request message (admi.006) and the FedNow Service will return a copy of payment messages sent. One retrieval request (admi.006) can be used to request up to 50 messages. Participants can use the retrieval request to retrieve messages for the current or prior seven business days.

RTP reports:

- Numerous reports available to RTP Participants, Funding Agents and Third Party Service Providers via the RTP Management Portal or transmission using a File Transfer Protocol (FTP).
  - Activity reports for value and non-value messages
  - Reconciliation Reports
  - User and Audit History Reports
  - Funding Agent specific reports

Same Day ACH reports:

ACH Operators provide detailed activity reports, such as FedACH's:

- **Notification of Change (NOC) report**, which provides information on NOCs received
- **ACH Return Item** report, which provides information for return items and operator-rejected entries received. (Also available in a machine-readable file.)
- **ACH Return Ratio** report, which helps participants monitor return activity
- **ACH Return Reason** report, which provides summary-level information on reason codes for returned entries

Sending FIs typically maintain a record of errors, with notes on their nature and frequency, to indicate basic deficiencies in the total system within the ODFI' internal controls. Other tools include the FedACH Exception Resolution Service which provides a 13-month archive to access and view all cases sent or received by a FI.

Mastercard Send reports:

- **Daily transaction report (DTR)** shows all approved and declined transactions processed by Mastercard Send APIs.
- **Mismatched and adjustment transaction report (MAT)** shows transactions that are settled in a different way (and for a different amount) than what was processed and reported by the Mastercard Send APIs.

Visa Direct reports:

- Participants can subscribe to receive the VisaNet Settlement Service (VSS) reports (provide reconciliation and settlement data at a summary level) and Single Message System (SMS) reports and raw data (which provide transaction-level research, participants can also create custom reports).
- **Reconciliation information through "Reports API"** provides reporting capabilities such as transaction reconciliation data in the API response. The data includes both push and pull transaction details and any exceptions such as chargebacks and reversals.

Methods used by Faster Payments rails to reduce exceptions processing

Methods to help participants reduce exceptions processing are available at several steps of the payment flow, from pre-message submission to final settlement. This section describes the methods used by each Faster Payments rail.

FedNow and RTP Network – Methods used to reduce exceptions processing

Mechanisms designed to reduce the need for exceptions processing are initiated from the start of the payment flow, where the sending FI conducts its own anti-fraud checks. This step serves to diminish the volume of payment return requests and, more importantly, minimizes the impact of fraud on end users. Subsequently, each participant in the payment flow undertakes its own validations as the payment message moves through the system. For example, all participants must ensure accuracy of message contents, that the participant profile has an active status and is enabled to send or receive the message, is connected to the service, ensures that the message has a unique ID, correct use of business application headers, ensure messages are sent to connected participants, etc. – all prior to payment submission.

The message is sent to the receiving FI once the payment message is processed by FedNow or RTP, and the receiving FI has the option to accept the payment, reject it, or accept it without posting. Opting for acceptance without posting grants the receiving FI additional time to perform its own checks and validations, allowing for the request of further information from the sending FI. Following this, the receiving FI may either reject the payment or proceed with posting it. Consequently, receiving FIs possess the flexibility to subject it to further review rather than immediately rejecting a payment. This approach enhances the likelihood of successfully processing more messages.

During the settlement stage, both FedNow and RTP enable participants to implement tools and processes to meet their liquidity requirements. This measure serves to decrease the cases of payments being rejected due to insufficient liquidity.

Other examples used by these Faster Payments rails include using broadcast messages, active participant lists, and providing participants with a message validations checklist within the rulebook and rejecting messages that appear to be duplicated.

TABLE 37. FEDNOW AND RTP NETWORK – METHODS USED TO REDUCE EXCEPTIONS PROCESSING

	INITIATION STAGE	VALIDATION STAGE	RESPONSE STAGE	SETTLEMENT STAGE	POSTING & NOTIFICATION STAGE
Sender					
Sending FI	Anti-fraud checks	Message validation prior to submission			Notifies sender
FedNow & RTP Network		Formatting and business rule validation		Validation and settlement	Confirmation to the sending and receiving FIs
Receiver FI		Message validation	Can respond with 'accept', 'accept without posting', or 'rejected' Receiver FI can request more information from the Sending FI.		Posts and notifies receiver

Same Day ACH – Methods used to reduce exceptions processing

Participants in ACH Network are advised to conduct anti-fraud checks to combat unauthorized returns and malicious users, and validate that the message is properly structured, is not duplicated, falls within the network limits, and contains the necessary information prior to submission. If the RDFI detected an error in the account information during the processing stage, it can send a Notification of Change (NOC), which helps the ODFI update the account information so future payments are sent correctly.

Reversals are also available in Same Day ACH where the ODFI can send reversals and correct files/entries when it detects an error.

Mastercard Send and Visa Direct – Methods used to reduce exceptions processing

Both Visa Direct and Mastercard Send provides participants with VAS through APIs that help reduce payment rejections and retries, reducing exceptions processing. For example, participants can use APIs such as Visa's Account Look-up API or Mastercard's Account Verification Service (AVS). These tools are usually utilized at the payment initiation stage. Participants must also validate other aspects of the payment message prior to submitting it to the network, such as validating that the card is eligible to receive push-to-card transactions and account validity.

At the processing stage participants can inquire about the status of the payment before initiating exceptions processing, meaning these APIs help participants understand whether or not the payment has already been rejected, the reasons for rejections, or if it can still be processed.

TABLE 38. MASTERCARD SEND & VISA DIRECT – METHODS USED TO REDUCE EXCEPTIONS PROCESSING

	INITIATION STAGE	VALIDATION STAGE	PROCESSING STAGE	POSTING AND NOTIFICATION STAGE	SETTLEMENT STAGE
Sender					
Sending FI	Anti-fraud checks	Account and payment verification (available APIs for this purpose)	APIs to inquire as to the status of the payment	APIs to inquire as to the status of the payment	
Push-to-card Network			Message validation		Settles payment
Receiver FI			Sends a response (Mastercard Send)	Message validation	

Potential reasons for using 'accept without posting' responses

Accept without post (ACWP) is one of the three available responses a receiver FI can use in response to payment messages (pacs.008 in both FedNow and the RTP Network. An ACWP message means that the receiving FI has not yet determined whether to send an "accept" or "reject" message and will not provide immediate funds availability to the receiver. Some receiving FIs conduct real-time transaction screening for legal or compliance reasons. If a receiving participant's screening software generates a match that could indicate the receiving user is not permitted to receive the payment, the participant may respond with an ACWP message. This allows the FI to investigate further.

Participant may only submit an 'accept without posting' message if:

- It requires additional time to review the message for compliance or legal reasons or
- The payment is returning funds to a closed account in response to the participant's request for return and
  - a) the participant has closed the receiver's account identified in the payment due to unauthorized activity; b the receiver has another account with the participant, or the participant is in the process of establishing a new account for the receiver; and
  - c) the participant cannot satisfy the immediate funds availability requirement of the "accept" response because it needs time to credit the payment to the receiver's other account.

Note that a payment message 'accepted without posting' is immediately and finally settled in the same way as all accepted payments. This means that the sending FI is obligated to pay the amount of the payment message even when the receiving FI sends an 'accept without posting' payment status message. If the receiving FI blocks or rejects the pacs.008, it should include a reason in a pacs.002 message for the sender FI to review. The receiving FI must also return the funds via a payment return (pacs.004) message.

Time frames for resolving 'accept without posting' responses

FedNow – Time frames for resolving 'accept without posting' responses

Participants must reject the transaction or make funds available to the recipient by midnight Eastern Time of the next standard business day (Monday through Friday, except holidays), unless the participant is concerned with breaching applicable law (e.g., compliance). In this case, FedNow requires the participant provide a status of pending (PDNG) response to the sending FI by the ACWP availability deadline described above. The receiving FI must notify the sending FI upon resolution of the investigation and in response to any inquiry from the sending FI.

RTP – Time frames for resolving 'accept without posting' responses

Participants must determine whether to accept or reject the message by 11:59 p.m. local time the next business day following its 'accept without posting' message, except in cases in which the payment is being reviewed for compliance with applicable sanctions laws.

Procedures for implementing 'requests for returns' across Faster Payments rails

Some types of faster payments are inherently irrevocable due to the speed of their payments processing. Therefore, the only way funds can be recovered in case of a mistake is through a payment return. However, for a payment to be returned the recipient of the original transaction must consent to the funds to be sent back to the originator. This process begins by the originator sending a return request through the payment system, which then must be accepted or rejected by the recipient of the original payment.

FedNow

1. Initiation stage: user initiates a return request (camt.056)
2. Validation stage:
   - The sending FI generates a return request and sends it to the FedNow service with a return request reason code and message details for the original customer credit Transfer (pacs.008) or a liquidity management transfer (pacs.009)
   - The FedNow Service receives return request message and runs through all applicable validations
3. Response stage:
   - Receiver FI receives return request message and begins message validations
   - The receiver's FI reviews and determines if it intends to accept the message
     i. Affirmative case (Full return)
        1. Sends FedNow Service a return request response with code IPAY indicating it intends to accept the return request
     ii. Affirmative case (Partial return)
        1. Sends FedNow Service a return request response with code PECR, indicating it intends to partially accept the return request
     iii. Negative case
        1. Sends return request response with code RJCR to FedNow Service along with the reason for rejection; or
2. Responds to the FedNow Service with code PDCR, indicating return request requires further review.
   - FedNow Service validates return request response from receiver FI and passes through to sender FI
4. Payment return stage: the Receiving FI initiates a payment return (pacs.004) message

TABLE 39. FEDNOW – PROCEDURES FOR PROCESSING 'REQUESTS FOR RETURNS'

	INITIATION STAGE	VALIDATION STAGE	RESPONSE STAGE	PAYMENT RETURN STAGE
Sender	Initiates return request			
Sending FI	Performs anti-fraud checks	Sends return request to the FedNow Service		
FedNow		Receives return request and runs through applicable validations; when message passes all validations, it is sent to the Receiver FI	Validates return request response from Receiver FI and passes through to Sending FI	
Receiver FI			Receives return request message and begins to validate the message	The Receiving FI initiates a payment return (pacs.004) message if it intends to return the payment

RTP

1. Initiation stage: user initiates a Request for Return of Funds (RFR) Message
2. Validation stage: the Sending FI generates an RFR Message that references the original CT Message's Instruction ID and passes it to the RTP Network.
3. Response stage:
   - The Receiver FI receives the RFR and responds with a Message Status Report either confirming that the RFR was received and processed or indicating a reason why the RFR was unable to be processed.
   - The Receiver's FI then determines if funds can be returned to the Sender and transmits the appropriate Message(s) to the RTP Network.
     i. Affirmative case (Full or partial return)
        1. Response to Request for Return of Funds (RFRR) Message indicating the Receiving FI has agreed to fully or partially return the funds
        2. CT Message with the amount of funds being returned
     ii. Affirmative case made outside of RTP Network
        1. RFRR Message indicating the Receiving FI has agreed to fully or partially return the funds
        2. References a unique ID associated with the external payment method used
     iii. Negative case
        1. RFRR Message indicating the Receiver/Receiving FI has denied the Sender's request
   - The RTP Network performs a series of formatting, process, and business rule validations on the messages and routes them to the Sending FI
4. Payment return stage: the Receiving FI generates an RTP Credit Transfer if it intends to return the payment

TABLE 40. RTP – PROCEDURES FOR PROCESSING 'REQUESTS FOR RETURNS'

	INITIATION STAGE	VALIDATION STAGE	RESPONSE STAGE	PAYMENT RETURN STAGE
Sender	Initiates return request			
Sending FI	Generates RFR message and passes to the RTP Network			
RTP		Performs a series of format and business rule validations and if successful, passes the RFR to the Receiving FI	Performs formatting, process, and business rule validations and routes the Messages to Sending FI	
Receiver FI			Receives return request message and responds with a Message Status Report. Then determines if funds can be returned to the Sender and transmits the appropriate RFRR Message to the RTP Network.	The Receiving FI generates a RTP Credit Transfer if it intends to return the payment

Same Day ACH

Like account-to-account payment transactions, an entry or a file of entries that has been transmitted to the ACH Network cannot be recalled but may be requested to be returned under limited circumstances. In such cases the ACH Network allows two scenarios where 1) a Sending ODFI attempts to correct an erroneous entry or file by transmitting a reversing entry or reversing file to the RDFI (reversal) or 2) a Sending ODFI requests the RDFI to return an erroneous entry (request for return). The following processes demonstrate the implementation steps for both reversals and request for returns.

Reversals (files and/or entries)

1. Reversal initiation stage:
   - The sender initiates a reversing entry or a reversing file.
   - The originator or its FI must submit reversals following the format and timing required by Nacha.
   - An originator or ODFI initiating a reversing file must concurrently initiate a correcting file, unless the erroneous file was a duplicate.
   - Note that the originator must make a reasonable attempt to notify the receiver of the reversing entry and the reason for the reversing entry no later than the Settlement Date of the reversing entry. Nacha does not specify the method (i.e.,, email, telephone, etc.)
2. Response/approval stage:
   - This does not always guarantee successful recovery from error since reversals, like other entries, may be returned/rejected by the RDFI for insufficient funds or other valid reasons.

Request for return

1. Request for return initiation stage:
   - Originator initiates a request for return by contacting a designated ACH contact at the ODFI.
   - The ODFI, not the originator, should make the request to the RDFI.

2. Response/approval stage:
   - The RDFI may agree to return the entry, but it is not required to do so. If the RDFI does agree, it may ask the ODFI for a letter of indemnification prior to returning the entry. The RDFI should use Return Reason Code R06 (Returned at the Request of the ODFI) to return the entry.

Push-to-card Network

Both Mastercard Send and Visa Direct enable senders and the sending FIs to use a funds transfer API to return funds. For push-to-card networks, if a push payment fails to complete, the sending FI must reverse the original transaction pulling funds from the sender's account. In the case where Mastercard Send funds transfers involve funding transactions before payment transactions, the Originating Institution/Transaction Initiator (OI/TI) requests the reversal of funding transactions that were originally transmitted to secure the funds to be transferred.

For Visa Direct, when the Original Credit Transaction (OCT) is declined, the Account Fund Transfer is reversed using the Adjustment API. The funds are pushed back to the sender's Visa account after a financial message called an Account Funding Transaction Reversal (AFTR) has been initiated. In this case, there is no 24-hour limitation in the AFTR. The original pull funds transaction can be reversed using either the ReverseFunds operation for a single account or the MultiReverseFunds operation for multiple accounts simultaneously. The following steps summarize the process of returning funds for push-to-card networks.

1. Initiation stage: OI/TI initiates a funding reversal API request when a payment transfer is unsuccessful.
2. Validation stage: the card network validates the request and, if accepted, transmits the request message to the sending FI for approval.
3. Response stage:
   - Sending FI receives the funding reversal transaction request and either approves or declines the request.
   - The response is passed to the card network and then the network forwards the API response to the OI/TI.
4. Payment return stage: OI/TI begins a payment transaction to transfer funds to the sending FI.

TABLE 41. MASTERCARD SEND – PROCEDURES FOR PROCESSING 'REQUESTS FOR RETURNS'

	INITIATION STAGE	*VALIDATION STAGE*	*RESPONSE STAGE*	*PAYMENT RETURN STAGE*
Sender				
Originating Institution/ Transaction Initiator	Initiates a funding reversal API request when a payment transfer is unsuccessful			Begins a payment transaction to transfer funds to the Sending FI
Push-to-card Network		Validates the request and if OK, transmits the request to the Sending FI	Receives the API response and forwards to the OI/ TI	
Sending FI			Receives the funding reversal transaction request and either approves or declines the request	Submits funding reversals

Reasons and circumstances to allow 'requests for returns' and 'reversals'

While most types of Faster Payments transactions are final and irrevocable once the payment is settled, financial institutions may submit requests for returns and reversals in case the transactions are deemed fraudulent or due to error.

FedNow

FedNow's credit transfers can be returned either when a FedNow participant decides to honor a return request (camt.056) message or when the FedNow participant cannot apply the funds it received, for example, when rejecting a payment after initially responding with an 'accept without posting' status. Once a payment return (pacs.004) is initiated the message follows the same end-to-end processing across the FedNow Service as a customer credit transfer (pacs.008). For that reason, the return request (camt.056) message also can be used to report fraud. Similarly, a liquidity management transfer (pacs.009) can be returned if a participant determines it is necessary.

RTP

The sending FI in the RTP Network may initiate a Request for Return of Funds if it encounters errors in the payment, claims unauthorized transactions, or experiences processing issues leading to incorrect payments. These include unauthorized RTP payments, erroneous RTP payments, or RTP payments made in response to fraudulent payment requests. A sender or sending FI issuing a Request for Return bears no liability to the receiving participant, as it does not constitute the cancellation of an RTP payment. The receiving participant is not obliged to return funds in these cases. All credit transfers are finalized upon the system's processing of the positive response message from the receiver FI. On the other hand, a Request for Return of Funds serves as an appeal to the receiver FI to investigate the possibility of fund retrieval, with no guarantee of successful recovery.

Same Day ACH

The ODFI in Same Day ACH can rectify an incorrect entry or file by sending a reversing entry to the RDFI. This addresses entries transmitted erroneously or without the proper authorization of the originator. When the sending ODFI has made processing errors it can request the RDFI to return an entry. These instances include duplications, incorrect payment orders, variations in payment amounts, unauthorized credit entries (e.g., corporate account takeover), and discrepancies in the timing of debit or credit entries. However, relying on reversals does not guarantee a successful correction of errors, as they may be rejected due to reasons such as insufficient funds or other valid causes. The decision to return an entry lies at the discretion of the RDFI. File reversals can be initiated to fix issues like duplicated files or files with a significant number of incorrect entries. It is the responsibility of the party that originated the duplicate or erroneous file to reverse that file. A failure by the ODFI to fund a credit file is not considered a valid reason for initiating a file reversal. Each reversing file must be in the format required by ACH specifications.

Push-to-card Network

For push-to-card networks the sending FI must reverse the original pull funds transactions when payment transactions fail for reasons that can include a lack of funds. For Visa Direct the original pull funds transaction can be reversed using the Adjustment API, which initiates a financial message called an Account Funding Transaction Reversal (AFTR). The ReverseFunds operation can be processed for a single account or multiple accounts simultaneously. A reversal of an originating credit transaction can be processed only when the reasons for the reversal include incorrect payment credentials, incorrect transaction amount, duplicate processing, or incorrect transaction code.

Timeframes for processing 'requests for returns'

When it comes to submitting requests for returns, each payment service operator requires a specific timeframe for senders and sending FIs to follow. The FedNow Service strongly recommends that receiver FIs respond as soon as possible, but no later than midnight of the next standard business day. Only FIs that decide to investigate for future information can take up to ten standard business days. The RTP Network also indicates that receiver FIs should respond within ten standard business days of receiving the request for return. When there is a need for further investigation of the claimed fraud or breach of warranty, FIs can take longer than ten days.

For Same Day ACH every reversing entry should be sent to the ACH Operator within a timeframe that ensures it reaches or is accessible to the RDFI within five banking days from the settlement date of the erroneous entry. Both the sender and ODFI are required to transmit each reversing file promptly and, when applicable, respond to a correcting file, to the ACH Operator within five banking days following the settlement date.

When returning funding transactions, Mastercard Send requires that funding reversals be submitted within 30 minutes of the funding transfer request. For Visa Direct, a reversal of an Original Credit Transaction (OCT) must be processed within one business day of the processing date of the transaction.

TABLE 42. FASTER PAYMENTS NETWORK – TIME FRAMES FOR PROCESSING 'REQUESTS FOR RETURNS'

FASTER PAYMENTS NETWORK	*TIME FRAME*
FedNow	Receiver FI should respond to the return request as soon as possible, but no later than midnight of the next standard business day.
RTP Network	Receiver FI should respond within 10 banking days of receiving the request for the return of funds, however may take longer than 10 banking days to investigate the claimed fraud or breach of warranty.
Same Day ACH	Sending FI/ODFI must transmit the file to the ACH Operator within 24 hours of the discovery of the error, and Receiver FI/RDFI should receive the reversal file within 5 banking days after the settlement date.
Mastercard Send	Receiver FI must submit funding reversals within 30 minutes of the funding transfer request.
Visa Direct	OCT reversals must be processed within one business day of the processing date of the transaction.

Post-transaction exception evaluation

Faster Payments utilize various methods to create, record, and retain information needed to evaluate issues that may arise after a payment has been initiated. These methods can involve monitoring, payment tracing, incident tracking, and using transaction IDs, though specific procedures for handling such transaction exceptions can vary depending on the Faster Payments rail.

FedNow

All return request (camt.056) messages contain a unique message identification, assigner, assignee, and creation date time. The case is assigned a unique identification and the creator of the investigation case when the investigation of the request begins. This can include contact details in case further discussion of the request or a request response is needed. Return request messages must also indicate a reference to the original customer credit transfer. This information comprises of original message identification, original message name identification, original creation date time, original instruction ID, original end-to-end ID, original interbank settlement amount and settlement date, etc., to help the other counterparty identify the transaction requested to be returned. Cancellation reason information is required, which specifies the reason for the return request and should contain a code from the ISO 20022 externalized

CancellationReason1Code list. Reason code FRAD, for instance, is be used to report fraud and meets the FedNow Service fraud reporting requirement.

RTP Network

A Request for Return of Funds message is initiated by the sending FI to the receiving FI to request the receiving FI investigate and return funds for any reason. The Response to the Request for Return of Funds message may be a positive response, indicating that funds will be returned, or negative (rejected), indicating that funds will not be returned. In either case, the response is not expected to be immediate, but after an investigation by the sending FI. The sending FI may returns funds either via the Real-time Payment System or another Payment System. If sending via the RTP Network, the receiving FI will create a new Credit Transfer message (pacs.008) that includes the original transaction's unique ID to the corresponding Request for Return of Funds in the Credit Transfer (pacs.008).

Same Day ACH

For Same Day ACH reversals it is necessary that the word "REVERSAL" is placed in the Company Entry Description Field of each Company/Batch Header Record. In addition, ODFIs must ensure that the sender or ODFI's name reflects the RDFI identified in the erroneous entry being reversed when initiating a reversal. The ODFI of the reversing entry may make negligible variations to the original content of the Company Name field, such as for accounting or tracking purposes, on condition that the name of the sender remains readily recognizable to the receiver. Same Day ACH further recommends that ODFIs set a reasonable level of monitoring and tracking parameters that look out for questionable transactions that are classified Code R17 (File Record Edit Criteria/Entry with Invalid Account Number Initiated Under Questionable Circumstances).

Mastercard Send

Mastercard manages and records the information needed for post-transaction exception evaluation by employing transaction identifier codes to track payments. These codes serve as unique identifiers, ensuring a connection between various authorization requests and the corresponding transaction. Transaction identifier codes for Mastercard require the following identifying data elements: transaction date, a Trace ID code, a BankNet reference number, and the date of the payment settlement. The unique identifier generated becomes a critical part of the transaction authorization and clearing process and is used in tracking payments as a part of exception processing.

Visa Direct

Visa Direct utilizes the status of Direct API calls which include various error codes, HTTP status code returned in the response, and other information in the response message to process exception handling. Any reverse funds request generally requires data elements such as the Systems Trace Audit Number, Retrieval Reference Number, Transaction Identifier, approval Code, Date and Time of Transmission, and Acquiring BIN, as well as sender's account information.

Knowledge of the types and purposes of back-office systems and functions for processing payments

Faster Payments rails' impact on OFAC screening processes

The adoption of Faster Payments Systems in the U.S. have presented new challenges to sanctions screening because of the shorter processing times. OFAC sanctions screening ensures that transaction parties are not on the U.S. Treasury Department's denied party list for international or domestic transactions. OFAC sanctions screening is conducted by FIs and third-party service providers via software. Faster Payments systems require processes to be completely automated so as to minimize the risk that funds are sent to a sanctioned entity. This in turn requires that FIs subscribe or otherwise access sanctions screening lists on a 24/7/365 basis rather than download batch-based files that can be checked prior to payment message submission.

To address this challenge, the Office of Foreign Assets Control has introduced a risk-based advisory approach for sanctions screening of Faster Payments. Compliance with OFAC regulations must focus on management commitment, risk assessment, internal controls, testing, auditing, and staff training. OFAC recommends developers integrate sanctions screening compliance into the design of Faster Payments systems by facilitating communication between FIs, exception processing, and establishing minimum compliance expectations for participants. Faster Payments systems rely on FIs to verify customers against the OFAC sanctions list during onboarding and at regular intervals, although screening is usually conducted on an individual basis for international payments.

Faster Payments rails' impact on AML compliance processes

All FIs are required to enact policies geared towards identifying and preventing instances of money laundering. Legal requirements for AML efforts springs from several pieces of legislation, including the Bank Secrecy Act of 1970 (requiring FIs to assist government agencies in the detection and prevention of money laundering), the USA PATRIOT Act of 2001 (which enhanced these provisions and required more advanced record keeping and reporting), and the Money Laundering Control Act of 1986 (which made money laundering a federal crime).

FI AML efforts include the establishment of AML policies, generalized sanctions screening, monitoring, and reporting unusually large Currency Transaction Records (CTR) (daily aggregate totals over $10,000) or Suspicious Activity Reports (SAR). Suspicious transactions could include frequent structuring of transactions to evade reporting requirements, abrupt changes in usual account activity, rapid movements of funds, or international transactions to high-risk jurisdictions. The limits for SAR are generally lower vary by state. FIs achieve compliance via monitoring by properly trained staff or machine learning-powered data analysis. However, SLAs around processing and settlement speed can hinder non-automated transaction analysis and FIs' ability of to conduct AML efforts for Faster Payments systems. Common responses to this problem have included the usage of advanced data analysis, such as artificial intelligence and/or machine learning, to quickly identify potential cases of money laundering as well as to identify and release false positive cases. FIs and operators can also increase training and staff size for the review and resolution of flagged transactions, though the 24/7/365 nature of many Faster Payments mean that some form of automation is required.

All Faster Payments systems require participant compliance with AML legislation, customer identification measures, and periodic due diligence checks against sanctions lists. Faster Payments systems also require participant FIs to have sufficient AML policies, programs, and units in place. Obligations around AML compliance also require these systems to account for the usage of third-party senders and verify their AML compliance. The card networks require participants to comply with network-specific AML programs which include standard monitoring and due diligence checks, and mandatory responses to periodic AML assessments and questionnaires.

Faster Payments rails' impact on fraud detection processes

The speed and irrevocability of Faster Payments systems drastically shrinks the time period in which stakeholders are able to spot and identify fraud, presenting opportunities for fraudsters. Fraud detection strategies for Faster Payments require more modernized approaches than traditional payment systems, and a variety of different fraud mitigation strategies are used.

'Smart friction' is often used in Faster Payments to help with fraud mitigation by creating artificial friction that forces users to think twice before following through on a payment initiation. Examples of this include confirmation-of-payee, in which the sending user is notified that the name on the beneficiary account does or does not match the name input by the sender, indicating the potential for fraud or a misdirected payment. Multi-factor authentication is another example.

Community-wide information sharing is another anti-fraud tool used in Faster Payments systems. Community-wide information-sharing involves system participants sharing fraud-related intelligence and information among each other; sometimes this sharing is done on a voluntary basis and sometimes on a mandatory basis. This can take many forms, including flagging accounts known to be used by fraudsters, alerting participating FIs of suspicious transactions or types of transactions, or forms of centralized reporting.

End user education measures also further strengthen fraud mitigation practices for Faster Payments systems by providing more information on the differences between individual payment rails and what this means in terms of liability recovery for funds. This information empowers users to make more informed choices, adopt more secure practices, and recognize potential red flags on fraud. This is important because consumer liability differs significantly between the different types of Faster Payments, something end users may not be aware of.

Additionally, the various Faster Payments systems employ VAS and anti-fraud tools to combat the threat posed by fraudsters. For the FedNow service, fraud prevention tactics including the use of participants-defined negative lists (akin to a suspicious entities list), mandatory fraud reporting measures, digital signatures for all transactions, and the use of data encryption. For the RTP Network, TCH requires the usage of MFA authentication for all payments. Fraud screening can be conducted by FIs using software to identify unusually high-value or frequent transactions, as well as those sent to unlikely beneficiaries for this purpose. For Same Day ACH payments services are offered aimed at fraud detection and OFAC screening, along with proxy identification services. Both MasterCard and Visa offer fraud detection solutions that utilize advanced data analysis and machine learning to identify potentially fraudulent transactions.

Faster Payments rails' impact on deposit systems

Financial institutions offering Faster Payments face challenges related to the speed and duration of payment processing, i.e., instead of having a cycle-based business day, posting to beneficiary accounts are now 24/7/365. Batch payments work in cycles, with specific windows for different steps of the processing flow, such as, collecting payment requests, file creation, validation, submission, clearing, settlement, and posting. However, faster payments require all these steps to happen within minutes – if not seconds. One consequence of this is that deposit systems must be able to move funds from settlement accounts to their customers' accounts much faster, requiring financial institutions to plan their liquidity needs in advance rather than being able to adjust them to the batch payments value at hand. Automation and updating business processes are key to handling these challenges as are having sufficient technical backups, enabling 24/7/365 operations and system maintenance.

While this is less of an issue with Same Day ACH, FedNow, RTP, Mastercard Send and Visa Direct all come with these challenges, though push-to-card is less intense than with the instant systems.

Reporting requirements necessary to process Faster Payments

Providing faster payments requires increased planning, and thus reporting from FIs, to limit risks arising due to the speed of payments. Internal reporting capabilities have to be tuned to prepare FIs to the uncertainty related to Faster Payments. Firstly, regarding the instant systems, banks must ensure sufficient liquidity 24/7/365 because they do not have time to adjust liquidity. This is less of an issue with push-to-card. Second, payments processing system failures are greatly damaging when it comes to Faster Payments because FIs have a much shorter buffer window: with the instant systems there is no buffer. Third, monitoring systems need to be enhanced so as to process AML, CTF, OFAC, and fraud checks all within a tight deadline for push-to-card and instant systems.

Back-office systems' controls for Faster Payments processing

Financial institutions need a number of back-office systems to support the seamless processing of Faster Payments. First, liquidity management systems are responsible for guaranteeing that an FI has sufficient liquidity at its disposal to handle all expected transfers. Fraud management systems help prevent unauthorized transactions from being initiated, thereby protecting customers from losing funds due to fraud. Third, sanction and OFAC clearing systems make sure that transactions initiated do not involve any individual or business that is on any sanction list.

Back-office systems needed to meet audit and reconciliation requirements

Participants of Faster Payments rails are expected to utilize internal reporting for improvement and are subject to external audits. This means FIs need to operate certain back-office systems that facilitate reporting. The audits are usually concerned with compliance with system rules, payments reconciliation, downtime, and fraud. Hence, back-office infrastructures ought to include systems to match transaction records with reports provided by the system operators, systems that collect data of liquidity usage and forecasts future needs, systems that keep track on fraud-related events and those that keep track of availability on payment rails, prevent connection failures and calculate adequate planned downtime to sort out arising system issues.

Knowledge of the interoperability considerations regarding foreign and domestic payment systems

Interoperability across Faster Payments rails

Interoperability between Faster Payments systems is rare in the US. Aside from the two push-to-card networks, which are able to send payment messages between one another, none of the other systems are interoperable.

The ACH Network can carry some ISO 20022 data, but does not have the infrastructure to enable the instant transfer of message by message routing. As for the ISO 20022 based solutions, at this time they have adopted slight variants of the standard and cannot be exchanged directly. At the time of this writing, there have been no initiatives by developers to enable any greater interoperability between systems, directly.

Data fields, format, and standards required for interoperability of foreign and domestic payments systems

TABLE 44. DATA STANDARD BY PAYMENT RAIL

PAYMENT RAIL	DATA STANDARD
FedNow	ISO20022
RTP	ISO20022
Same Day ACH	Proprietary legacy
Mastercard Send	ISO8583
Visa Direct	ISO8583

TABLE 45. DATA STANDARDS OF SELECTED INSTANT PAYMENT SYSTEMS ACROSS THE GLOBE

MARKETS	INSTANT PAYMENT SYSTEM(S)	DATA STANDARD
Australia	NPP	ISO 20022
Canada	RTR (under development)	ISO 20022
India	IMPS	ISO 20022
Mexico	SPEI	Proprietary legacy
SEPA (Europe)	TIPS, RT1, various national systems	ISO 20022
United Kingdom	FPS	ISO 8583

Having similar messaging formats makes it much easier for payment systems to be interoperable, either in the domestic case or internationally. Without this, each system has to decode and translate messages from other systems to their own standards, which can lead to data truncation and fragmentation issues, which is inefficient in terms of time and cost.

A common messaging standard is beneficial and promotes greater interoperability, easing cross-border payments and enabling a consistent service delivery for end users. SWIFT developed ISO 20022 to create a single standardized approach for all payment systems and their participants. ISO 20022 is an open standard providing a common language for multiple types of transactions, providing richer and more structured data compared to proprietary standards. Despite ISO 20022's introduction, there are differences between markets in terms of implementation, which creates inconsistences.

While only FedNow and RTP in the U.S. use ISO messaging standards to send payments, card networks also utilize APIs to enable participants to initiate transactions. Some participants might find APIs more beneficial than ISO connections as it is to implement and use. Furthermore, Mastercard and Visa have incredible reach, so joining their networks via API is the easiest way for FIs to connect with thousands of other institutions around the world.

Risk factors related to cross-border payments

Financial institutions are faced with a multitude of risk factors that must be managed when processing cross-border payments. These risks encompass a broad spectrum ranging from regulatory compliance risk (AML, CTF, and sanctions screening), correspondent relationships, currency fluctuations, liquidity, and general payment infrastructure stability. The stability of general payment infrastructure depends on a country's physical and technological conditions. A notable instance is the recent service disruption in the Japanese national clearing system. In this case, inadequate risk management by the clearing network resulted in a failure of backup measures, causing a two-day delay in all payments both domestic and cross-border. Managing these risks is critical to ensuring institutional integrity, transaction fulfillment, legal compliance, and business risk.

Screening compliance risks

FIs facilitating cross-border transactions must ensure compliance with pertinent regulatory considerations, especially sanctions screening, CTF, and AML concerns. These regulations ensure that FIs are not a party to terrorism financing or money laundering activities. Compliance with varying regulations across different jurisdictions poses significant challenges, as anti-terror financing legislation varies across jurisdictions. KYC, transaction monitoring, and monitoring correspondents' due-diligence procedures can also be insufficient or not up to US standards, making it difficult to know the ultimate beneficiary of a cross-border transaction or identify suspicious transaction patterns. Adding in the time pressures inherent to Faster Payments also means that these processes need to be done quickly, which presents its own challenges.

Foreign exchange & liquidity risks

Currency fluctuations and liquidity issues present a risk factor to financial institutions for cross-border payments because currency fluctuations impact profitability and create operational challenges, especially when time changes and operating hours of FX markets are taken into account. Unpredictable exchange rates can complicate financial planning and risk management, which in turn can lead to losses and operational inefficiencies for FIs. Liquidity in these currencies also needs to be managed. Currency fluctuations can also increase settlement risk, as FIs can end up needing to settle for larger amounts than accounted for in liquidity management programs.

Infrastructure stability risks

Infrastructure stability can significantly impact cross-border payments. The resilience and reliability of the technological and logistical systems that underpin payment systems are paramount to the fulfillment of cross-border payments. Disruptions to supporting infrastructure in the form of natural disasters, cyber-attacks, or system failures can lead to transaction delays, unmet SLAs, financial losses, and even compromised data security. Examples of this would include Japan's national clearing system experiencing delays due to failures in backup measures, showcasing the vulnerability of the nation's payment infrastructure. Similarly, disruptions in the UK's RTGS system CHAPS in 2023 resulted in temporary payment (domestic and cross-border) delays, emphasizing global repercussions when critical payment infrastructure faces instability. Examples such as this demonstrate the importance of robust risk management and resilient financial infrastructure for the functioning of cross-border payments.

Barriers to cross-border interoperability of Faster Payments rails

Cross-border interoperability for Faster Payments rails face numerous challenges, including differing payment messaging standards, AML and sanctions screening, foreign exchange issues, settlement methods, and other factors. Of the Faster Payments systems covered in this handbook, only the push-to-card networks currently allow for cross-border payments, though the RTP Network is working on a cross-border system with the Eurozone instant system RT1.

Barriers to cross-border payments

Screening procedure barriers

AML and sanctions screening represent obstacles to cross-border interoperability of Faster Payments systems. This is because receiving institutions have to trust that screening and onboarding procedures in other markets are accurate and meet the same requirements as in the home market. Furthermore, the nature of Faster Payments means that time to verify transactions is limited, meaning that automation is key as well as a source of risk. A lack of universal AML and sanctions compliance requirements across jurisdictions, varying regulatory approaches

and onboarding requirements, and fears of reputational/financial loss by FIs lead to significant risk in the processing of cross-border payments.

This is one reason why the correspondent-respondent relationship has been so successful – trust can be difficult to establish and is hard to do on a market-wide basis.

Fragmentation in International Data Standards

International data standards present significant barriers to cross-border faster payments, primarily arising from fragmentation and uncertainty within data frameworks. One significant challenge lies in the inconsistency across jurisdictions regarding the implementation of rules and standards. This fragmentation extends to jurisdiction-specific requirements for messaging formats and legal documentation, resulting in manual interventions and higher rates of payment rejections. Moreover, payment service providers grapple with balancing various obligations under different data frameworks, such as regulatory and supervisory requirements, third-party provider usage, and data security and privacy mandates. The uncertainty surrounding these obligations hinders efficient cross-border Faster Payments transactions and raises concerns about compliance with regulations like anti-money laundering and combating the financing of terrorism (AML/CFT).

Additionally, restrictions on data flow across borders adds on to the existing challenges. Mandates for data localization impose significant costs on FIs in terms of overhead and maintenance, impeding their ability to identify fraud, comply with regulations, and manage risks effectively. These measures not only hinder innovation within the Faster Payments system but also degrade data security and system resilience. Innovations that could enhance payment efficiency, like full payment traceability and prepayment validation, are stifled by the lack of a cohesive global approach to data standards. Consequently, these barriers not only impede the optimization of cross-border Faster Payments systems but also limit competition and innovation, ultimately affecting the accessibility and cost-effectiveness of such transactions.

Foreign exchange barriers

Foreign exchange can present obstacles to cross-border instant payments due to currency conversion costs, transparency concerns, tight SLAs, fluctuations in exchange rates, and FX market operating hours, among other factors. These factors collectively increase the cost, uncertainty, and risk associated with these transactions as well as potentially harming the end user experience. Currency conversions can introduce hurdles to payment fulfillment due to the additional transaction costs involved, fluctuating exchange rates, and desire to conclude relationships with FX service providers. Furthermore, restricted operating hours for FX markets contribute an additional obstacle, as transactions may be delayed or affected by the limited availability of trading opportunities. The costs involved with these payments can also be uncertain and untransparent, creating ambiguities for the FIs involved as well as end users. Stipulations around SLAs can be difficult to maintain, increasing the risk of timeouts, while also making AML, CTF, and sanctions screening more difficult, increasing legal risk.

Settlement barriers

Settlement for Faster Payments can present special issues depending on the type of Faster Payments system being used. While instant settlement systems, such as FedNow and RTP, are more likely to use a form of synchronized settlement – whereby transactions settle in both currencies synchronously – this is purely theoretical as neither system is live with cross-border solutions. Push-to-card systems do offer cross-border payments and are therefore the only Faster Payments system to do so. Each transaction settles, but typically does so after posting, meaning that net positions and currency issues are solved after the transaction posts to the beneficiary's account. Same Day ACH is not available for cross-border at all.

Messaging standard barriers

Messaging standards are an issue for cross-border interoperability. ISO 20022 in its current application represents a common base for formatting payment messages, but it is not used by all Faster Payments systems, neither in the U.S. nor abroad. Furthermore, its use varies consistently across different markets even that do use it for domestic payments, which can lead to confusion and friction in cross-border payments as well as data loss and misinterpretation of messages. This can, in turn, lead to payments not being accepted, being wrongfully flagged by AML or sanctions screening software, and lead to additional work in the form of exceptions handling on both sides of the transaction. Information in ISO 20022 messages can be structured differently across markets. For example, TaxIDs in Germany, Social Security Numbers in the US, and Aadhaar numbers in India serve the same purpose but their use in ISO messages can vary, complicating the translation of messages.

Solutions to cross-border interoperability for Faster Payments rails

This section explores key avenues to alleviate friction and complexities associated with cross-border payments, focusing on screening procedure harmonization, the establishment of cross-border system links, foreign exchange friction reduction, and messaging standard harmonization. By delving into these potential (and mostly unrealized) solutions, this section aims to illuminate how frictions to cross-border payments can best be navigated.

Screening procedure harmonization

Harmonizing sanctions screening lists and AML procedures, as a part of more generalized information-sharing and standardization program, presents a major challenge and opportunity to cross-border interoperability. A more unified approach to sanctions screening lists and AML protocols could streamline compliance efforts across financial markets. If FIs were able to adhere to more uniform AML procedures and screening lists, discrepancies could be reduced and a more cohesive approach to risk management of cross-border payments could be cultivated. Collaborative efforts among regulatory bodies could also involve heightened information sharing among FIs and regulatory bodies, which could reduce friction and cultivate a more streamlined cross-border payment experience.

Establishing cross-border system links

One solution to settlement issues for cross-border transactions is the establishment of bilateral systems. This can best be exemplified by initiatives like UPI establishing settlement links with countries like Singapore and the United Arab Emirates, or the pilot program IXB, which links TCH RTP to EBA Clearing's RT1 system in Europe via synchronized settlement. Bilateral systems offer a targeted solution to alleviate settlement-related frictions in cross-border payment interoperability. By establishing direct connections between systems, these bilateral arrangements bypass intermediary channels, reducing delays and potential errors that tend to be prevalent in cross-border payments facilitated by correspondent banks. This approach enhances the speed and efficiency of cross-border settlements. Transactions can be executed more directly between participating FIs, leading to a more seamless and interoperable international payment landscape. The downside is the sheer number of connections needed given the number of currencies around the world.

Foreign exchange

One way to reduce friction involved in foreign exchange (FX) for cross-border payments is through innovative measures such as implementing a centralized module listing FX prices of various financial institutions for specific currency corridors. This centralized platform would provide transparency and real- time information on exchange rates, minimizing uncertainties, enable price shopping, and reduce negotiation hurdles. Alternatively, the introduction of a centralized body to facilitate FX transactions could streamline processes, offering a standardized framework. Introducing a centralized body to offer foreign currency exchange would cut down on friction associated with converting payments into different currencies. However, potential trade-offs, such as concerns about price gouging by a potentially monopolistic entity, would have to be carefully considered.

Messaging standard harmonization

Standardizing the usage of ISO messaging standards and harmonizing sanction screening lists and procedures can offer a robust solution to alleviate challenges in the fulfillment of cross-border payments. Entities like the Committee on Payments and Market Infrastructure (CPMI) have developed voluntary harmonization guidelines set to be adopted by 2027. These guidelines encompass recommendations on messaging types, standardized time conventions, transparent currency conversions, and globally accepted payment codes. Harmonization, specifically for ISO 20022, has the potential to mitigate challenges in cross-border payments arising from different usage of data formats leading to loss of data or incorrect data placement and interpretation This harmonization enhances information transparency, enabling operators to swiftly analyze crucial components of international payment messages and streamline transaction processing.

Knowledge of the types of user transparency and remittance information provided by Faster Payments rails' stakeholders

Faster Payments rails facilitate the transmission of both value and non-value information through messages. Participants and the system can inquire as to and provide information on the status of each payment message, ensuring that participants remain informed at every step of the process flow and can take appropriate actions. Besides the information contained within the payment messages themselves, participants in Faster Payments rails utilizing the ISO 20022 data standard can share extensive remittance information. This facilitates multiple business processes, benefiting participants and both the sending and receiving users.

This section provides an overview of the types and functions of status messages within Faster Payments rails and outlines how remittance information is utilized and shared within each payment network.

Types and functions of status messages

FedNow and RTP – Types and functions of status messages

Messages exchanged between participants, and the Faster Payments system

Both participants and the system itself use messages to share the status of payment messages, return messages, share additional information about the payment, and share a participant's connection status. The table below shows the type of messages available to FedNow and RTP participants.

TABLE 46. FEDNOW AND RTP – STATUS MESSAGES

ISO MESSAGE	USED BY	MESSAGE FUNCTIONALITY
Participant payment status (pacs.002)	Participant (FedNow & RTP)	Inform the system about the processing status of a previously received payment, (pacs.008 /pacs.004). e.g., in FedNow the receiving participant responds with • (ACTC) Accept • (ACWP) Accept without post • (RJCT) Reject A pacs.002 message can also be used by the receiving FI to notify the system (and in turn the sending FI) that funds have been made available to the receiving user. The receiving FI sends a pacs.002 message with the code ACCC.
RFP response (pain.014)	Participant (FedNow & RTP)	Processing status of a previously sent RFP
Information request (camt.026)	Participant (FedNow & RTP)	Request the participant to provide further information on a previously sent customer credit transfer (pacs.008), payment return (pacs.004), or request for payment instruction (pain.013).
Information request response (camt.029)	Participant (FedNow & RTP)	Processing status of a previously received information request (camt.026). Information request response options: • Pending (PDNG): Information request is being reviewed. • Payment instruction is duplicate (IDUP). • Additional information will be sent (INFO) in an additional payment information message (camt.028) • Payment Initiated (IPAY)
Additional payment information (camt.028)	Participant (FedNow & RTP)	Respond to an information request (camt.026) with additional information about the initial customer credit transfer (pacs.008), return of funds (pacs.004) or request for payment instruction (pain.013).
Return request response (camt.029)	Participant (FedNow & RTP)	Processing status of a previously sent return request (camt.056).
Payment status request (pacs.028)	FedNow or RTP participant; or FedNow system	Request the processing status of a previously sent payment message from another participant or the system.
FedNow payment status (pacs.002)	System (FedNow)	Informs the participant of the processing status of a payment, e.g. the system sends a payment status report indicating that the message was accepted. Another scenario when this message is used is when the receiving FI notifies FedNow that the funds were made available to the receiving user, then FedNow notifies the sender.
Message reject (admi.002)	Participant (FedNow & RTP)	Inform the participant that the system has rejected the message.
Message reject (admi.002)	System (FedNow & RTP)	Inform the participant that the system has rejected the message.
Receipt acknowledgement (admi.007)	FedNow	When sent by the system this message informs the participant that a previously sent message was successfully processed by the system and delivered.
Receipt acknowledgement (admi.007)	FedNow Participant	When sent by a participant this message informs the sending FI that a previously sent message has been received by the receiver's FI.
Payment Acknowledgement by receiver (camt.035)	RTP Participant	Sent by receiving FI to a sending FI if the receiving user wishes to acknowledge and/or transmit information related to a received payment to the sending user.

Payment status messages shared with the end user

Notifying the sending user of a successful or unsuccessful posting of funds to the receiver's account is not mandatory but is highly encouraged; it is also considered a best practice within FedNow. Receiving FIs are also encouraged to notify receiving users of funds availability.

Same Day ACH – Types and functions of status messages

Messages exchanged between participants and the Faster Payments system

In the ACH Network ODFIs can learn the status of a payment (i.e, whether it was rejected). They must notify the ODFI the reason for the rejection if the message was rejected by the RDFI or the ACH Operator. There are defined reason codes for each reason. Other types of messages notify participants of the status of a payment, such as payment acknowledgment messages. These non-monetary entries are sent to acknowledge a corporate credit or debit (CCD) payment entry.

TABLE 47. SAME DAY ACH – STATUS MESSAGES

ACH ENTRY / MESSAGE	USED BY	MESSAGE FUNCTIONALITY
"Return Entry" or "Return"	Participant or ACH Network	A credit or debit entry initiated by a RDFI or ACH Operator that returns a previously originated credit or debit entry to the sending FI.
ACK: ACH Payment Acknowledgment	Participants (receiving FI)	Identifies a non-monetary entry initiated by a RDFI to provide an acknowledgment of receipt of a corporate credit payment.

Another way in which participants are made aware of the status of a submitted file is through ACH file acknowledgments. ACH Operators have information on the processing status of an originated file shortly after receipt of the file from the ODFI. The status information indicates whether or not the files have successfully passed editing and validation routines. Acknowledgment is in the form of a message or report made available to the ODFI.

Mastercard Send – Types and functions of status messages

Messages exchanged between participants and the Faster Payments system

In the case of Mastercard Send the receiving FI needs to approve the transaction, which is then shared with the sending FI through the Mastercard Network. Participants of both Mastercard Send and Visa Direct can obtain information about the transaction by sending a request message via API.

TABLE 48. MASTERCARD – STATUS MESSAGES THROUGH API

API REQUEST	USED BY	MESSAGE FUNCTIONALITY
Payment Transaction (POST request)	Participant	The API will respond with 'approved' for successful transactions. If the transaction is declined an API 402 error response with the reason code 'decline' is sent. Error responses are also returned for API requests that fail Mastercard Send validation or transaction processing, such as when a required field is missing or a value is incorrectly formatted. Mastercard Send provides a list of failure/ decline reason codes
Payment Transaction (GET request)	Participant	Participants can use this API to retrieve transfer details, including status, including whether a payment transfer had errors ('error') or was declined ('declined').

Visa Direct – Types and functions of status messages

Messages exchanged between participants and the Faster Payments system

Visa Direct participants can determine the status of any Visa Direct API call through the HTTP status code returned in the API response, the error code from Visa Direct or the platform, and other information included in the response message. This information tells participants why a payment failed, to apply the correct processing mechanism, and communicate with customers.

TABLE 49. VISA DIRECT – STATUS MESSAGES THROUGH API		
API REQUEST	***USED BY***	***MESSAGE FUNCTIONALITY***
Funds Transfer API (POST request)	Participant	The API response provides the payment status, e.g., approved, declined, or unknown. In cases of payment failure, participants can also obtain the reason for decline. Visa provides a list of error codes and descriptions.
Funds Transfer API (GET request)	Participant	Used to obtain further information about the status of the payment.

An originating FI can submit an inquiry through Visa Resolve Online to obtain confirmation or proof from a receiving FI that funds have been posted to a receiving user. The receiving FI must respond to the originating FI confirmation request through Visa Resolve Online within 5 calendar days from the date of inquiry.

Remittance information

Remittance information refers to the supplementary details or data associated with a payment, offering additional insights into the payment's purpose, the payment receiver, etc. Remittance information can be transmitted together with the payment or in a separate non-value message. This information can improve payment reconciliation processes for businesses and individuals, fostering transparency and streamlining financial processes.

RTP – Remittance information

The RTP Network allows the sending user to send unlimited characters via extended remittance information about the payment to the receiving user in addition to credit transfer messages. The sending FI may send a standalone message called a remittance advice (ISO 20022 message ID remt.001) message to the receiving FI if the information does not fit into the limited remittance fields within the credit transfer message. A remittance advice message may also be sent in conjunction with an RFP as an invoice.

- The sending FI ensures that the remittance advice message properly references the credit transfer message by including the credit transfer's unique ID.
- The receiving user may send a message confirming receipt of funds and provide additional information. For example, the payment acknowledgment by the receiver (ISO 20022 message ID camt.035) message may include a note or information from the receiving user such as:
  - "Thank you for your payment. Goods have shipped with tracking number: 123456789."
  - "Your payment has been received and credited to your account. Your available credit following payment is $5,000.00."

FedNow – Remittance information

When it comes to FedNow, participants may provide optional remittance information within customer credit transfer messages and RFP messages via a short free-form text field or detailed with designated elements, such as specific fields for an invoice number, the amount due, due date, and others. FedNow also supports reference remittance information that is made available separately, e.g., including a hyperlink to a document posted on a web portal.

Same Day ACH – Remittance information

In the ACH Network an ACH addenda record is an optional record attached to an ACH payment carrying remittance information. It can be described as a note or a memo attached to the payment that's used to

provide supplemental information needed to identify the account holder or to provide additional payment information to the receiving user/business.

As an example, Corporate Credit or Debit (CCD) and Corporate Trade Exchange (CTX) entries are distinguished by the ability of the sending and receiving users to exchange various amounts of remittance information. Companies that need to transmit remittance data with the CCD entry may add an addenda record to carry the payment related information (CCD+ transactions), such as invoice and order numbers. The CTX format supports the transfer of extensive addenda records, up to a maximum of 9,999 records each carrying 80 characters of payment-related data, along with the transfer of the payment. The CTX format is typically used to pay multiple invoices that are listed in the addenda records, although it may be used for a single invoice too.

Addenda records may use the American National Standards Institute (ANSI) Accredited Standards Committee (ASC) X12 transaction sets and data segments to relay payment-related information. However, CTX transactions may also include ISO 20022 remittance advice messages (remt.01 and remt.02) if both the sending and receiving FIs have opted-in to participate in Nacha's XML-ACH Remittance Information Opt-in Program.

Push to card – Remittance information

Visa Direct and Mastercard Send messages are processed with a different structure than A2A Faster Payments rails. The message standard and the required fields to process a payment vary from those used by FedNow and RTP, where ISO 20022 is used.

Remittance information in push-to-card networks is additional information carried with the payment such as the sender reference, the sender identification number for businesses, etc. Both networks determine the type of data fields and information that should or must be included and the information that may optionally be included. Both networks enable VAS APIs that help participants obtain additional information about the payment.

Remittance information disclosure requirements

It is expected that participants pass along the remittance information to receiving users. However, Faster Payments networks usually do not determine how this information is passed along. This decision is left to participants. This section describes the requirements that Faster Payments networks have (if any) with regard to disclosing remittance information.

ACH Network – Remittance information disclosure requirements

In the context of the ACH Network (particularly relevant to CCD and CTX entries carrying addenda records), the Nacha Operating Rules require RDFIs provide all payment-related information contained within the addenda records transmitted with CCD and CTX entries upon the request of the receiving user (in this case a business). RDFIs must provide the receiving user with the requested information by the opening of business on the second banking day following the settlement date of the entry to which the payment-related information relates. Nacha Operating Rules don't dictate how this information is shared with the receiver (e.g., in human-readable or machine-readable format).

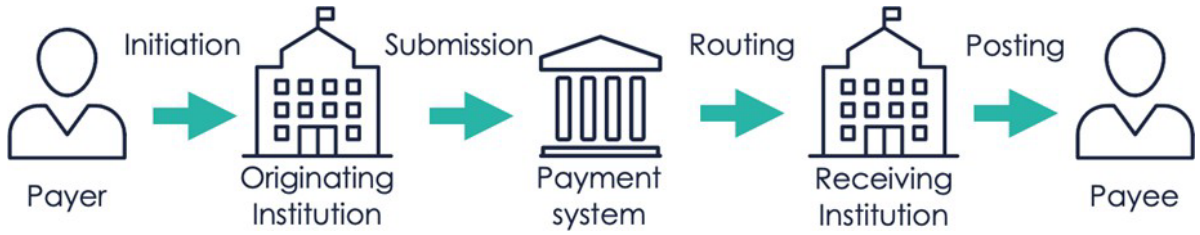
Other networks do not provide guidelines.

Knowledge of the payment network process flows by participation level

Payment process flows

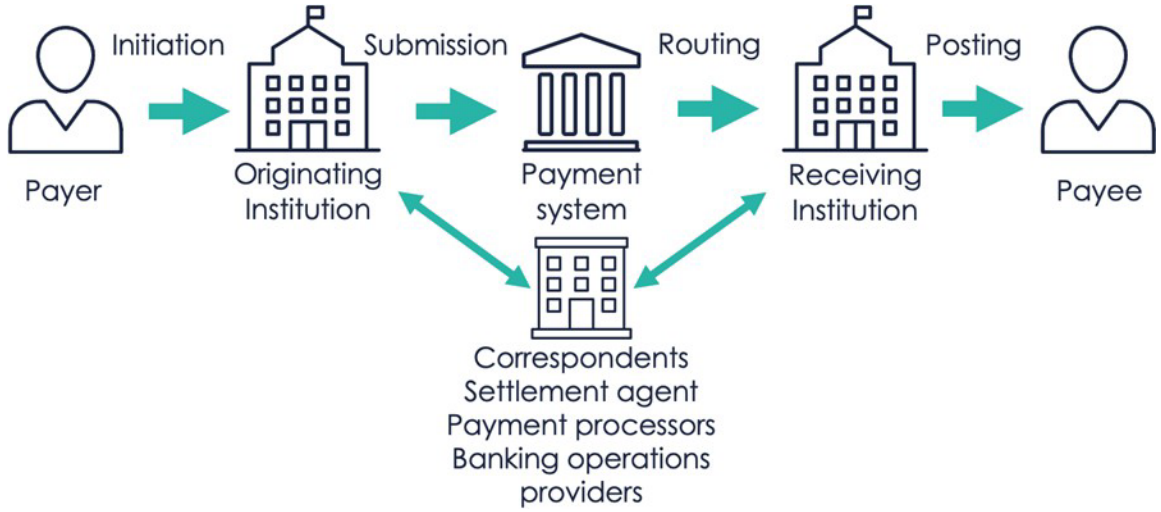
The payment process flow is fairly similar for all Faster Payments systems available in the United States. They include five stakeholders: the payer, the originating institution, the payment system, the receiving institution and the payee; and four steps: payment initiation, submission, routing and posting. While the number of stakeholders can expand when technical service providers, initiation service providers, and settlement agents are used, these processes were simplified for the sake of clarity. Regarding the four steps involved, these, too, were simplified to only took at the payment flow rather than all of the potential checks and potential errors that could occur along the way.

Figure 7. Generic payment process flow



One can differentiate between the type of third parties involved in the payment processes. Same Day ACH and instant payment systems have the same type of third parties that help participants mainly with banking operations, corresponding banking, and settlement.

Figure 8. Payment process flow for instant payments and Same Day ACH



Card networks allow participation for third parties that provide switching services to support payment processing in addition to ones that facilitate payment initiation between payers and originating financial institutions.

Figure 9. Payment process flow for card networks

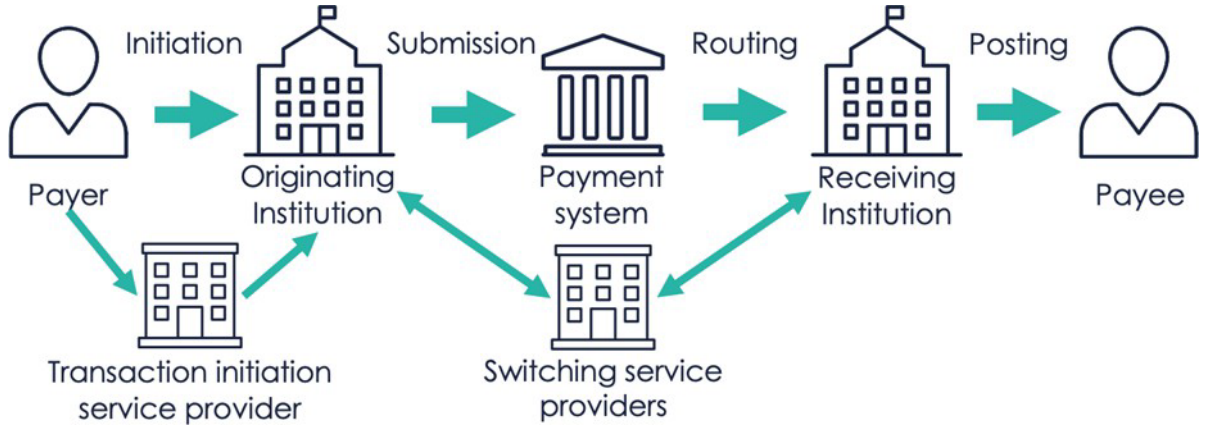


TABLE 50. TERMINOLOGY BETWEEN DIFFERENT FASTER PAYMENTS RAILS

NETWORK ENTITY	VISA / MC / ETC.	SAME DAY ACH	RTP	FEDNOW
Party initiating payment instruction	Merchant	Originator	Payer	Sender
Initiating processor	Acquirer	Originating Depository Financial Institution (ODFI)	Payer's FI	Sender's FI
Receiving processor	Issuer	Receiving Depository Financial Institutuion (RDFI)	Payee's FI	Receiver's FI
Party receiving payment instruction	Card holder	Receiver	Payee	Receiver

Differences in clearing process

Clearing describes what happens to a payment message between initiation and the settlement of obligations arising from a payment transaction. This usually includes the transmitting and exchanging of data and reconciling and confirming of transfers in addition to fraud and sanction screening.

The speed of Faster Payments shortens the clearing periods compared to traditional payment methods, as instant payments and card networks must post funds in a matter of minutes or even seconds. These systems have integrated clearing systems that validate payment messages based on their compliance with operating rules and formatting standards, accept or reject payment messages from originating financial institutions and route eligible payment requests to the receiving institutions. Due to their settlement timeframes card networks have additional time to facilitate data exchange and clarify inconsistencies between participants, which can be up to 30 minutes.

Same Day ACH has a longer clearing window (though shorter than traditional ACH). This means that there is more time to for ODFIs to run checks on payment messages. Payments rejected by either the RDFI or the receiving operator can be recreated and resubmitted.

Knowledge of the availability, permissible maintenance windows, and outage notifications by rail

Every participant must ensure they meet the system's SLAs as per participation requirements. Faster Payments rails acknowledge participants' needs to conduct system maintenance that may impact availability. The systems also recognize the possibility of unforeseen failures affecting system connections. Each Faster Payments rail determines the procedures to follow in cases of system maintenance, outages, and the implications for message transactions through its operational and technical guidelines. This section provides details on how each of these topics is addressed for the various Faster Payments rails.

Availability expectations for accessing Faster Payments rails

Faster Payments rails in some cases are available 24/7/365, though there are some exceptions. For example, some networks allow for short non-connectivity periods (permissible downtime) or make the 24/7/365 availability requirement for receiving messages but not necessarily for sending messages (e.g., participants can set their own sending windows). The ACH Network is open for processing payments 23¼ hours every business day (i.e.,, participants can submit payments to the Network within these operating hours). However, payments are only cleared and settled during Same Day ACH cycles. Moreover, Same Day ACH payments can only be settled when the Federal Reserve's settlement service is open (currently, the Federal Reserve's settlement system is closed on federal holidays and weekends, and business days from 6:30 p.m. ET to 7:30 a.m. ET).

System maintenance and outage notification requirements

Faster Payments rails acknowledge that participants may require planned downtime to conduct system maintenance. This section covers how, when, and for how long participants should carry out planned system maintenance for each Faster Payments rail as well as how the network and participants should be notified of unplanned outages.

TABLE 51. SYSTEM MAINTENANCE REQUIREMENTS BY FASTER PAYMENTS RAILS

SYSTEM MAINTENANCE REQUIREMENTS PER FASTER PAYMENTS RAILS	
FedNow	Participants planned downtime should not exceed two sequential hours or 24 hours total per quarter, and any planned downtime must be scheduled on Sundays between 2 a.m. and 6 a.m. ET. Participants and Service Providers are expected to take steps to reduce their need for planned downtime that better accommodates continuous operations over time.
RTP Network	Participants must satisfy a standard of 99.5% continuous connectivity to RTP measured on a calendar month basis, excluding permissible downtime; and be available to accept or reject messages while maintaining this connectivity standard. The permissible maintenance window is each Sunday from 2:00 am to 6:00 am (ET)
Same Day ACH	There are no specific requirements. Participants typically conduct system maintenance when the processing windows are closed.

TABLE 52. OUTAGE NOTIFICATION REQUIREMENTS BY FASTER PAYMENTS RAILS

OUTAGE NOTIFICATION REQUIREMENTS PER FASTER PAYMENTS RAILS	
FedNow	Participants should contact the FRB Services Support Center for guidance on unplanned downtime and any action to be taken. Participants are also recommended to sign off from the system so that other participants are informed (FedNow notifies other participants through a broadcast message). It is also recommended that participants implement internal processes to disable sending messages during the unplanned downtime.
RTP	Participants must send a participant sign-off (ISO 20022 message ID = admn.003) message to RTP.
Same Day ACH	There are no specific requirements

Straight-Through Processing expectations of Faster Payments rails and implications of transaction rejects (e.g. transaction timeouts)

Straight-through processing (STP) refers to when a payment process is completely electronic and automated from beginning to end. STP helps reduce the time and cost of processing transactions, minimizing the risk of errors. This is particularly relevant in the context of Faster Payments due to expectations of immediate funds delivery. The higher the Straight-Through Processing (STP) rates, the faster payments can be processed end-to-end and delivered to the receiving user as defined in the network's SLAs.

One crucial aspect of meeting system SLAs is to prevent transaction timeouts. FedNow has established a timeout clock that sets expectations for processing speed and capability for both the sending and receiving FIs. The timeout clock begins when the payment message is forwarded by the sending financial institution to the network and has a duration of 20 seconds and ceases counting just before settlement occurs. This means that receiving FIs can determine the number of seconds needed to be reserved for them based on the time it takes for them to process payments. A receiving FI can reserve up to five seconds of the clock, ensuring this time is guaranteed as part of the flow. The timeout clock will expire prematurely if, based on the configuration set by the receiving FI, there is insufficient time remaining for the receiving FI to provide a response (i.e., process the payment).

Faster Payments processing results in a higher rate of successful transactions, as the receiving FI reserves fewer seconds, allowing more time for the transaction to be processed end-to-end.

TABLE 53. TRANSACTION TIMEOUT BY FASTER PAYMENTS RAIL

TIMEOUT PERIOD BY FASTER PAYMENTS NETWORK	
FedNow	Timeout clock expires after 20 seconds
RTP	Messages time-out at 15 seconds, starting from when the message reaches RTP for processing
Mastercard Send	For a Payment Transfer API, if Mastercard Send does not receive a response from the receive network within 40 seconds a timeout will occur
Visa Direct	Timeout will occur in 30 seconds if the API is not able to complete the transaction

Knowledge of the procedures for reporting fraudulent and/or suspicious activity related to Faster Payments

Fraud classifications for Faster Payments

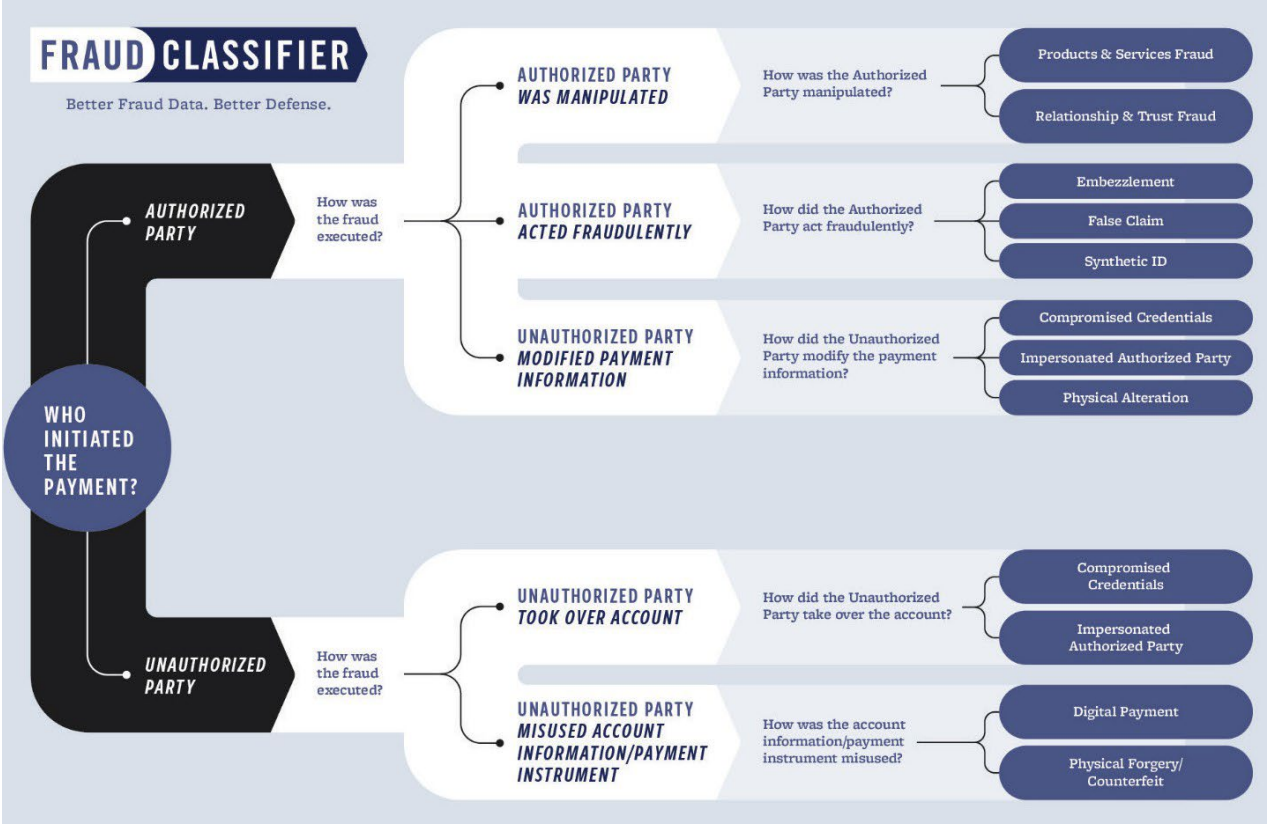
Faster Payments present a unique fraud risk due to the speed in which the transaction is processed and the subsequent reduced timeline for fraud checks. Fraud can be classified into two main forms, authorized or unauthorized, and often involves sophisticated techniques such as identity theft, account takeover, and/or phishing. The unique challenge for Faster Payments is to develop robust anti-fraud practices without compromising the speed and convenience that these payments provide. Addressing fraud in Faster Payments therefore requires a multidimensional approach to mitigation.

The Federal Reserve published its FraudClassifier as a tool to enable FIs to better understand how fraud occurs in the payments industry. The FraudClassifier allows FIs to classify fraud independently of payment type, channel, or characteristics.

The FraudClassifier utilizes several parameters, the first of which is whether the fraud was authorized (in which the account owner authorizes fraudulent transactions due to deceptive tactics) or unauthorized (in which the transaction transpires without the account owner's knowledge or approval). Authorized fraud is further divided into whether the party was manipulated, acted fraudulently, or whether an unauthorized

party manipulated the payment information. Unauthorized fraud is where the unauthorized party took over the victim's account or the victim's payment information was misused.

Figure 10. Federal Reserve FraudClassifier



Fraud reporting obligations

FedNow

FedNow participants are required to report to the central service when fraudulent transactions occur. However, the particulars of FedNow fraud reporting requirements are confidential.

Same Day ACH

Consumers are required to notify their financial institutions within 60 days of when fraud occurs, though businesses only have 24 hours. Regulation E requires that the fraud must be reported to the ODFI or the defrauded party to be compensated. FIs are required to investigate instances of fraud and report it to the relevant authorities, with overall responsibility lying with the ODFI.

RTP

Fraud reporting by RTP participants is mandatory under the RTP Technical Specifications and Risk Management/Fraud Controls requirements. Financial institutions must notify both the counter-party FI and The Clearing House via the RTP Network as soon as possible when fraud occurs or is suspected. According to the RTP Operating Rules participants must also act promptly based on any alerts of fraud or suspected fraud.

Mastercard Send

Mastercard has established clear reporting requirements in cases of fraud for stakeholders involved. FIs are obligated to promptly report fraudulent activity to the relevant authorities as well as to Mastercard's Fraud and Loss Database. APIs are also available to enable real-time reporting of fraud for participant entities in the Mastercard scheme.

Visa Direct

Any entity within the Visa payment value chain that suspects or confirms fraud has taken place is required to comply with network reporting requirements. FIs are required to report instances of fraud to Visa within 3 business days and subsequently must submit an incident report to Visa following an independent investigation. All other relevant parties involved with the fraudulent or suspected fraudulent transaction must also be notified. Visa's Risk Investigations group must also be provided with all account data pertaining to the transaction, after which Visa may conduct its investigation.

Methods and messages used for reporting fraud

Each Faster Payments network utilizes its own methods and message types for communicating cases of fraud to both the system itself and other participants. Some networks have specific message types or reason codes according to the type of fraud (e.g., authorized or unauthorized), while other networks notify cases of fraud more generally.

The information below summarizes the messages used for reporting fraud per network.

FedNow and RTP – Methods and messages used for reporting fraud

- Fraud reporting with "Return request (camt.056)": The return request message camt.056 is used to report fraud along with a request to returning funds from a previously sent message. For example, if a participant sends an instant payment and later identifies it as having been fraud, the participant sends a camt.056 including the reason code "FRAD" to notify the receiving FI.
- According to RTP rules the FRAD code cannot be used in a request for Return of Funds message (camt.056) unless the sending FI has determined that a payment is an unauthorized payment. For payments that a sender claims were made under false pretenses (e.g., authorized fraud), the "CUST" code must be used.

Same Day ACH – Methods and messages used for reporting fraud

Participants must use the appropriate reason code to reflect the reason for payment return/rejection. The information below describes the reason codes used in cases of fraud or suspected fraud.

- R03 "No Account/Unable to Locate Account": This reason code is used when the account number structure is valid and it passes the check digit validation, but the account number does not correspond to the individual identified in the entry or the account number designated is not an existing account (e.g., reflecting a case of potential authorized fraud).
- R06 "Returned per ODFI's Request": This reason code is used when the sending FI has requested that the receiving FI returns an erroneous entry, or a credit entry originated without the authorization of the sending user (e.g., potentially reflecting unauthorized fraud or error)
- R10 "Customer Advises Originator is Not Known to Receiver and/ or Originator is Not Authorized by Receiver to Debit Receiver's Account": This reason code is used when the receiving FI has been notified

by the receiving user that it does not know the identity of the message originator, has no relationship with the message originator, or has not authorized the originator to debit his or her account.

- R17 "File Record Edit Criteria/Entry with Invalid Account Number Initiated Under Questionable Circumstances/Return of Improperly-Initiated Reversal": This reason code is used when field(s) cannot be processed by the receiving FI, the entry contains an invalid account number e.g., account closed/no account/unable to locate account/invalid account number and is believed by the receiving FI to have been initiated under questionable circumstances, or the receiving FI or the receiving user has identified a reversing entry as one that was improperly initiated by the originating user or originating FI.

A practical example of the reason code R17 can be described as follows: A sending user (in this case a business/company) receives an R17 return that includes 'QUESTIONABLE' in the addenda information. The payment is being returned for more than an account number error and it is possible that the company may be a victim of fraud. An R17 in conjunction with this description will enable these returns to be differentiated from other returns with more routine account number errors and flag them for closer examination.

Push-to-card – Methods and messages used for reporting fraud

Push-to-card networks provide participants with error codes when transactions fail so that participants react to transaction errors, including those errors related to fraud.

Mastercard Send

When a sending FI sends a payment transfer request to the network (through an API call), the sending participant receives an API response indicating whether the transaction was approved or declined (i.e.,, those that resulted in a 402 error response). Error response messages may contain specific Error Detail Codes. The code 130002 is used when the transaction was declined because fraud was detected. Participants can obtain further information (i.e., the network response code) on declined transactions using GET requests. The ISO Code 83 is used when the transaction was declined for fraud/security reasons.

Visa Direct

Similar to Mastercard Send, participants use the combination of the HTTP status code returned in the response, error code from Visa Direct or the platform, and other information included in the response message to determine the status of any Visa Direct API call. These error codes and response messages indicate the nature of the declined transaction, including in cases of suspected fraud.

Chapter Three

Faster Payments Risk Management

SECTION BLUEPRINT COMPOSITION: 19%

Types of risks present in Faster Payments

The high velocity of payments handled by Faster Payment systems may lead to a number of risks for stakeholders. As users increasingly adopt Faster Payments, understanding and mitigating risk grows in importance. These risks, if not addressed, can affect not only the smooth functioning of the payment system, but also the stability and trust in Faster Payments themselves. The upcoming sections will elaborate on several risks that Faster Payments face, including operational, cross-channel, compliance, fraud, liquidity, technology, and credit risks.

Operational risks present in Faster Payments

Operational risk represents the amount of loss related to unintentional errors, which may result in delayed, duplicated, or otherwise erroneous payments. All financial market infrastructures face operational risk that originate due to deficiencies in information systems or internal processes, human error, management failures, or disruptions from external events that result in the reduction, deterioration, or breakdown of services.

Operational risk poses a huge challenge particularly in many Faster Payments systems because of the need for continuous system availability 24/7/365. This imposes significant demands on operational reliability for Faster Payments and participants. Speed, on the other hand, magnifies the impact of any operational incidents leading to delays or interruptions in Faster Payment services, and these interruptions are immediately noticeable to end users. The repercussions of an operational incident in Faster Payments may manifest much sooner in comparison to traditional retail payments, where service interruptions or slowdowns could go unnoticed.

A Faster Payment system being unavailable for a few minutes can cause many payments to fail; the consequences of operational downtime can therefore have even more serious implications than with traditional systems. As users become accustomed to utilizing Faster Payments for time-sensitive payments, any interruptions could affect users and result in penalties for late payments or insufficient funds for other transactions.

Faster payment systems are subject to fraud and security risks, forming a large subset of operational risks. These risks can originate from internal or external sources and are closely linked to each other.

Cross-channel risks present in Faster Payments

Cross-channel risk occurs when the movement of fraudulent or illegal payment transactions from one payments channel to another is met with inconsistent risk management practices and lack of information sharing across

payment channels about fraud. A financial institution offering faster payment products in addition to traditional payment products (e.g., checks, wire transfers) is exposed to increased cross-channel risk due to the multiple payment channels available.

The speed and finality of faster payments reduce the time available for fraud detection and prevention. This can make it easier for fraudsters to potentially move funds quickly through different channels before an institution can review and detect suspicious activity.

Compliance risks present in Faster Payments

Compliance risks in Faster Payment systems pose a significant challenge for FIs and other stakeholders involved in processing these payments. The major factor contributing to this increase in compliance risks arises from the increased speed of most types of Faster Payments. Concerns around insufficient time to adhere to regulatory guidance, anti-money laundering (AML) protocols, KYC due diligence, and fraud prevention are increasingly being solved by automation, which leads to its own problems. The high velocity of Faster Payments provides a smaller window of opportunity for stakeholders to conduct thorough due diligence and real-time monitoring, potentially increasing the risk of non-compliance and fraud.

FIs must navigate the delicate balance between meeting account holder expectations and satisfying regulatory requirements around financial crime and risk controls. This poses a considerable challenge for FIs who are accustomed to lengthier time periods with the possibility of manual intervention. Despite the challenges associated with compliance, the cost of non-compliance can be very steep. Penalties for non-compliance can include fines, suspension, and even exit of the FI from the Faster Payment systems.

Fraud risks present in Faster Payments

Fraud risk is the risk that a payment transaction will be initiated or altered in an attempt to misdirect or misappropriate funds. Risks related to fraud are affected by internal as well as external factors. Fraudulent activities may be the work of dissatisfied, disgruntled, or dishonest employees, third-party processing personnel, originating company personnel, or other outside parties (e.g., cyber thieves, fraudsters). Some common fraud scenarios in Faster Payment include:

- End user manipulation by a fraudster, resulting in the issuance of a payment instruction bound for the fraudster by the payer acting in good faith (i.e., credit-push fraud)
- Payment initiation by a fraudster that has fraudulently obtained the payer or payee's sensitive payment data
- The modification of a payment attribute (such as the account number, transaction amount, or name of the payee or payer) of a genuine payment instruction intercepted by the fraudster

Fraud risk has the potential to affect all actors in the payment value chain, including end users, FIs, and the overall Faster Payment system, and these risks are common across both Faster Payments and traditional retail payments. As with operational risks, the end-to-end speed of Faster Payments, especially the immediate availability of funds, makes Faster Payments more lucrative and susceptible to fraud risk than traditional payments.

The Federal Reserve's FraudClassifierSM model provides some guidance on how to systematically classify fraud based on three key questions:

- Whether an authorized or an unauthorized party initiated the payment
- How the fraud was executed
- What tactic was used to commit fraud

Using this model and adding payment attributes (such as payment amount, date, type, and channel) to these classifications can provide additional insights to help aid the organization in identifying fraud trends.

Liquidity risks present in Faster Payments

Liquidity risk is the current and potential risk to earnings or capital arising from a financial institution's inability to meet its obligations when they come due. That is, liquidity risk occurs when a financial institution cannot settle an obligation for full value when it is due even if it may be able to settle at a future time.

Liquidity risks in Faster Payment systems depend on the type of settlement model being used by the system. For systems such as FedNow and RTP, which utilize real-time settlement, liquidity risk is minimal because settlement is pre-funded and each transaction is settled in real time during payment processing. For push- to-card systems such as Mastercard Send and Visa Direct, which use prefunded settlement, even though the settlement happens same day or next day, the liquidity risk is mitigated as the funds are blocked for settlement at the time of processing. However, Same Day ACH FI relies on deferred net settlement so there is liquidity risk that builds up between the settlement cycles. Liquidity is primarily required at the end of each settlement cycle, introducing potential risks if inter-PSP settlement (Inter-PSP settlement refers to the process of settling funds between Payment Service Providers (PSPs) after a transaction has occurred between their respective customers. This is a critical component of fast payment systems and cross-border transactions.) occurs beyond normal business hours. To mitigate such risks, tools like prefunding, liquidity pools, collateral agreements, or arrangements with liquidity providers are employed, ensuring ample funds for settlement. Furthermore, posting only occurs after settlement.

Technology risks present in Faster Payments

Technology risk refers to any potential for technology failures to disrupt business. There are a number of different types of risks that can expose the vulnerabilities of the technology of Faster Payments systems.

1. External attacks, most likely cyber-attacks rather than physical attacks on the system's premises, can cause significant issues related to the loss or theft of sensitive data, damage in the operating technologies, and/or the integrity of the system.
2. Mistakes in the programming can result in operational errors and thus unplanned downtime when system updates take place.
3. While the modern payments landscape is more and more automated, there is still room for human error in the process of building and maintaining information technology systems.
4. Technology systems run on electricity and are vulnerable to outages.
5. While most Faster Payments systems have modern infrastructures, there is a theoretical possibility that they reach their processing limits and overload, and therefore would not be able to fulfill their responsibilities related to the speed of processing.
6. System owners often outsource their infrastructure operations to technology service providers, thus all the above-mentioned risks can also affect the system itself even if they are not directly connected to the system owner.

Credit risks present in Faster Payments

Credit risk is the risk that a party to a transaction will not be able to provide the necessary funds, as contracted, for settlement to take place on the scheduled date (i.e., it arises when a party will not settle an obligation for full value). Each Faster Payment instrument has a specific settlement process that depends on the entities involved.

As it relates to Same Day ACH processing, both ODFIs and RDFIs are exposed to credit risk; however, ODFIs have the greatest credit risk. Their losses generally occur due to failure or bankruptcy by their Originator. ODFIs are exposed to different levels of credit risk for Same Day ACH credit and debit entries.

An ODFI's credit risk associated with Same Day ACH credit entries is the period of time between when the file is initiated to the time the Originator funds their account for settlement. For example, if a Same Day ACH credit transaction is transmitted to the ACH Operator by the 10:30 AM ET processing deadline, it will not settle until 1 PM ET. Thus, the ODFI's credit exposure would be a few hours.

For Same Day ACH debit entries, credit risk is from the time the funds are made available to the Originator to the time entries cannot be returned by the RDFI. For creditworthy Originators, it is customary for the ODFI to provide the Originator with availability of funds for the total amount of the debit file on Settlement Date. The ODFI of a debit file is exposed to the risk that entries may be returned by the RDFI and that it may be unable to recover the amount of those funds previously released to the Originator. The shorter the time in which an RDFI must return an entry, the lower the risk to the ODFI that the Originator will have gone out of business or have insufficient funds in the account to cover the chargeback of the debit. So, the ODFI's credit risk can range from 2 banking days (items that cannot be posted) to 60 calendar days (disputed ACH debit entries). And since ACH debits are generally sent on a recurring basis, the credit risk associated with them can compound making ACH debits a higher risk than credit entries.

An RDFI's potential credit risk and loss occurs when it (1) posts a Same Day ACH credit entry and makes funds available for withdrawal prior to receiving final settlement, or (2) mishandles the return of a Same Day ACH debit entry (e.g., returns the debit outside of the prescribed timeframes or uses an incorrect return reason code).

Much like an ODFI in the ACH Network, a provider of RTP or FedNow origination services could experience credit risk or a loss due to failure or bankruptcy by the entity initiating the payment. Of course, they can easily manage this by using a "good funds model." That is, ensure they debit the Sender's account prior to transmitting the payment into the network. But they may decide for some business clients, they are willing to allow an intraday overdraft and will transmit the payment. However, if the business client's cash position remains overdrawn, then they could suffer a financial loss.

There is no credit risk associated with receiving an RTP or FedNow credit payment as both networks require funds be made available to the account holder by the receiving institution upon receipt of a Message Status Report. The Message Status Report is sent after settlement has been processed to the receiving institution.

Types and application of internal controls (user access controls) related to Faster Payments

Evolving technology and the increased volume of payment transactions have made fraud increasingly more relevant. Faster Payments must uphold internal controls encompassing administrative controls and internal accounting controls, which cover account types and limits, access controls, the segregation of duties, and the reconcilement of input to output as well as management review. Faster Payments use various fraud detection and mitigation techniques to ensure a safe experience for consumers. Though the implementation of internal controls has its own barriers and limitations, diverse methodologies and techniques on risk assessment as well as various requirements guarantee a secure and undisturbed transaction journey.

Application of fraud detection and mitigation techniques for different types of fraud

As described in the "Fraud classifications" section of this handbook, fraud can be generally classified into authorized and unauthorized fraud. Accordingly, there are specific fraud detection and mitigation techniques that are particularly appropriate for each of these fraud types.

An example of authorized fraud is where fraudsters manipulate the account holder to make a payment to the fraudster by tricking them into believing the fraudster is someone they trust. Combatting authorized fraud requires focus on mitigation techniques such as consumer education and fraud awareness in addition to using account validation tools such as Confirmation-of-Payee (COP), an account name verification service designed to reduce fraud and misdirected payments by ensuring that the name entered by a payer matches the name on the recipient's account. It has become a key fraud prevention tool in the payments industry, particularly in combating Authorized Push Payment (APP) fraud.

Looking at unauthorized fraud, fraudsters use sophisticated techniques to take over accounts or initiate fraudulent payments by misusing account information. Strong account holder authentication is a key preventative tool along with fraud monitoring systems. Authorized Push Payment (APP) fraud is a growing threat in the payments industry, particularly as real-time and faster payment systems become more widespread. APP fraud occurs when a consumer is tricked into authorizing a payment to a fraudster—often under false pretenses such as investment scams, romance scams, or impersonation of trusted individuals

The table provided below offers a non-exhaustive breakdown of mitigation and detection techniques tailored to the distinct characteristics and challenges of each type of fraud.

TABLE 54 FRAUD MITIGATION AND DETECTION METHODS AND TECHNIQUES

APPROACH	METHODS AND TECHNIQUES	TYPE OF FRAUD	
		AUTHORIZED	UNAUTHORIZED
Technology	Confirmation-of-Payee (CoP)	X	
	Fraud monitoring system / trend analysis and anomaly detection	X	X
	Biometric and behavioral authentication	X	X
	Digital ID		X
	Fraud negative lists (fraudulent individual database)	X	X
Regulation and scheme rules	Transaction value & volume limits	X	X
	Transaction hold for analysis	X	X
	Strong customer authentication / multi-factor authentication	X	X
	Fraud reporting transparency (e.g., industry-wide reporting with clear, standardized, and transparent fraud classifications)	X	X
Education and collaboration	End user education (e.g., fraud awareness campaigns) and staff training	X	X
	Industry-wide fraud classification models (e.g., Fed's FraudClassifier model)	X	X
	Cross-industry collaboration (e.g., information-sharing practices, fraud awareness campaigns, etc.)	X	X

Sources: The World Bank "Fraud Risks in Fast Payments" October 2023; Faster Payments Council "Examining Faster Payments Fraud Prevention" July 2020; and Lipis Advisors research .

Commercially reasonable methods and techniques used to detect fraud

Commercially reasonable is associated with an organization's specific set of facts and circumstances. The term is used to describe the method that should be used by FIs in a number of circumstances, such as fraud prevention techniques/methods, data security, and others.

Methods and techniques used to detect fraud in the payments' ecosystem have evolved as technology progresses, new fraud typologies emerge, and payment methods and use cases continue to evolve.

Trend analysis and anomalous transaction detection involves examining historical data to identify patterns and anomalies that may indicate fraudulent behavior. Payment system operators are increasingly implementing trend analysis at the central infrastructure level to leverage system-wide data and information-sharing practices. Individual financial institutions also carry out their own efforts in this regard prior to payment submission.

Machine learning has become a powerful tool in fraud detection, leveraging algorithms and predictive models to analyze large datasets and identify potential fraudulent transactions. These systems continuously learn and adapt based on new data, improving their accuracy over time. Predictive and behavioral modeling goes hand-in-hand with machine learning, as they involve creating models that anticipate potential fraud based on historical patterns and user behavior. Machine learning models with a combination of tactical rules and behavioral biometrics can contribute to a robust and more proactive fraud detection system.

There has been an increased focus in recent years on preventing Faster Payments fraud during payment initiation and when accessing the related transaction account. This emphasis is due to the increasing proficiency of fraudsters in exploiting end users. The use of strong account holder authentication is being prioritized by financial institutions.

Confirmation-of-Payee is another relevant pre-payment initiation tool that aims to verify the identity of the payee in a transaction, ensuring that payments are directed to legitimate recipients. This involves cross-referencing information provided by the payee with external data sources to confirm their identity, reducing the risk of fraudulent payments where fraudsters pretend to be someone or something they are not.

Enhanced due diligence is a proactive methodology involving investigations to authenticate the legitimacy of clients during the processes of opening bank accounts or enrolling in new services. This is a preventive measure against fraud, specifically aiming to deter individuals with malicious intent from establishing accounts at FIs for the receipt of illicit funds. Enhanced due diligence can act as a constraint, impeding a fraudulent actor's ability to carry out illicit activities.

Same Day ACH

According to the Nacha Operating Rules, originators of WEB debit entries must use what the rules define as "commercially reasonable fraudulent transaction detection system" to screen entries for fraud. The organization using the ACH Network should determine what is commercially reasonable to them when choosing solutions to comply with the rule. 'Account validation' is part of a 'commercially reasonable fraudulent transaction detection system.' Examples of methods to validate an account may include, but are not limited to, the use of a prenotification entry, ACH micro-transaction verification, use of a validation service provided by either an ODFI or a third-party, and the use of account validation capabilities or services enabled by APIs.

Another example is how, according to the Nacha Operating Rules, an originator of micro-entries must conduct "commercially reasonable fraud detection" on its use of micro-entries, including monitoring of forward and return volumes of micro-entries to minimize the incidence of fraud schemes using micro-entries and establishing a baseline of normal micro-entry activity for the originator.

Another example is Nacha's rule on ACH data security, wherein each FI , originator, third-party service provider, and Third-Party Sender must adopt and implement "commercially reasonable" policies, procedures and systems to receive, store, transmit and destroy consumer-level ACH data in a secure manner and to protect against data breaches.

Applications of internal controls used to mitigate risk related to Faster Payments

Internal controls include the policies and procedures that financial institutions establish to reduce risk and ensure they meet operating, reporting, and compliance objectives. Many of the internal controls that are effective with other existing payment instruments are also applied by participants of Faster Payments and by Faster Payments networks.

As described in Nacha's Operating Guidelines, internal controls encompass both administrative controls and internal accounting controls, though these internal controls can be generally implemented for any Faster Payments network.

Administrative controls are those that establish parameters for transaction authorization, including:

- **Organization controls**: organization of the department, staffing requirements, and placement of the Faster Payments department within the company's organizational structure
- **Training program:** a formal program designed for all levels of personnel to be trained
- **Operating procedures:** formal, detailed, documented procedures that regulate schedules and workflow
- **Analysis of transaction errors:** maintenance of a record of errors, with notes on their nature and frequency, to indicate basic deficiencies in the total system
- **Risk controls:** control over processing risk to determine the overall exposure to the institution, including all operating, credit, systems, and processing areas

Internal accounting controls increase the reliability of financial records, including:

- **Segregation of duties:** duties assigned to prevent an individual from both initiating and concealing errors, either intentionally or accidentally i.e., one person inputs the payment, and another person verifies and approves the payment
- **Reconcilement of input to output:** reconciling and balancing batch totals and run totals
- **Physical security:** control over the physical movement of logs, transmission registers, etc., whether received from other institutions or transferred internally
- **Management review:** formal review that enables management to determine whether operating and accounting procedures are being followed

The reputation of the Faster Payments networks depends on the efficiency of automated systems and the operating procedures developed to support them, so participants must work to minimize human error. Furthermore, the receiving FI is responsible for the proper posting of transactions to its customers'/ members' accounts, while the sending FI is responsible for submitting items to the Faster Payments network. Both parties should have sound internal controls and procedures that will help ensure that their responsibility is fulfilled.

Barriers and limitations to the implementation of internal controls for Faster Payments

Implementing internal controls for Faster Payments can face barriers and limitations depending on the organizational context, internal expertise, and the characteristics of the Faster Payment system. For

instance, process issues may arise, including challenges related to workflow adjustments, integration with existing payment methods, and the need for standardized procedures. Service Level Agreements (SLAs) can also pose hurdles, involving concerns about meeting performance expectations and ensuring timely transactions while managing risks.

Some common challenges may include:

- **Resource allocation and costs:** organizations may face budget constraints that limit their ability to allocate the right resources needed to carry out tasks. Staff with knowledge of Faster Payments may need to be sourced and trained.
- **Integration with existing payment methods:** Integrating the processes of new payment systems with existing processes and technologies may require significant changes. Additionally, the presence of silos in payment processes can create barriers to the implementation and design of internal controls.
- **New security concerns and challenges:** Faster Payments may introduce new security risks, such as fraud and cybersecurity threats. The lack of expertise to deal with these issues and set the necessary internal controls poses an additional barrier. In certain organizations, employees may be unfamiliar with the new technologies and challenges associated with Faster Payments, requiring the allocation of additional resources to staff training.

The impact of service disruptions on Faster Payments system integrity

Service disruptions can be more damaging for Faster Payments than for traditional payment systems because one of the selling points of Faster Payments is their (near) 24/7 availability. While traditional systems have planned downtimes, most Faster Payment systems do not; they typically operate on a continuous basis whereas traditional systems clear and settle transactions over a longer time.

Service disruptions can damage the value of Faster Payments in multiple ways. First, maintaining end user and participant trust in the Faster Payments system's services is very important, otherwise end users may stop using the system and participants may withdraw from it, lowering system utility for everyone. Second, end users – both consumers and businesses – utilize Faster Payments to make important payments. Service disruptions, especially extended disruptions, can complicate an end user's ability to execute important financial transactions, potentially disrupting supply chains, important business-to-consumer disbursements, bill payments, etc. These disruptions can have ripple effects throughout the economy if they are not taken care of quickly.

Risk assessment methodologies applicable to Faster Payments

A risk assessment provides a way for an FI to get a handle on their current risk exposure and identify types and levels of risk they are willing to accept to achieve their strategic goals. It is a proactive approach that helps them reduce the potential for harm, improve safety, and ensure compliance with relevant payment system rules, laws and regulations. As the volume of payments rises, payment system operators have strengthened their risk management systems and controls to maintain secure practices in transferring funds.

FedNow®

The Federal Reserve has several compliance-related mandates on FedNow participants, necessitating the establishment of compliance and customer due diligence programs along with procedures for scrutinizing account holder data against sanction lists. These requirements, irrespective of legal obligations, align with the standards applicable to federally supervised financial institutions. The Federal Reserve Banks retain authority to terminate or limit a participant's service access for non-compliance with these requisites or applicable laws. In addition, managing third-party risk is crucial in payment processor relationships, requiring risk assessment,

policies and procedures, due diligence, and oversight to mitigate associated risks. The Office of Foreign Assets Control (OFAC) oversees and enforces economic sanctions programs and U.S. persons, including financial institutions, which must adhere to these requirements. As such, the FedNow Service assists financial institutions in risk management through various methodologies employed as well as additional features such as customizable transaction limits and the creation of negative lists to prevent payments to specific accounts.

RTP®

RTP rules and regulations outline the essential risk management and fraud control measures that a participant must follow. For instance, sending participants must employ multi-factor authentication. Participants must also conduct thorough fraud monitoring before submitting a payment message to RTP. Participants are required to regularly evaluate the effectiveness of their fraud monitoring systems and make necessary updates to address emerging fraud risks. The fraud and risk-screening measures must be available on a 24/7 basis. Participants must promptly investigate transactions or activities associated with fraud alerts from TCH's fraud-monitoring program and integrate alerts related to confirmed fraud into their standard fraud-detection practices and report them to the system operator.

Same Day ACH

Both FIs and their service providers should establish and maintain systems and controls to minimize the associated risks with ACH activities, which have clearly defined objectives, business strategies, and clear risk parameters. An effective vendor management program should include a due diligence process for selecting third-party service providers and an oversight process. All these management programs should be produced and captured in periodic reports that allow for various analyses and evaluations. Moreover, financial institutions should implement credit risk controls that establish underwriting standards that require an analysis of originators' creditworthiness and set appropriate credit exposure limits. Similarly, when the FI requires pre-funding before transactions originate through the ACH network, they should ensure funds are collected before an ACH file is sent to the ACH Operator.

Push-to-card

Card networks require their participants to follow KYC and AML procedures to assess risks related, at a minimum, to suspicious activity monitoring and the participants' risk tolerance before transaction initiation. First, participants are required to monitor cardholder and merchant transactions based on their risk ratings. These ratings can be provided by credit rating agencies, the participant's own calculation, or the card networks' value-added services. Second, participants are required to monitor their clients' transactions based on the types of transactions they send, the products they use, the geographies transaction are sent to, and the changes in their behaviors over time, such as volume, value, and frequency. Third, all participants must apply some sort of validation of both the originators and receivers of P2P transactions and the receiving entity that issued the receivers 'cards. Additionally, receiving entities ought to apply similar monitoring procedures to screen for suspicious activities before posting funds to the beneficiary.

The use of process flows in risk assessment for Faster Payments

Faster payment systems employ structured process flows for comprehensive risk assessment in managing the intricate landscape of Faster Payment transactions. These systems identify and map various risks, including fraud, liquidity, operational, legal, settlement, sanctions compliance, and systemic risks, throughout the payment lifecycle. Thus, risk assessment is crucial for managing diverse risks, including liquidity, third-party management, fraud, and information security. By integrating multi-factor authentication, real-time transaction monitoring, and compliance checks, Faster Payments bolster security and ensure adherence to

regulatory standards before the payment initiation stage. Thorough due diligence on participants, continuous review and updates, and well-defined response mechanisms contribute to proactive risk mitigation. The documentation and reporting of the risk assessment process, along with regular communication of outcomes to stakeholders, regulatory bodies, and internal decision-makers, further enhance the reliability and robustness of Faster Payment ecosystems. Payment process flows are integral to the overall risk assessment and management within the realm of instant payment systems.

Reasons for implementing prefunding requirements

The choice of a settlement model has important consequences for the safety, efficiency, and liquidity costs/processes of a Faster Payments system and its participants. There are two major settlement models used in Faster Payment systems: real-time gross settlement and deferred net settlement. Each model has its own potential benefits and downsides, and the most appropriate choice depends on the specific context of each Faster Payments system, such as the regulatory framework, the number and type of participants, the degree of adoption of Faster Payments, transaction value limits, and others.

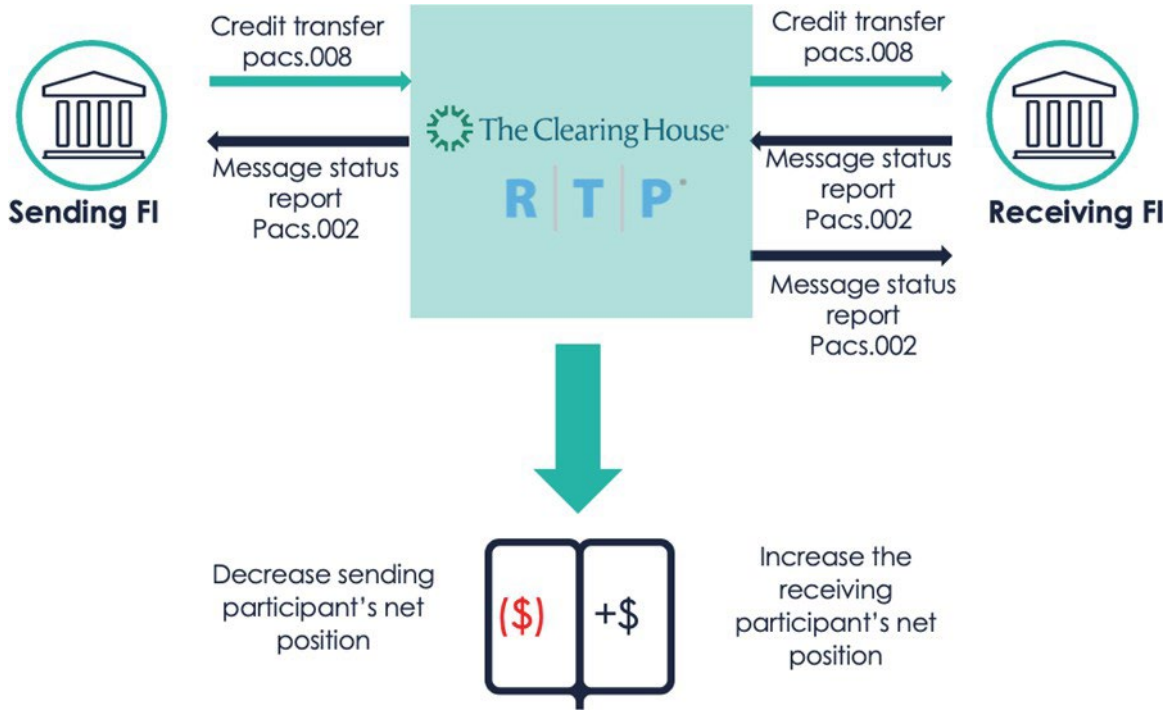
The main advantage of real-time gross settlement is avoiding the buildup of credit risk between participating FIs. This settlement model requires higher liquidity management efforts, such as liquidity forecasting, particularly when the National Settlement Service (NSS) is not operating. There are also differences in terms of the type of account used for settlement, the funding mechanisms, and the liquidity management tools available to participants even between Faster Payment systems using real-time gross settlement.

Looking at Faster Payments systems in scope of this handbook, FedNow uses real-time gross settlement via transactions between Federal Reserve master accounts. In contrast, RTP operates through a fully prefunded account (the "RTP Prefunded Balance Account") jointly owned by all the participating entities and held at the Federal Reserve Bank of New York. TCH is the sole agent of the account. When a transaction in RTP is settled, ownership shares of the transacting parties is shifted, so no value is transferred into or out of the TCH account as part of settling the transaction.

All payments are prefunded by sending participants. Participants fund the RTP account according to their prefunded requirement, which is determined by The Clearing House. In general, TCH will determine the Prefunded Requirement for a Participant based on its total U.S. Transactional Deposits based on public call report data. Participants – directly or through a Funding Agent – must transfer funds in an amount equal to or greater than its prefunded requirement to the Prefunded Balance Account via Fedwire before it can originate any RTP payments. FIs that do not provide their own funding must have an arrangement in place with another FI.

RTP verifies the sending participant's Current Prefunded Position (CPP). RTP will reject the payment if the sending participant has insufficient liquidity to cover the payment. RTP records entries by decreasing the net position and current prefunded position of sending participants (or its funding provider) and increases the net position and current prefunded position of receiving participants (or its funding provider) when transactions are successful. These current prefunded positions are backed dollar-for-dollar by funds held in the prefunded balance account.

Figure 11. How an accepted credit transfer impacts the net positions of sending and receiving participants in RTP



Participants must fund the prefunded account according to their RTP payment needs because RTP's settlement account is separate from the participant's Federal Reserve master account, ensuring the presence of sufficient liquidity to avoid payment rejections.

Onboarding programs, policies, and procedures

Knowing your customer/member is vital for financial institutions to provide products and services. It is also important to gather information that can be used to verify their identity to ensure that payments are initiated by legitimate account holders for lawful reasons. Rules and requirements have been put in place by regulators and payment system operators for the collection and retention of account holder information by financial institutions, which can vary based on the FI's risk profile, size, and the type of services it offers. All financial institutions are bound by federal regulations to collect customer data, use them for KYC, AML and sanction list checks, and to regularly verify their account holders' identities.

Importance of onboarding programs for managing risk related to Faster Payments

There are a number of different risks related to Faster Payments, though legal and security risks generally originate from the account holder side. Proper onboarding of account holders is vital for financial institutions and third parties in order to minimize these risks.

This leads us to legal risks related to preventing fraudulent activities and providing payment system access to individuals who should not have it. FIs can be penalized by financial regulators for non-compliance with rules and regulations with requirements related to these topics. As Faster Payment methods, especially those that are irrevocable, are easier to abuse by fraudsters, the number of cases can increase significantly and thus the legal risks and costs along with them. Therefore, participants of Faster Payment systems ought to

pay increased attention to applying fraud prevention and mitigation methodologies, including but not limited to, adequate customer onboarding procedures.

Customer Identification Program (CIP)

Customer information collection

Under the USA PATRIOT Act, financial institutions are required to have a customer identification program (CIP) in place to verify an account holder's identity and prevent fraudulent account openings and other financial crimes. It is intended to enable an institution to form a reasonable belief that it knows the true identity of each person opening an account. Each institution should have a written CIP that is appropriate for its size and type of business and that also includes minimum requirements set by the BSA.

CIPs require financial institutions to complete the following prior to opening a new account:

- Verify the identity of any person seeking to open an account,
- Maintain records of the information used to verify the person's identity, and
- Consult government known or suspected terrorist lists to determine whether the person appears on any such list.

Verification

Again, verification of a person's identity must be completed by the FI prior to account opening. The first step in this process is to collect the person's name, date of birth, address, and identification number (e.g., Social Security number for U.S. citizens, or a taxpayer identification number for non-U.S. citizens). They are expected to review a government-issued form of identification proving the person's nationality and/or residence, and a photograph or similar safeguard, such as a valid passport or driver's license. FIs are also encouraged to review multiple documents to limit fraud risk. FIs are allowed to verify identification without documents, for instance, contacting the person, using independent verification services such as credit reporting agencies, or to check references from other FIs.

Record keeping and retention

As part of its CIP program, a financial institution must maintain records of the information used to verify a person's identity for a period of five years after an account is closed. They also need to keep the descriptions of the documents used for verification, the methods used for verification in the absence of a document, and the resolution of any substantive discrepancy discovered when verifying customer information. All requirements are set out in 31 CFR 1010.430, "Nature of records and retention Period".

Comparison with government lists

Additionally, FIs are required to include procedures in their CIP processes that determine whether or not the person appears on any government sanctions list (e.g., OFAC's Specially Designated Nationals (SDN) List or government known list of suspected terrorists). These procedures must take place within a reasonable time following the account opening. FIs must follow federal directives in connection with such lists. Once an account is established, FIs must periodically check account holders against OFAC's SDN List, which is a separate requirement from the CIP.

Responsibilities of parties involved in a customer identification program (CIP)

TABLE 55. FASTER PAYMENTS STAKEHOLDER RESPONSIBILITIES IN THE CUSTOMER IDENTIFICATION PROGRAM

STAKEHOLDERS	*RESPONSIBILITIES IN CIP*
Financial institutions	• Set appropriate CIP based on the size and type of business • Obtain person's information upon account opening • Verify the identity of each person through risk-based procedures • File suspicious activity report upon CIP failure • Comply with record keeping requirements • Cross check registered account with government lists • Provide adequate notice before verification procedures • When relying on another FI for a person's information ensure that it is regulated by a federal regulator and hasadequate AML procedures • Ensure its third-party providers adhere to CIP
Third parties	• Carry out services to FIs, regardless of data collection or management, according to the FI's CIP
Customer	• Share required information upon account opening • Update information upon changes

Additional requirements for Faster Payment customers that use Request-for-Payment functionality

In addition to the standard CIP responsibilities of Faster Payments stakeholders, specific requirements are set for the users of the Request for Payment service in RTP. TCH requires participants that send Request for Payment messages to ensure the sender of the request has a legitimate purpose to request funds from the receiver of the message, such as:

- The message is initiated by a non-consumer sender and is sent for a current sale or transaction, or the amount is due, owed, or otherwise agreed to be paid to the sender.
- The message is initiated by a consumer sender to a receiver who is known to the sender and who is reasonably expected to receive the request from the sender.

Types and applications of authentication channels and methods for transactions performed via Faster Payments solutions

Authentication is the means of recognizing an account holder's identity. In the landscape of Faster Payment transactions, security hinges on robust authentication practices.

This section will delve into authentication methods used in these systems and shed light on Strong Customer Authentication, biometric authentication, device binding, and risk-based authentication. The subsequent section will examine the various parties that must be authenticated in a Faster Payment transaction. The section will conclude with an examination of the authentication methods used in each of the various Faster Payment rails examined throughout this text.

Types of authentication methods used for faster payments transactions

Strong Customer Authentication (SCA)

Settlement finality and processing speed make authentication a critical component of Faster Payment security. The most common tactics around authentication revolve around some combination of Strong Customer Authentication (SCA) measures, with account holders using one or more factors to authenticate their identity before initiation. The factors that make up SCA may include a combination of the following:

- **Possession:** something a person physically possesses, such as a card, key fob, or phone.
- **Inherence:** verification using something unchangeable to the nature of a person. Biometric verification via fingerprint, iris scan, or voice falls into this category.
- **Knowledge:** something a person knows, such as a PIN, passcode, or memorable information.

Some FIs only allow for a payment to be initiated once SCA is completed. While SCA practices are present in the United States, there is no nationwide mandate requiring its usage. FIs and payment service providers (PSPs) either define individually or through industry agreements what sort of authentication methods they intend to use. SCA is used widely in the sign-in process for mobile banking and payments apps, and consumers can opt into it for certain actions such as high-value payments.

Biometric authentication

Biometric authentication uses physical features that are unique to an individual and can be easily retrieved and verified from a database. Biometric authentication is inherently more secure, accessible, and convenient than knowledge-based verification, as it cannot be forgotten, easily shared or stolen, or guessed. Common types of biometric verification include fingerprint scanning, voice identification, facial recognition, and eye scanning. Users will often verify themselves biometrically when logging into the channel through which they access Faster Payments, usually via smartphone or laptop, or for certain transactions such as high-value payments. Biometric authentication ensures that payments are associated with a specific person while minimizing the risk of fraud or unauthorized usage. However, generative AI poses a challenge to biometric authentication by generating highly realistic synthetic images, videos, and voice that can mimic genuine biometric features, enabling potential fraudsters to create convincing replicas for unauthorized access, thereby undermining the security measures implemented by biometric authentication systems for Faster Payments.

Device binding

Device binding is a method of authentication that links a user's identity to the possession of a specific device, such as a smartphone or laptop, from which authorized payments can be made. In this authentication approach, payments can only be made from devices that are linked to the user, mitigating the risk of unauthorized payments in the event of a compromised password or data breach. Device binding enhances security by requiring both the proper login credentials as well as access from a registered and trusted device. However, device cloning or theft of the physical device associated with a Faster Payments account mitigates this if not promptly reported, as the thief can directly send fraudulent payments from the trusted laptop or phone.

Risk-based authentication (RBA)

Risk-based authentication is a method in which the risk associated with a transaction is assessed, determining the level of security necessary for fulfillment. This method analyzes risk using factors such as the user's transaction history, intended payment beneficiary, geographic location, IP address, and transaction value. RBA compares these factors against behavioral data gathered on users to identify anomalies that could signal that a payment requires additional identification. If any of these factors raise alarm with the payment service provider, additional verification may be needed. RBA adapts security methods based on perceived risk, allowing low-risk transactions to be processed seamlessly while prompting additional authentication steps for higher-risk transactions.

Parties involved in Faster Payments transactions who need to be authenticated

Individuals

Individual payment senders must be authenticated, usually during the login process for the channel through which they access the service. This process is usually performed via some method of SCA, using something

the sender knows (such as a password or PIN), something they are (biometric verification), and something they have, such as a trusted device through which they receive authentication codes. These measures help ensure that the person initiating the payment is the legitimate account holder, enhancing security and reducing the risk of unauthorized access.

Businesses

Authentication for businesses is a slightly more rigorous process than for individuals using a Faster Payment service. In the initial signup process for Faster Payment services, authentication may require legal documentation. Authorized representatives within the organization can also be assigned specific access credentials and often employ digital signatures to authenticate payment instructions. Robust access controls, such as role-based access can also be implemented to restrict system entry, and risk-based authentication can also be applied to business payments.

Financial institutions

Authentication for financial institutions connecting to a Faster Payment system involves rigorous measures to ensure the security and integrity of transactions. Institutions undergo thorough identity verification, often requiring legal documentation to confirm their regulatory standing. Secure access controls and SCA methods are implemented, including cryptographic keys or digital certificates. The integration of advanced monitoring systems helps detect and respond to unusual patterns in transaction behavior, enhancing fraud prevention. Compliance with industry standards and regulatory requirements is mandatory, ensuring that financial institutions maintain a robust authentication framework, protecting the overall stability and trustworthiness of Faster Payment systems.

Features and functions of authentication types by Faster Payments rail

Same Day ACH

Authentication for Same Day ACH payments is generally performed through the channel by which the ACH Network is accessed. Nacha requires that ODFIs utilize a commercially reasonable method to verify the identity of payment originators when the origination agreement is entered into. Beyond the basic requirements set out by Nacha, FIs may require more stringent verification policies for Same Day ACH transfers made via their channels, such as SCA and risk-based analysis for anomalous payments.

FedNow

FedNow relies on the same authentication solutions and processes as the Fedwire Funds Service. For direct access to this service by FIs, SCA is used, including a physical USB security token device and the usage of a passphrase known by the user accessing the service. For consumer payments sent via a third party or payment app, there is no general requirement for SCA, but it is implemented in many of the login processes for channels through which consumers access the service. FIs handling FedNow payments will often utilize additional authentication methods as a part of industry standards and individual FI standards. These methods may include biometric authentication, Strong Customer Authentication, utilizing passwords or phrases, device binding, SMS or email codes, or risk-based authentication, which analyzes transaction patterns, device information, and user behavior for dynamic security adjustments.

RTP

Sending institutions must at minimum utilize SCA (specifically inherence and knowledge-based) to authenticate the identity of senders before payment submission. This can be performed via biometric verification during the initial payment channel login payment process, and text/email verification before payment submission. Processes

must also be in place to detect anomalous activity around the initial login and authentication of a sender as part of the authentication procedure. Strong Customer Authentication is also commonly used by FIs when facilitating RTP payments via mobile banking app or online channels. Zelle, which uses RTP rails to facilitate some of its transactions, also requires the use of multifactor authentication, though not for all transactions made using the service.

Network specific authentication practices: Push-to-card payments

For push-to-card payments authentication procedures are carried out via the channel through which the user accesses the service. FIs are responsible for authenticating originating account holders in all transfers in accordance with their own authentication compliance policies and the applicable state and federal regulations. The card networks also place an emphasis on secure authentication for transfers made via mobile app, mandating that FIs use some sort of a secure authentication method for service access such as user ID/password authentication or biometrics..

Chapter Four

Technology Considerations for Enabling Faster Payments

SECTION BLUEPRINT COMPOSITION: 15%

Types and purposes of technology components of faster payments rails

The payment industry has witnessed significant change in recent years thanks to various technological components that power the Faster Payments rails. Numerous technology factors including APIs, tokenization, network connectivity, and many more have changed the payments industry. The core banking system, APIs, directory services, secure authentication processes, and network connectivity all work together—each playing an equally vital role—to deliver seamless, reliable, and enriched payment experiences for every user.

Critical technology components of Faster Payments rails

Faster Payments rails are empowered by various technology components that have begun transforming the payment landscape, enabling faster, more secure, and highly available, or "always on" payment experiences.

APIs

Application Programming Interfaces (APIs) are sets of protocols and tools that enable different software applications to interact with one another. They define the methods and data formats that applications can use to request and share information and enable smooth communication and interaction between different players in the payment ecosystem, including financial institutions, payment processors, third-party service providers, and even payment infrastructures. APIs permit the development of flexible and customizable solutions within the payment system and enable participants to integrate with various services. APIs allow for the real-time exchange of data and support the seamless flow of information and transactions.

QR codes

Quick Response Codes (QR Codes) are a versatile type of bar code that represents structured information and can enable the visual exchange of that information. One such usage is the encoding of information needed to process payment transactions between accounts through either cloud-based or local merchant systems. QR code payments can be initiated by either a merchant or a customer. Some of the potential benefits of QR codes include cost-effectiveness, simplicity of use, and flexibility, as QR codes can contain both static and dynamic information.

Static QR codes contain information that remains constant and can be printed and exhibited in different locations like the point of sale, menus, bills, etc. Conversely, dynamic QR codes can carry up-to-date

information along with optional transaction-specific details. Dynamic QR codes can help with security, fraud control, and traceability.QR codes enhance the user experience by streamlining the payment initiation process, proving especially beneficial in situations such as PoS transactions, self-service kiosks, invoices and bills and peer-to-peer (P2P) payments.

The Accredited Standards Committee X9 is developing a new QR code payments standard. The new standard will establish the content of a QR code for payments, emphasizing interoperability, in order to process a payment digitally, and will include both merchant-presented and customer-presented QR Codes. Encoded in a payment QR code will be all the information necessary to execute a payment without the customer or merchant having to share financial account details. Further information can be found here: https://x9.org/x9-launches-effort-to-create-payment-qr-code-standard/).

Directories

Directories are databases of payment-related information and simplify payment routing by enabling payment initiation using social aliases such as email, cell phone number, etc whereas routing tables serve as the fundamental directory service, making payment routing possibly. These aliases are linked to payment account information and replace the necessity of memorizing, sharing, or storing sensitive and complicated information, thereby facilitating payment initiation and assisting in fraud prevention. Faster payment directories support multiple use cases such as P2P, C2B, B2C, and others.

Encryption & Tokenization

Encryption and tokenization are two methods that provide additional layers of security to transmitted data. Encryption and tokenization reduces the risk of data exposure by replacing sensitive data is replaced with values that have no meaning outside a system.

Cryptographic protection is a process of converting data into a coded value which is indecipherable protected by a secret via an algorithm which enables the recipient of the coded value to recreate the original data provided they have the secret. **Encryption** is the application of the algorithm to encode the original data using the algorithm and secret. **Decryption** is the process of using the algorithm and secret to recreate the original data. Asymmetric cryptography uses a pair of keys, called private and public keys, which are different. Information encrypted with a private key can be decrypted with the paired public key and vice versa. Asymmetric cryptography is typically used for digital signature applications. Symmetric cryptography uses one key to encrypt and the same key to decrypt. Symmetric cryptography is typically used for protecting data at rest an in transit.

Tokenization is a security technique that replaces sensitive payment information, such as account numbers or credit card numbers, with a unique, random string of characters called a "token." Tokenization usually involves a party, called a Token Service Provider, who takes the sensitive data, turns it into tokens and later can take tokens and turn them back into the sensitive data. The tokens are meaningless to bad actors who may acquire them, but authenticated participants in a tokenization scheme can present the tokens to the Token Service Provider and retrieve the sensitive information for processing.

Network connectivity

FIs connect to Faster Payments rails both directly or indirectly through other entities like third-party service providers or just service providers, correspondent banks, or corporate credit unions. These connections occur on each payment rail network, which ensures secure exchange of payment and non-payment messages among various participants. Third-party providers can include a core processor, a hosted gateway, corporate credit union, etc. Generally, FIs are responsible for services they offer, even if some of them involve 3rd parties.

Pursuant to various rules of each payment rail, certain participants are obligated to validate their abilities to meet requirements.

Authentication

Authentication is the process of digitally proving that a party presenting a credential is the rightful owner of that credential. A credential can represent a person, a company, a server, etc. This safeguard is especially vital in Faster Payment systems, where it blocks unauthorized parties from initiating transactions and thus protects end users from fraud. Faster Payments manages security risks by protecting transactions with robust security measures augmented by technology, such as two-factor or multiple-factor authentication, biometric verification, and real-time monitoring. These technologies provide an additional layer of security, ultimately contributing to the speed and efficiency of the overall payment system.

FI core banking platforms

A responsive, efficient core banking system is essential for handling Faster Payments, supporting key functions such as accounting, CRM, risk management, operations, and reporting. Technology has advanced the core banking systems to support payment processes to better manage account information and balance updates continuously 24/7/365. Core banking systems have traditionally been built for a batch-based world which is no longer fit-for-purpose, so it is important that FIs participating in one or more Faster Payments systems have a core banking platform that is up to the challenge. The OCC has identified outdated core banking systems as a threat to financial institutions (https://www.pymnts.com/news/banking/2025/occ-finds-outdated-core-systems-threaten-future-of-us-regional-banks/),

Application Programming Interfaces (APIs) in the context of Faster Payments

Quick availability is a fundamental characteristic of Faster Payments in the United States. Traditional payment-origination methods are adapting not only to meet faster posting times, but also to satisfy 24/7 customer expectations, tighter regulatory and compliance requirements, real-time fraud-prevention demands, richer data standards such as ISO 20022, and competitive pressure from agile fintech alternatives. Payment providers, core platform, financial institutions and others have adopted application programming interfaces (APIs) to facilitate seamless and instant programmatic connectivity between corporate originators, merchants, consumers, and technology providers. APIs involve establishing a trusted programmatic connection between two parties, such as financial institutions and third-party payment processors. APIs enable participants to build adaptable solutions within each payment system and to support diverse use cases.

APIs and Faster Payment solutions are often developed by each payment provider, meaning APIs are not likely based on an open framework, making integrating other service providers more challenging. This complicates and slows the FIs' ability to add new services and innovate. Within the payments ecosystem, Faster Payment APIs have five core functions as categorized and defined by the Faster Payments Council's API Working Group: registration and onboarding, log-in and API access management, payment initiation and payment processing-related, directory functions, and embedded and additional functions.

TABLE 56 CORE FUNCTIONS OF INSTANT PAYMENT APIS

API CATEGORIES	CORE FUNCTIONS OF INSTANT PAYMENT APIS
Registration & onboarding	Registration APIs enable corporations to enroll in a provider's Faster Payments API service, while onboarding APIs streamline tasks like KYC validation and user authentication for employees or customers accessing the service.
Log-in & API access management	Maintenance APIs facilitate the administration of tasks like adding or adjusting account settings, including managing log-in/API access, usernames, passwords, and channel access via methods such as mobile phones, digital wallets, email, and online platforms.
Payment initiation & payment processing-related	This extensive category of APIs supports various Faster Payment processing functions, encompassing activities such as initiating payments, request-for-payment initiation, receiving acknowledgment or response codes, and querying and reporting payment status and account balances.
Directory functions	A directory API grants access to information related to customers or entities, including bank routing numbers, account details, aliases, and other data essential for payment or transaction processing, usually maintained by a central authority or organization like a payments network.
Embedded & additional functions	This category of APIs encompasses a wide range of supplementary services including fraudulent transactions prevention controls, sanction screening features, and transaction data reporting features.

Methods for protecting data

Payment systems employ diverse methodologies to secure data during transmission and storage. Among the most utilized technologies are tokenization and encryption (see sections above). Specific rule-based guidance governs the protection mechanisms which must be used for certain systems as outlined below.

FedNow

The FedNow Service requires that all messages are digitally signed utilizing asymmetric cryptography as described above. Digital signing provides a layer of security controls, helping verify the integrity and authenticity of messages exchanged through the FedNow Service. All messages in the FedNow Service are therefore signed with the sender's private key and verified by the receiver using the sender's public key. The FedNow Service rejects messages that lack a signature, use an expired key, or carry an unrecognized key, and returns a specific error code for each condition. Participants, and Service Providers acting on their behalf, are encouraged to maintain multiple active key pairs, each with a unique one-year expiration date. Digital signing of messages is required for RTP as well.

RTP

The RTP Network introduced enhanced security through a capability called Secure Token Exchange, providing account tokenization for financial institutions. This feature issues tokens for account numbers, reducing vulnerability from storing information outside the banking system and thus preventing information theft and fraud. RTP payments using tokens operate similarly to those with real account numbers. Use of the Secure Token Exchange for RTP participants is optional. Like FedNow, the RTP Network further employs Transport Layer Security (TLS), s a cryptographic protocol that encrypts data while it is in transit, and each message is digitally signed to ensure authenticity in accordance with ISO 20022.

Same Day ACH

The ACH Network defines information security as protecting data from unauthorized access or modification. The key components of information security consist of confidentiality, data integrity, and availability. Furthermore, the Nacha Operating Rules require protecting the security and integrity of certain ACH data throughout its life cycle by establishing access controls, self-assessment, and verifying the identity of third-party senders and originators. All non-consumer originators, DFIs (Depository Financial Institutions), TPSPs, and third-party

senders must also establish, implement, and update security procedures, policies, and systems related to the initiation, processing, and storage of entries.

Each Non-Consumer Originator that is not a Participating DFI, each Third-Party Service Provider, and each Third-Party Sender whose ACH Origination or Transmission volume exceeds 2 million Entries annually must protect DFI account numbers used in the initiation of Entries by rendering them unreadable when stored electronically. This obligation applies no later than June 30 of the year immediately following the year in which such volume first exceeds the 2 million entry threshold, and consistently thereafter regardless of annual volume.

Mastercard Send & Visa Direct

The Payment Card Industry Data Security Standard (PCI DSS) was developed by Visa, Mastercard, and other payment card industry leaders to set a baseline level of protection for customers, merchants, and service providers reduce fraud and the impact of data breaches. PCI DSS applies to all entities that store, process, or transmit cardholder data and/or sensitive authentication data, enhance consumer confidence, and protect the overall integrity of the payment system.

Every transaction moving on Visa's trusted global network is supported by a multi-layered system of risk and compliance controls available, which consists of tokenization, fraud controls, and robust program governance. Visa Direct APIs are required to support Message Level Encryption (MLE), which serves the dual purpose of securing information and communication with other parties by preventing unauthorized parties from understanding the information or communication.

Communication between client applications and card networks is safeguarded through TLS/ SSL (Transport Layer Security/Secure Sockets Layer), ensuring automatic data encryption during transmission over networks. Moreover, certain services incorporate end-to-end payload encryption. For instance, services handling sensitive information, such as cardholder Personally Identifying Information (PII), must adhere to the PCI DSS by encrypting the transmission of cardholder data across open, public networks.

Connecting with Faster Payments platforms

Financial institutions connect to Faster Payments platforms directly or indirectly via third-party service provider (TPSP) including correspondent banks, and corporate credit unions, and technology companies. Further detail on participation types and service offerings as well as eligibility requirements for each payment rail is available in Chapter 1.

FedNow

Financial institutions have the option to connect to the FedNow Service either by utilizing a FedLine Solutions connection (WAN or VPN) for message exchange, profile administration, and other functions, or through a third-party service provider like a payment processor, correspondent bank, or corporate credit union. Service providers that act on behalf of participating FIs establish connections to the FedNow Service using a FedLine Solution. These providers can process transactions for multiple financial institutions through a single connection, be it WAN (FedLine Direct) or VPN (FedLine Command or Advantage). For fintech companies or other providers exclusively offering instant payment solutions to consumers and businesses (e.g., payroll providers, payables and receivables vendors, treasury management platform enablers), a relationship with a FedNow participating FI is required. Depending on the services they provide, third-party service providers may or may not have a direct connection to the Federal Reserve.

RTP

FIs can connect to the RTP Network directly or using a third-party service provider, such as a core processor, a hosted gateway, a bankers' bank, or a corporate credit union. RTP connects via TCH proprietary connections.

Same Day ACH

To participate in the ACH Network parties connect to one of the two ACH Operators – Electronic Payments Network (EPN) provided by TCH or FedACH provided by Federal Reserve Financial Services. The ACH Network allows direct access to originators, third-party senders, or third-party service providers who act as an intermediary to transmit credit or debit entries using an ODFI's routing number and settlement account. However, a participant depository FI must either on their own or through a correspondent bank maintain a settlement account at a Federal Reserve Bank. The ACH Network requires ODFIs to register certain direct access relationships with Nacha. For example, specific relationships with third parties and originators must be defined and registered accordingly.

Visa Direct

The sender FI must either be a licensed Visa acquirer or a third-party originator that is sponsored by a licensed Visa acquirer for a Visa Direct program to use the Visa Direct APIs. They must also be able to validate their ability to meet the requirements of a full push payments processing service.

Mastercard Send

Sender FIs can integrate with Mastercard Send directly via a single connection or work with Mastercard-approved partners (known as Transaction Initiators), such as digital players. While a single connection is provided by Mastercard's network, connectivity to multiple networks is available in certain markets.

Faster Payments Directories

Payment directories are informational databases that support the exchange of payment-related information. Directories facilitate payment initiation by linking a payment alias(es) (email, cell phone number, etc.) to payment account information, such as the bank, account number, routing number, PAN, etc. This means that the payee can share their alias with the payer rather than sensitive information that can lead to misrouting a payment. Aliases also tend to be easy to remember in contrast with the information required to route a payment over a Faster Payment rail, thus facilitating payment initiation.

Directories are often overlay services and are not necessarily directly related to the underlying payment system infrastructure. Zelle, for example, is an overlay service operated by Early Warning and is not formally tied to TCH's RTP or Same Day ACH. On the other hand, Visa Direct and Mastercard Send do have their own directories for use in their respective systems. As of now no directory uses the FedNow system, though this will likely change.

Directories, whether they are centralized or federated, have a number of responsibilities. This includes securing information and ensuring that only authorized parties access the directory and only to parts they have authorization to access. Directories have to strike a balance between openness and security: directories want to ensure widespread adoption and therefore may be open to third party access but must also ensure that they are safe and secure against abuse. The convenience directories provide enhance data security and accuracy, allowing individuals and/or businesses to make and receive payments without sharing sensitive information, simplifying the payment process.

Faster payment directories can be broadly categorized into two types according to the way they look up payee information:

- **Centralized Directory:** This type of directory maps an alias to its PSP and the corresponding account number (or other type of account-related number such as a PAN). The mapping of alias is a one-step exercise in case of centralized directories.
- **Federated Directory:** This type of directory maps an alias to its corresponding PSP. The PSP is responsible for mapping the account information locally to the alias provided, using its own directory, making this a two-step process.

Zelle operates the largest Faster Payments alias directory for bank-initiated transactions in the United States, while Venmo and PayPal are two non-bank payment service providers (PSPs) that provide alias-driven payment services. All three services allow users to link their account information to an email address or phone number, while Venmo and PayPal also allow users to initiate payments using a user-generated identifier. The three directory services are not interoperable, though Venmo and Paypal are owned by the same company.

Purpose, functions, and benefits of directories used to support Faster Payments

Directories link aliases, typically a short identifier that is easy to remember and share, to a user's transaction account information. This allows simplified payment initiation without the need to know and input the payee's payment details. For the payee, using an alias removes the need to memorize and share payment information, which can be sensitive. In the context of bulk payments such as government benefits or other types of disbursements (salary payments, dividends, etc.), the use of aliases simplifies the maintenance of beneficiary information. Aliases mask a transaction account number preventing the theft of transaction-account information, reverse lookup attacks, and the automated skimming of customer information.,

The U.S. Faster Payments Council's Directory Models Working Group has outlined the essential functional characteristics of directories, classifying them into core and value-added features. The core characteristics encompass several key aspects: Directories must possess the capability to validate the accuracy of routing information. Additionally, directories need to ensure that payment aliases remain unique within the directory, allowing for efficient and unambiguous identification. The support for multiple routes linked to a payment alias would enhance flexibility in payment processing, though is not considered essential. Furthermore, directories should minimize the storage of sensitive information, emphasizing the importance of safe-guarding user data. Directories must prevent the exploitation of the directory for mining payment details, reinforcing the need for secure and ethical use. Moreover, they should allow the establishment of end-user-controlled profiles, offering users a degree of autonomy in managing their directory-related preferences.

In addition to the core characteristics, the Faster Payments Council identifies value-added features that directories should incorporate: A directory should support some variant of confirmation-of-payee, which would enhance the security and reliability of payment transactions. This feature helps verify the accuracy of payee information, reducing the risk of errors or fraudulent activities. Additionally, directories should be equipped to support request-for-payment transactions, facilitating a broader range of payment interactions and ensuring a comprehensive and versatile payment ecosystem.

For senders and recipients of faster payments, the use of aliases improves the user experience making it more seamless by:

- **Making it easier to share payment details:** It is usually easier for the payee to share a phone number or email address than a bank identifier and account identifier

- **Reducing the sharing of sensitive account-related data:** A payment can be addressed to the payee without the need for the payee to share sensitive account details (or even information about where they hold their accounts)
- **Providing confirmation-of-payee:** Directories may provide some forms of confirming the identity of the payee, for example, by providing the real verified name of the account holder (as provided by their PSP) or a nickname defined by the user to the payer. This gives the payer confidence that funds are being sent to the correct account as well acting as a tool against fraud.

Faster payment directories can support multiple use cases. The use cases almost always begin from P2P payments and then expand to include C2B, B2C, B2B and others. Zelle, Venmo, and PayPal, are example of a directory-based payment system,, The initial release of FedNow did not include a directory service, limiting financial institutions' ability to offer convenient P2P payment services. FIs can use a private-sector directory to access routing information to allow alias-based payments on FedNow. Visa Direct offers their Visa Alias directory service that links different aliases to a payment credential.

Comparison of the use of directories in other industries

The use of directories has streamlined payment initiation and helped prevent errors and ensure swift transfer of funds. Apart from the Faster Payment directories that link to the payee's account information, there are also closed loop directories such as the ones used in Venmo and PayPal that help process payments within their systems. Payment directories in the United States are highly fragmented and not interoperable, operating in closed loop ecosystems as payment service providers treat their network and associated information as a valuable asset and are not willing to share this information with other networks.

Need for governance around directories that support Faster Payments

The payments ecosystem operates within a framework of business expectations, customer contracts, regulations, and scheme rules. Within this context, directories supporting Faster Payments play an important role and require a well-defined governance structure to establish the criteria for participation, outline participant obligations, and allocate rights and responsibilities to the entities utilizing the directory. All participants in the Faster Payments ecosystem should adhere to a common set of rules and guidelines. This ensures a standardized and consistent approach across the board, contributing to the reliability and efficiency of the payment processes facilitated by directories.

There are many governance structures for directory models, distinguished by platform/framework ownership and eligibility rules for participation. One model is consortium-led/bank-led, where participating banks collaborate to define the governance framework. Challenges may arise in implementing governance changes swiftly due to diverse stakeholder opinions and this model can be difficult in highly decentralized markets such as the U.S.

Another model is mobile operator-led, where mobile operators take the lead in platform development, and banks join as participants. This model has the potential to include non-bank players, fostering competition and innovation. However, it may also lead to market fragmentation as more diverse players, such as financial institutions, mobile wallets, and apps, enter the market. Implementing this model would require changes to scheme rules, which may be challenging in the short-to-medium term.

Security considerations related to the use of directories

The use of directories entails several security standards to ensure the trust and safety of end users. If a directory is perceived as unsafe, users are likely to avoid its use altogether. Directories such as Venmo and Zelle have their own governance systems and safety procedures. These mechanisms secure access to the

directory, facilitating secure payment initiation. The issue of fraud in the United States necessitates that any directory, from its inception, must prioritize safety. Several key factors contribute to the safety of a directory: the extent of payment data visible to the sending party, sender authentication methods, and the type of directory employed.

A major security consideration for Faster Payment directories is to prevent their misuse for data mining. To prevent data mining, various directories adopt different approaches. Some directories restrict the sender's view to part or all of the recipient's name (i.e.,, confirming what the sending party inputs) while others provide the sender with the recipient's full name, confirming the recipient. Notably, none of the directories reveal the receiving party's account information to the sender.

Sender authentication methods present another security challenge as there is no consensus on the best approach. Zelle utilizes the authentication from the respective banking app, others utilize thumbprint or facial recognition or rely on each bank's app authentication measures. Recommending multi-factor authentication, emerges as a potential solution to enhance security against unauthorized payment initiation. Passkeys, as way to provide a secure way to perform authentication without passwords. are gaining traction.

Launching and operating a directory service demands robust data security and privacy measures. Consent from customers for the use of a specific social alias identifier is essential. Payment system operators must establish rules governing fraud risk management and liability. Moreover, stringent verification processes should be in place during customer registration of an alias. Notably, the chosen social alias identifier should not function as an authentication mechanism, such as a username for internet or mobile banking, nor should it compel users to disclose sensitive information, like using a primary account number as an alias.

Purpose and application of messaging standards for Faster Payments platforms

Messaging standards in Faster Payments serve a dual purpose: enhancing interoperability and ensuring seamless communication among financial institutions and payment system participants. This facilitates swift and secure transmission of payment information, reducing errors and accelerating transaction processing. In essence, messaging standards are crucial for creating a robust and efficient Faster Payments infrastructure, enabling a uniform and reliable exchange of information.

ISO 20022 and its application to faster payments

ISO 20022 is an international data standard that establishes a framework for payments, delineating message specifications for, and facilitating the transmission of consistent, comprehensive, and structured data elements.

The messaging standard has multiple applications for Faster Payments. Firstly, it enables richer data elements enabling more accurate payment processing. Second, its structured data supports straight-through processing, which reduces the need for manual interventions, streamlining the payment flow and facilitating quicker transaction times. Third, its increased data granularity and standardized format contribute to fewer errors and misdirected transactions. Fourth, the additional information it can carry enables real-time tracking and confirmation of payment, which is helpful with reconciliation and compliance with regulatory requirements related to the speed of transactions. Lastly, while most Faster Payments systems in the U.S. are not interoperable with either each other or systems in other markets, the use of ISO 20022 can facilitate future interoperability.

ISO 8583 and its use for push-to-card

ISO 8583 is an international messaging standard developed for card-initiated transactions and is currently used by Mastercard and Visa" ISO 8583 is used for the payment card network products (e.g., credit, debit,

Mastercard Send, Visa Direct).ISO8583 is a mature standard used for card-based payments (credit, debit, prepaid, ATM, POS, and mobile). First published in 1987 by ISO with updates in 1993 and 2003. It defines message layout and communication flow. It establishes a structured framework including key data elements for transactions creating a standardized language for card payment messages.

ISO 8583 ensures standardized communication, minimizes errors in processing push-to-card transactions, and facilitates speedy processing. ISO 8583 can enable interoperability when different card networks use it, such as Mastercard Send and Visa Direct. Furthermore, ISO 8583 has been used by the card networks for decades, so using it for push-to-card transactions means FIs have to do less work to implement it, thus lowering costs.

Messaging standards used with Faster Payments rails

Different Faster Payments rails utilize different messaging standards as illustrated in this table:

TABLE 57 MESSAGING STANDARDS BY FASTER PAYMENT RAILS

PAYMENT RAILS	*MESSAGING STANDARDD*
FedNow	ISO 20022
RTP	ISO 20022
Same Day ACH	Nacha
Mastercard Send	ISO 8583
Visa Direct	ISO 8583

ISO 20022 message categories and usage

Purpose of the two key components of ISO 20022

The ISO 20022 standard has two main components, its message structure and data dictionary. The message structure defines the format and organization of messages exchanged between financial institutions. It includes fields and segments with a specific purpose and defines the data type to ensure consistency. ISO 20022 uses a flexible XML-based syntax, allowing for the inclusion of a wide range of financial information in single messages, leading to clarity in communication, reduced ambiguity, and facilitating automation and straight-through processing.

The data dictionary serves as a comprehensive catalog of data elements and their definitions, providing the financial industry with a standardized set of terms. Both financial institutions and payment system developers can refer to this catalog to interpret and use data elements consistently across markets. It includes both business and message concepts, as well as various data types. The dictionary is dynamic and new items are added regularly to avoid misinterpretation continuously.

Faster payments value-added services

Overlay services are specialized offerings that enhance payment services by utilizing existing infrastructure to provide value-added services (VAS) to users. Financial institutions seek to connect to multiple networks and switch between them based on various factors such as transaction volume, pricing, and reach. This is becoming more important as the number of faster payment services grows. Overlay services include liquidity management, tokenization, biller directory(ies), fraud prevention, and customer directory(ies). These services benefit all participants in the payment ecosystem and facilitate the expansion and interoperability of instant payment services.

Collaboration between networks is essential to identify common overlay services, leading to faster adoption, increased efficiency, and innovation. For financial institutions, utilizing common overlay services reduces duplication of effort, enhances flexibility, lowers overhead costs, and promotes innovation. Common messaging standards and guidelines facilitate interoperability between networks. By leveraging these overlay services and establishing collaborative frameworks, the payment ecosystem can achieve seamless interoperability, driving adoption and improving the overall user experience.

Types of value-added and overlay services for Faster Payments

Alias Directories

Alias directories are a particularly useful VAS enabling widespread adoption of Faster Payments by easing payment initiation. Directories simplify the initiation process and can help ensure that payments are sent to the intended payee. The convenience aspect, when combined with security and speed, makes alias directories powerful tools to promote Faster Payment adoption.

Request-for-Payment (RfP)

A Request for Payment is a standardized network message that payees can leverage to send electronic requests to payers through a faster payment network. Unlike direct debits, where the payment itself is initiated by the payee, the paying party controls when and if they will respond to an RfP request with a payment. The underlying payment is still a credit transfer even if the RfP itself is generated by the payee. RfP messages may include reconciliation data for bill and invoice presentation, allowing for easier presentation and information collection. The ease with which billers and consumers can send RfP messages could become a driver of use case expansion by providing more utility to end users.

Confirmation-of-Payee

Confirmation-of-Payee (COP) is a service designed to cut down on misidentified or misdirected payments by providing the sending party with the name associated with the beneficiary account prior to a payment being initiated. Confirmation-of-Payee services can be provided by third parties or by the central payment infrastructure provider itself. COP helps inspire user trust by providing identity verification services, enhancing overall transparency, and cutting down on error rates, therefore making the system more reliable for end users. This added layer of identity verification not only promotes transparency in the payment process but also empowers users to actively confirm their recipient's payment details, reducing the likelihood of accidental fund transfers.

Tokenization

Tokenization is an available service to enhance the security of financial transactions. This protects users from fraud by eliminating the need to transmit sensitive information in each transaction. Tokenization services increase user trust and offer a secure environment for Faster Payment transactions.

Digital IDs

Digital IDs are another trust-enhancing value-added service that allows end users to verify their identity in a digital environment, preventing unauthorized parties from initiating payments. This service takes different forms in different markets, with governments issuing IDs usable in digital environments in the form of chips in some countries whereas others have embedded digital identity services into mobile banking apps. In some markets, bank-held information is used to verify consumer identities in digital environments such as online banking or payment initiation. Due to the complexity involved in issuing and verifying digital IDs, government

entities are most often the providers of this sort of service, with Faster Payment system infrastructure operators making use of this potential functionality. Digital IDs are enormously useful for enhancing security in Faster Payments and help combat unauthorized party access to accounts. The rise of generative AI, however, poses a risk for digital IDs that rely on facial or voice recognition.

Fraud Scoring

Fraud scoring is yet another trust-enhancing service and is increasingly used by Faster Payment systems across the globe. Fraud scoring provides insights at the central infrastructure- and FI-level by employing advanced data analytics to assess a transaction based on various risk indicators. Fraud scoring involves analyzing transaction frequency, amount, location, end user behavior patterns, and other data points to determine whether a transaction is likely fraudulent in nature. These measures allow financial institutions to identify risky transactions, enabling them to intervene and investigate transactions further. Faster Payment infrastructure operators are best suited to manage fraud scoring systems due to their ability to monitor a large volume of payments from a centralized view. Operators can then more easily identify patterns and anomalies, enabling the development of advanced fraud detection algorithms, and allowing for the swift communication of risk alerts to FIs via the network's telecommunication rails. Alerts can be sent to FIs, who can then use this data to supplement their own data before deciding how to proceed.

Liquidity Management

Liquidity management do exactly what they sound like they do – they help Faster Payment participants manage their liquidity. These services are offered by the central infrastructure provider and allow participants to make internal adjustments or source funds from other FIs to support their liquidity needs stemming from payment activities. These services provide insights into the liquidity positions of participant FIs and allow them to monitor their cash flows, forecast future liquidity needs, and make proactive transfers to prevent service interruptions stemming from liquidity constraints.

Exceptions Handling

Financial institutions and payment system participants must have policies, processes and systems to handle payment exceptions and communicate about same with their customers. Exception handling tools allow FIs to identify and manage scenarios such as errors, anomalies, and potentially fraudulent activities that occur during the payment process. Once exceptions have been identified and/or reported, Faster Payment systems can trigger predefined workflows to resolve them, including predetermined responses to mitigate risks, alerts to relevant stakeholders, and in some cases, fund retrieval and/or return.

Value-added and overlay services

Within the U.S., Faster Payment system VAS and overlay services are quite common, barring the exception of Digital ID services. The table below showcases the various services offered and available in connection with Faster Payment services. While not all VAS are offered by the operators, some are still available via third-party service providers (i.e., alias directory services via Zelle).

TABLE 58: FASTER PAYMENT OVERLAY SERVICES IN THE US

	SAME DAY ACH	FEDNOW	RTP NETWORK	VISA DIRECT	MASTERCARD SEND
Alias directories	Possible through Zelle	No	Possible through Zelle	Possible via Visa's Alias Directory Service	Possible via Mastercard Consumer Mapping Service
R2P	No	Yes	Yes	No	No
COP	Possible through Zelle	No	Possible through Zelle	Yes	Yes
Tokenization	No	No	Yes, optional via Secure Token Exchange service	Yes, via Token ID service	Yes
Digital IDs	No	No	No	No	No
Fraud scoring	No	No	Available via 3rd party	Yes, via Visa Advanced Authorization	Yes, via Mastercard Early Detection System
Liquidity management tools	No	Yes	No	No	No
Exceptions handling*	Yes	Yes	Yes	Yes	Yes

*Note: This tables describes the VAS that is supported by the infrastructure itself, not whether third parties or FIs perform the VAS (i.e., fraud scoring). In the case of exceptions handling, "yes" here refers to operator- provided guidelines for exceptions handling.

Chapter Five

Governance Framework

SECTION BLUEPRINT COMPOSITION: 20%

Rules and guidance applicable to Faster Payments rails

Day-to-day operations of Faster Payment rails are governed primarily by the operating rules of the individual Faster Payment Systems i.e., the Nacha Operating Rules, RTP rules, FedNow rules, Visa Direct rules, and Mastercard Send rules. Visa and Mastercard also ensure compliance with the PCI/DSS rules since processing cardholder data is subject to PCI/DSS rules. Regulatory guidance about Faster Payments plays an important role in safeguarding the secure and streamlined processing of transactions. Several regulatory entities, including FinCEN, FFIEC, FDIC, OCC, and others, issue and revise guidance. It is imperative financial institutions and participants in the Faster Payments ecosystem stay informed of the latest regulations and guidance, as these updates can significantly impact their operational procedures.

Rule sets that govern Faster Payments rails

Faster Payments operate in a highly regulated space with multiple guidance and compliance arising from legislation, regulation, and network specific rules. Following is a non-exhaustive list of some of the major rules that govern Faster Payment rails:

TABLE 59: REGULATIONS GOVERNING FASTER PAYMENT RAILS	
REGULATION / RULE	***FOCUS***
Network-specific Rules: Same Day ACH, FedNow, RTP, Visa Direct, Mastercard Send	Each Faster Payments network has its own set of operating rules that govern the entire payment chain.
Federal Reserve Banks Operating Circular No. 8	This circular complements the FedNow Service Operating Procedures, which provide operational details on key areas such as participant and service availability expectations, connection profiles, fraud mitigation and reporting, and ISO 20022 messaging used within the FedNow Service.
PCI-DSS rules	FFIEC (Federal Financial Institutions Examination Council) develops uniform reporting systems for federally supervised financial institutions, their holding companies, and subsidiaries. It publishes guidance for the industry in the form of handbooks on cybersecurity, risk management, and other IT systems safety.
FFIEC	FFIEC (Federal Financial Institutions Examination Council) develops uniform reporting systems for federally supervised financial institutions, their holding companies, and subsidiaries. It publishes
OCC	The Office of the Comptroller of the Currency's (OCC) Committee on Banking Supervision (CBS) supervises the payments function of a bank to determine the potential operational, compliance, credit, liquidity, strategic, and reputation risks and how these are incorporated into bank-wide risk assessments. OCC examiners also assess bank risk management practices, including governance and controls of change management, information technology, information security, compliance, and fraud forFedNow participants.
The Bank Secrecy Act (BSA)	Legislation aimed at preventing criminals from misusing financial institutions to hide or launder money. FIs are required to provide evidence of necessary and reasonable actions taken to identify and prevent money laundering and criminal incidents.
OFAC	Office of Foreign Assets Control (OFAC) is responsible for administering and enforcing economic sanctions based on US foreign policy and is responsible for developing, administering, and managing US sanctions programs.
Uniform Commercial Code 4A	A set of laws that provide standardized guidance and regulations governing commercial transactions and fund transfers, particularly electronic fund transfers, including ACH transactions. Article 4A specifically focuses on fund transfers, establishing rules related to security, liability, warranties, and dispute resolution in a transaction.
USA PATRIOT Act	Establishes standards for identifying consumers at account opening utilizing a Customer Identification Program (CIP). The CIP must be clearly documented and should include risk-based procedures for verifying the identity of each customer to a reasonable and practicable extent.
Regulation E	Regulation E implements the Electronic Fund Transfer Act (EFTA), which establishes a basic framework of the rights, liabilities, and responsibilities of participants in the electronic fund and remittance transfer systems.
Regulation J	Regulation J provides the legal framework for depository institutions to collect checks and other items, and to settle balances through the Federal Reserve System. This regulation is supplemented by operating circulars issued by the Reserve Banks.
EFAA and Regulation CC	The Expedited Funds Availability Act (EFAA) is a law that deals with the problem of banks taking too long to make deposited money available to customers. It has three main rules for banks: they must make money deposited in regular accounts available quickly, pay interest on certain accounts promptly, and tell customers about their policies as to when funds will be available. The EFAA is enforced by a set of rules called Regulation CC, which has four parts. Part A explains terms and who enforces the rules. Part B sets the schedules for when banks must make funds available and includes exceptions, disclosure rules, and interest payment rules. Part C has rules to speed up the processing of checks, and Part D covers rules for substitute checks.
31 CFR, Part 210	Federal government payments are governed by 31 CFR, Part 210. In 1999, Part 210 adopted the Nacha Operating Rules as the rules governing all federal government payments, with certain exceptions.

Comparison of RTP and FedNow rules

FedNow and the RTP Network are governed by distinct structures. FedNow is operated by the Federal Reserve, with its functionality shaped by public feedback and insights from the FedNow community, consisting of private sector stakeholders. The RTP Network is managed by The Clearing House, which is owned by a consortium of large banks, and governed by the RTP Business Committee, which includes representatives from both owner banks and non-member financial institutions, ensuring a diverse and inclusive governance model.

The FedNow Service Operating Procedures along with Federal Reserve Banks Operating Circulars No.5 and No. 8 provide operational details for funds transfers through the FedNow Service, while the RTP Network Operating Rules define the rights and responsibilities of participants and TCH concerning the RTP Network. RTP participants

are also obligated to comply with the RTP Technical Specifications, which include the messag- ing specifications and terminology.

In terms of transfer limits, as of 2024, FedNow defaults at a limit of $100,000, though participants can adjust this to a ceiling of $10,000,000 as of November 2025 based on their business needs and risk preferences. The RTP Network allows credit transfers up to $10 million. Both systems offer settlement in real-time on a bilateral gross basis, but FedNow settles through debit and credit entries to participants' reserve bank accounts whereas the RTP Network's settlement is backed by pre-funded balances in a joint account at the Federal Reserve Bank of New York.

Non-payment messaging functionality is slightly different between the two systems. FedNow provides additional messaging features, including requests-for-payment, request for returns (for payments sent in error), status, information, confirmation of posting, account balance, and activity reports, all adhering to the ISO 20022 standard. In comparison, the RTP Network supports functionalities such as requests for payment (RfP), acknowledgment of receipt, information requests, returns of funds, and remittance advice, with the ability to include links to external documents. Payment routing methods are very similar for both FedNow and the RTP Network, both systems route payments based on the account number and routing number of the receiving bank, with the RTP Network providing additional support for routing using alpha-numeric domain-controlled tokens.

Both systems emphasize the importance of preventing fraud and promoting security, though they take different approaches. FedNow rules allow participants to set lower limits and conditions for transaction rejection and have plans to introduce enhanced fraud prevention tools in subsequent releases while the RTP Network rules require participating financial institutions and payment service providers to implement strong authentication, fraud detection/prevention, fraud reporting (to the network), and consumer protection policies. All transactions are required to be digitally signed and encrypted. The Clearing House tracks reported fraud and participating financial institutions are required to investigate suspected fraud cases. In terms of alias or directory support, FedNow permits participants using external alias directories for P2P services, whereas the RTP Network allows independent third-party networks to facilitate alias or directory-based initiation of payments routed over the network. Neither network has any type of embedded nor overlay directory services.

KEY RTP RULES AND REQUIREMENTS

TOPIC	REQUIREMENT
Continuous operations	99.5% continuous availability, 24/7, measured monthly is required (with a Sunday maintenance window allowance, 2-6am ET)
Capacity	Participants and their Third-Party Service Providers must be able successfully to read and respond to a minimum of 50 transactions per second
Receiving participant payment response time.	The Receiving Participant must respond (Accept, Reject, or Accept without Posting) to an incoming credit transfer message within 5 seconds of receiving the incoming credit transfer message from the RTP Network.
RTP time out reject	The RTP Network will automatically timeout transactions after 15 seconds, measured from the time the RTP Network receives the credit transfer message from the sending participant to the time the RTP Network receives a response from the Receiving FI.
Transaction limit	An RTP payment cannot exceed $10,000,000. Sending participants may establish lower transaction limits for their senders. Receiving Participants must accept transactions up to $10, 000,000.00
Multi-factor authentication	Sending participants must, at a minimum, utilize multi-factor authentication to authenticate the identity of customers who transmit payment instructions to the sending participant
Fraud/Risk Monitoring	Sending participants must perform appropriate fraud monitoring prior to submitting a payment message to the RTP Network and perform periodic reviews and updates or their fraud monitoring procedures.
Respond to Requests for Return of Funds	Receiving participants must respond to a Request for Return of Funds message within 10 banking days of receiving it from the RTP Network (longer is allowed if the reason for the request is suspected fraud, to provide time for investigation). The Receiving participant is under no obligation to return the funds.

KEY FEDNOW RULES AND REQUIREMENTS	
TOPIC	***REQUIREMENTS***
Continuous operations and capacity	FedNow Participants and Service Providers are expected to be available 24-hours each calendar day, including weekends and Federal Reserve holidays. • Planned downtime can only occur on Sundays between 2-6 AM ET • No more than 2 consecutive hours of downtime • No more than 24 hours of downtime in a quarter
FedNow time-out	20 seconds is the maximum allowable time for a transaction to settle or be rejected by the FedNow system.
Customer credit transfer transaction limit.	The FedNow transaction send limit for a customer credit transfer is defaulted to $100,000. Sending participants can configure their limit up to a maximum of $10,000,000 as of November 2025. Sending participants may establish lower transaction limits for their senders. Receiving Participants may not establish lower transaction limits for their receivers. If needed, a return request (camt.056) *should* be sent within 60 calendar days from when the Customer Credit Transfer (pacs.008) settled. Camt.056 messages used to report fraud or an RFP warranty breach are not subject to the 60-day guideline.
Requests for Return of Funds	Response to Requests for Return of Funds should be transmitted as soon as possible, but no later than midnight (ET) of the next standard business day (M-F, except holidays). If initial response was "pending" (PDCR) then final response must be as soon as the FI has concluded the investigation for further information, or within 10 standard business days (M-F, except holidays)

Visa Direct

Visa Direct's Original Credit Transactions (OCT) transactions are used to send funds to an eligible Visa card. Visa governs the Visa Direct service.

The Visa Direct Original Credit Transaction (OCT) Global Implementation Guide (v2.16, effective Jan. 8, 2025) provides the following:

- An overview of OCT
- OCT implementation requirements and considerations for acquirers, service providers, merchants, and issuers
- OCT implementation requirements and considerations for recipient issuers
- Data element requirements for each type of OCT
- A reference table of key Visa Rules and Requirements that apply to the Original Credit Transaction (OCT) types.

Examples of Visa Direct services that use the Visa OCT (push to card) include money transfers to consumers, business to consumer funds disbursements (such as insurance claims), merchant settlement, cash out of a digital wallet, real-time check deposits, loyalty rewards, loading prepaid cards.

Appendix L of the Visa Direct Original Credit Transaction (OCT) Global Implementation Guide provides a reference table of key Visa Rules and Requirements that apply to the Original Credit Transaction (OCT) types. For access to the full rule set, refer to the Visa Rules, *Visa Core Rules and Visa Product and Service Rules* on Visa Access (www.visaonline.com).

Mastercard Send

Mastercard Send's MoneySend Payment Transactions Program enables both domestic and cross-border payment transactions to eligible Mastercard, Maestro, and Cirrus branded card accounts. Mastercard governs Mastercard Send and the MoneySend Payments Transactions Program.

After obtaining confirmation of the availability of the sender's funds, the originating Institution initiates the "Payment Transaction" (push credit to card) to transfer the funds to the receiving institution. If the receiving institution authorizes the Payment Transaction request, the receiving institution then posts the funds to the receiving within 30 minutes of the authorization approval for debit and prepaid cards, with limited exceptions.

The Mastercard MoneySend and Funding Transactions Program Standards, referred to as the "Main Guide" sets the standards related to the Mastercard MoneySend Payment Transactions Program. The following documents provide additional rules and guidelines:

- The Mastercard MoneySend and Funding Transactions Program Standards Addendum for Geographic Availability of Service lists the countries and territories where Mastercard MoneySend Payment Transactions are available.
- The Mastercard MoneySend and Funding Transactions Program Standards -Addendum for Coding Standards Sets out Standards for the coding of network messages
- The Mastercard Gaming and Gambling Payments Program Standards
- Mastercard Gaming and Gambling Payments Program describes the standards for domestic and cross-border gaming Payment Transactions to eligible Mastercard and Maestro® branded card accounts.

Faster Payments stakeholders responsible for PCI/DSS rule compliance

The PCI Security Standards are a set of technical and operational criteria established by the PCI Security Standards Council (PCI SSC) to safeguard cardholder data. The PCI SSC includes PCI Data Security Standards (PCI DSS) applicable to all entities that store, process, and /or transmit cardholder data. These standards cover technical, and operations system components included in or connected to cardholder data. Oversight and management of these security standards falls under the purview of the Council. Enforcement of PCI standards is carried out by the founding members of the Council, namely, American Express, Discover Financial Services, JCB, Mastercard, and Visa Inc.

Faster Payment systems such as Visa Direct or Mastercard Send that enable push-to-card payments to process transactions need to ensure compliance with the PCI DSS requirements. The following list includes key stakeholders:

1. **Payment processors:** Any entity involved in the processing, transmission, or storage of payment card data for Faster Payments needs to comply with PCI DSS requirements.
2. **Financial institutions**: Banks and other financial institutions that are part of the Faster Payments ecosystem and handle payment card data in various forms need to ensure that their systems and processes comply with PCI DSS if they engage in card payment activities.
3. **Payment service providers:** Payment service providers, including those offering services for Faster Payments, are responsible for PCI DSS compliance if they are involved in processing or transmitting payment card data.
4. **Merchants:** Merchants that accept payment cards, even in the context of Faster Payments, are subject to PCI DSS compliance if they store, process, or transmit cardholder data.
5. **Technology and solution providers:** Companies providing technology solutions, software, or infrastructure supporting Faster Payments may also be stakeholders responsible for PCI DSS compliance, especially if their systems handle payment card data.
6. **Networks and switches:** Payment networks and switches that facilitate the transfer of funds in Faster Payments need to ensure that their systems comply with PCI DSS if they handle payment card information during the transaction process.

Specific PCI DSS compliance requirements for each stakeholder can vary based on the role and functions of each stakeholder in payment processing. Stakeholders need to regularly assess their specific responsibilities and compliance obligations based on their role in the Faster Payments system.

Relevance of FFIEC guidance to Faster Payments

The Federal Financial Institutions Examination Council (FFIEC), an interagency body, holds significant influence in shaping the regulatory landscape for Faster Payments systems. Empowered by federal financial regulatory agencies, including the Board of Governors of the Federal Reserve System (FRB), the Federal Deposit Insurance Corporation (FDIC), the National Credit Union Administration (NCUA), the Office of the Comptroller of the Currency (OCC), and the Consumer Financial Protection Bureau (CFPB), the FFIEC prescribes uniform principles, standards, and reporting forms for the examination of financial institutions, including Faster Payments systems. Responsibility for developing uniform reporting systems extends to federally supervised financial institutions, their holding companies, and relevant subsidiaries. The FFIEC, through its supervisory role, conducts examiner training programs for federal and state agency personnel overseeing financial institutions.

In the context of Faster Payments, the FRB acts as the lead examiner of The Clearing House for examinations conducted under the Bank Service Company Act through an arrangement among the federal financial regulatory agencies through the Federal Financial Institutions Examination Council. Its authority over The Clearing House and its operation of the RTP Network is extremely broad. The Federal Reserve entity that directly regulates the RTP Network is the Federal Reserve Bank of New York, under a regulatory framework established by the FRB. The FFIEC's supervisory oversight extends to other Faster Payment networks such as Visa Direct, Mastercard Send, and the Same Day ACH network under federal banking laws. ,

In the regulation of Faster Payments, the FFIEC issues essential guidance in the form of handbooks, booklets, and statements. These materials cover operational, risk management, and system security aspects, providing a framework for standardized supervision. They serve as valuable tools for examiners, ensuring consistent evaluation of financial institutions and service providers involved in critical financial services within the Faster Payments systems.

Laws and regulations applicable to Faster Payments rails

Interaction between private sector rules and other banking regulations

Private sector rules and banking regulations in the payments industry mutually reinforce each other. On the one hand, banking regulations frequently establish legal frameworks that guide the private sector; on the other hand, private sector payment rules have been developed to enable and support the changes and development of new payment products. Thus, private sector rules and banking regulations complement each other, providing valuable support across various domains such as compliance, risk management, and consumer protection while fostering competition and innovation.

Compliance requirements

Financial regulators such as the Federal Reserve and the Consumer Financial Protection Bureau (CFPB) establish compliance standards for financial institutions such as Anti-Money Laundering (AML) processes, Know Your Customer (KYC) procedures, and other data privacy laws. This was strengthened by the Patriot Act which was enacted to ensure national security by preventing terrorist financing via money laundering and any other financial crimes that may be related to terrorism. As such, banks are subject to regular compliance examinations and are held to rigorous standards to ensure financial stability. These regulatory requirements

have been adopted for FIs and payment processors to ensure adherence to industry standards and best practices in the form of merchant compliance requirements and anti-fraud measures.

Risk management

Concerning risk management, FIs seek to identify, assess, and mitigate risks that may arise from various sources within the payment system. This can include fraud prevention via cybersecurity measures, liquidity management, and contingency planning to prevent fraud. Private sector actors may implement various authentication and authorization measures that comply with banking regulations present in the industry while adapting diverse measures for monitoring transaction activities. Some individual financial institutions may implement transactional limits though this can also be supported by banking regulations.

Consumer protection

The Federal Reserve and Congress strive to protect consumers, financial institutions, and the payment system itself by strengthening laws and policies. The rules set by the Federal Reserve Board for consumer payments and related federal laws typically don't cover the payment systems between banks. These rules do not usually spell out who is responsible if something goes wrong with interbank payments, even if a bank is legally required to protect its customers. For example, Regulation E provides strong protections for consumers if there are mistakes or unauthorized payments made from their accounts.

Competition and innovation

Rules set by private entities and governmental regulations can have either a positive or negative impact on innovation and competition within the Faster Payments sector. Private sector rules often support innovation by encouraging the use of new technologies, business strategies, and services. However, banking regulations must carefully manage innovation to maintain financial stability, market fairness, and consumer protection. Regulatory bodies may choose to have adaptable approaches to regulation, allowing for emerging payment innovations while ensuring adherence to essential regulatory principles and goals.

Payment rails governed by laws and regulations applicable to Faster Payments

The various Faster Payment rails are governed by several prevailing laws and regulations, including the Uniform Commercial Code (UCC) Article 4A, which governs funds transfers and establishes the rights and responsibilities of parties involved in such transactions while also covering payment order obligations of sending and receiving liability for late or improper execution, etc. Each state has adopted it. According to UCC Article 4A, if a funds transfer is not authorized by the customer the sending customer is not responsible to the sending bank for that transfer. Moreover, UCC 4A primarily applies to wholesale or large-value funds transfers between financial institutions, often involving business-to-business or interbank transactions. For this reason, UCC Article 4A does not apply to transactions governed by the EFTA or Regulation E. The EFTA and Regulation E address consumer electronic fund transfers and are the primary federal laws in the consumer payments area.

The Federal Reserve included the rules from UCC Article 4A into Subparts B and C of Regulation J, which pertain to FedNow transactions. This means that payments made by consumers through FedNow are subject to both UCC Article 4A and Regulation E.

The Office of Foreign Assets Control (OFAC) requirements generally support compliance efforts with sanctions regulations and mitigating the risks associated with instant payment systems. In addition, the Consumer Financial Protection Bureau (CFPB) broadly enforces Unfair Deceptive, or Abusive Acts or Practices (UDAAP) rules to prevent actions that may cause financial harm to consumers. The CFPB has outlined guiding principles to make sure that consumer protections are integrated into new payment systems from the outset, emphasizing the

importance of developing secure transparent accessible and affordable payment systems to provide protections against fraud and errors.

Many other requirements were reinforced for financial institutions by the Patriot Act. The customer identification program (CIP), for instance, mandates that financial institutions employ a program to verify the identity of customers who open accounts. Suspicious Activity Reporting (SAR) requirements mandate financial institutions to report suspicious activity that may be related to money laundering or terrorist financing to the Financial Crimes Enforcement Network (FinCEN).

In addition to current banking regulations, Faster Payment solutions are regulated by specific rules associated with each system. For instance, Same Day ACH operates under the guidelines outlined in the Nacha Operating Rules, while still complying with broader regulations like Regulation E and UCC 4A. Similarly, The Clearing House sets forth Operating Rules and Participation Rules for RTP transactions, although various Service Level Agreements might be established among transaction partners. Mastercard Send and Visa Direct also comply with Regulation E as well as PCI DSS for data security on top of their own network rules and policies.

TABLE 60: LAWS AND REGULATIONS GOVERNING PAYMENT RAILS

PAYMENT RAILS	*LAWS AND REGULATIONS*
FedNow	Regulation E, Regulation J Subparts B and C, UCC4A, FedNow Operating Rules,
RTP Netowrk	Regulation E, UCC4A, Service Level Agreements, RTP® Operating Rules, RTP® Participation Rules
Same Day ACH	Regulation E, UCC4A, EFTA, Nacha Operating Rules
Mastercard Operating Rules	Regulation E, PCI DSS, Mastercard Operating Rules
Visa Direct	Regulation E, PCI DSS, Visa Operating Rules

Governance of laws and regulations applicable to Faster Payments

The laws and rules governing Faster Payments are managed by various government agencies to ensure they're efficient, secure, and fair. Regulatory bodies at the federal level, including the Federal Reserve, the CFPB, and other agencies oversee and enforce regulations that govern Faster Payment solutions. These regulations encompass a wide range of areas, including consumer protection, risk management, anti-money laundering (AML) compliance, and interoperability standards. Congress enacted the EFTA and Regulation E with the specific aim of regulating consumer electronic fund transfers, ensuring a high level of consumer protection in electronic payment transactions. The evolving Faster Payments landscape requires constant vigilance and adaptation from regulators to address emerging risks and promote innovation while safeguarding the financial system's integrity.

Comparison between consumer and non-consumer regulations

Regulations that govern electronic transfer funds can be largely divided into consumer and non-consumer regulations. Consumer regulations are laws that focus on and provide consumers with protection from fraud or unauthorized electronic payments. Consumers are defined as those who hold bank accounts for "personal, family, or household" purposes and are differentiated from business customers. The Electronic Fund Transfer Act and Regulation E issued by the Consumer Financial Protection Bureau are some of the main federal laws that protect consumers. Under these regulations, consumers have limited liability for fraudulent electronic payments, which are subject to prompt notice of requirements. Banks are held more responsible for verifying

whether payments from consumer accounts are authorized. As such, consumer regulations mandate consumer disclosures, limit consumer liability for unauthorized transfers, and maintain procedures for resolving errors.

In contrast, non-consumer regulations govern electronic transfers between businesses. These may include various operating rules established by the payment networks, interbank network rules as well as broader financial regulations such as the Bank Secrecy Act and Anti-Money Laundering (AML) regulations. Regulatory oversight, industry self-regulation mechanisms, and contractual agreements are used to enforce non-consumer regulations. Violating non-consumer regulations can lead to contractual disputes, reputational damage, or financial penalties, compared to refunds and damages relief under consumer regulations.

Comparing and contrasting private rule sets and federal regulations

Both private rule sets and federal regulations play important roles in the Faster Payments industry, albeit with distinct approaches and priorities. Federal regulations are enacted through a structured process, involving legislators and regulators who solicit input through hearings and public comment. Although this approach guarantees a democratic process in rulemaking, it can be challenging to balance different responsibilities and priorities, which might hinder timely regulatory adaptation to market demands and technological advancements.

For private rule sets, immediate access to industry experts is an advantage. This helps in developing relevant interbank rules and procedures that are customized to the evolving needs of the payment landscape. This accessibility fosters innovation and fair competition in the marketplace as payment service providers differentiate themselves through their products and rules.

Federal statutes do not always cover payment systems and liabilities between financial institutions. Private- sector payment systems have developed a detailed system for allocating losses associated with fraud and unauthorized transactions among payment participants. This allocation depends on various factors, allowing payment systems to encourage fraud reduction or other policies deemed important by their participants.

Federal regulations are created through a thoughtful process of mandating standards as a foundation of the regulatory environment. However, they may face obstacles in keeping up with the rapid advancements of payment systems and industry practices. Private rule sets, on the other hand, demonstrate speedy adaptation and innovation due to their close engagement with industry experts and the competitive dynamics of the marketplace. The synergy between these two realms can create a robust and balanced regulatory framework, wherein the strengths of each complement the limitations of the other, ultimately enhancing the efficiency, security, and fairness of the payment ecosystem.

Objectives of the data privacy laws

Data privacy laws are significant for US Faster Payments systems to protect consumers from fraud, scams, and identity theft. The immediacy of funds availability in fast payment systems makes them attractive targets for fraudsters who could exploit the system before fraudulent activities are detected. Data privacy laws play a crucial role in safeguarding sensitive information and preventing breaches of privacy and data security that could lead to identity theft, financial harm, and fraud. These laws establish rules on data security and privacy, requiring institutions to obtain consumer consent before sharing data, anonymize data, and set minimum standards on data storage and transmission across borders. In the context of faster payments, where real-time transfers are not eligible for repudiation, adjusted rules and processes for remedial actions are necessary, highlighting the importance of robust security frameworks to protect consumers' privacy and identity in the face of emerging fraud risks.

Obligations of Faster Payments Stakeholders under data privacy laws

FedNow

In the interest of data privacy, stakeholders participating in FedNow, the Federal Reserve's instant payment service, are subject to stringent requirements to ensure the security and integrity of transactions. Compliance with Federal Reserve Operating Circulars 8 is mandated, encompassing regulations regarding access to Federal Reserve Financial Services and data transmission protocols. Additionally, stakeholders accessing the FedNow Service via FedLine Solutions must maintain robust information security programs, employing measures such as message signing using public/private key pairs, encryption of data in transit and at rest, and adherence to authentication protocols outlined by the Federal Reserve.

TCH RTP

The Clearing House requires a robust information security program for its RTP Network, restricting physical and logical access to customer data and ensuring its secure destruction. This program includes detailed specifications for system architecture, firewalls, and intrusion detection systems, enhancing overall security measures. TCH requires regular participant inspection of its security program and conducts regular audits by independent auditors, aligning with regulatory standards and ensuring continual improvement. TCH reviews the effectiveness of its security measures annually and commits to making necessary adjustments to address any identified vulnerabilities. TCH requires participants to have contingency plans for data recovery in the event of a breach and requires thorough background checks on all employees of participants that may have access to customer information.

Same Day ACH

Participants in Same Day ACH payments must adhere to rigorous data security standards outlined in the Nacha Operating Rules, emphasizing the protection of sensitive data throughout its lifecycle. Compliance entails implementing access controls and encryption measures to safeguard account information used in ACH transactions, alongside conducting self-assessments and verifying the identities of Third-Party Senders and Originators. Additionally, participants must consider state and federal laws governing the secure storage of data and sensitive documents, ensuring their storage methods align with regulatory requirements to uphold the commercial reasonableness of their practices.

Mastercard Send

Mastercard Send mandates participants maintain a robust written information security program encompassing technical, physical, and administrative safeguards. These measures are designed to safeguard the security and confidentiality of personal data, protect against potential threats to data integrity, and prevent unauthorized access or acquisition of personal data. Additionally, the program ensures proper disposal of personal data and requires regular testing or monitoring to assess the effectiveness of implemented safeguards, reflecting a commitment to ongoing security enhancement.

Visa Direct

Visa Direct imposes stringent security and authentication requirements on users of its APIs, including Two-Way SSL (Mutual SSL) authentication and channel encryption. This entails providing a username, password, and installing an X509 security certificate issued by Visa. Additionally, message-level encryption is mandated to further enhance data security.

Application of the Bank Secrecy Act and Anti-Money Laundering regulations to Faster Payments

The Bank Secrecy Act of 1970 was implemented to prevent criminals from using financial institutions to hide or launder money including for financial terrorism to ensure financial integrity. All Faster Payments FI participants are subject to this law. The BSA specifically imposes an AML program, record keeping, and reporting requirements on all financial institutions designed to prevent and detect money laundering and other crimes. It further specifies what an AML compliance program must include such as internal controls independent testing of responsible persons and training. These regulations mandate customer due diligence, transaction monitoring, and reporting of suspicious activities to regulatory authorities.

These regulations become even more crucial to ensure the integrity and security of financial transactions with the advent of Faster Payment systems. The speed inherent to Faster Payments presents challenges for regulatory authorities in detecting and preventing illicit activities, as traditional monitoring systems may struggle to keep pace with the volume and velocity of transactions.

To address these challenges, financial institutions implementing Faster Payment systems must adhere to stringent BSA and AML regulations. This involves implementing robust monitoring and reporting mechanisms to identify suspicious transactions, conducting thorough customer due diligence to verify the identities of individuals and entities involved in transactions, and implementing strong internal controls to mitigate the risk of illicit financial activities. These monitoring programs must include measures to detect payments that are suspicious in their value (i.e., frequent payments that are just under reporting thresholds or transaction limits), and their frequency. Monitoring programs must also account for other factors, such as the age of the account making these transactions. For example, accounts that have been opened recently, making payments just under reporting or transaction limit thresholds would be suspicious. This requires proper staff training and the establishment of proper investigation protocols to ensure that suspicious payments are identified, without the generation of too many false positives. Additionally, collaboration between financial institutions, regulatory agencies, and law enforcement entities is essential to effectively combat money laundering and ensure compliance with regulatory requirements in the rapidly evolving landscape of Faster Payments.

The error resolution protections, rights, and liabilities of participants

Error on a faster payment

An error refers to an unintended deviation or mistake that occurs in the processing of a faster payment. There are different types of errors depending on which stage of the payment the error happens including authentication, processing, authorization, settlement, and reconciliation errors. For instance, TCH defines payment errors related to RTP payments as when the payment amount is erroneous, a payment is sent to an unintended Receiving Participant or Receiver, or an RTP payment is sent twice (duplicate payment).

Error resolution includes the process of investigating whether or not an erroneous, unauthorized, or fraudulent faster payment transaction has been made and the corrective measures taken by users and financial institutions. Error resolution will differ based on the type of error as well as the rail involved.

Applicability of error resolution protections under Regulation E and UCC4A

Regulation E governs electronic fund transfers initiated to debit or credit a consumer account while the Unified Commercial Code (UCC) governs commercial transactions in the U.S.

In terms of error resolution, Regulation E provides specific rights and protections for consumers when errors occur in electronic payments, including specifications as to the timeframe for reporting errors and

investigation procedures for financial institutions. Regulation E also defines instances when consumers are entitled to provisional credit for disputed amounts during investigations and ensures the timely resolution of errors, thus safeguarding consumers' interests and protects them from errors, unauthorized transactions, and fraudulent activities.

Article 4A of the UCC provides a framework for the rights and obligations of banks and businesses involved in funds transfers. It focuses on defining the responsibilities of entities involved in commercial transactions such as giving notice to counterparties of erroneous or unauthorized transfers and investigating and resolving errors for recipient entities when receiving notice. Article 4A also specifies liability allocation, enforcement mechanisms, and remedies.

Payer liability for a misdirected payment

Once a payment is determined to have been misdirected, payers have certain liabilities under regulations and/or network rules. The table below outlines these liabilities for each faster payment rail in the U.S.

TABLE 61: LIABILITIES OF PAYERS RELATED TO MISDIRECTED PAYMENTS BY PAYMENT RAIL

PAYMENT RAILS	PAYER LIABILITY RELATED TO MISDIRECTED PAYMENTS
FedNow	Under Regulation E payers must give an oral and, upon request from its FI, a written notice of the error made within 60 days of receiving a periodic statement or other documentation first reflecting the alleged error from its FI. The notice must include the customer's name and account number, an explanation for why the error exists as well as the error's type, date, and amount.
TCH RTP	
Same Day ACH	
Mastercard Send	Under UCC4A, payers must disclose an error related to misdirected payments and notify their bank within 90 days of receiving payment confirmation from their bank. If the payer fails to do so, he or she is liable for the amount sent and not more.
Visa Direct	

Liabilities and error resolution responsibilities of financial institutions

TABLE 62: LIABILITIES AND ERROR RESOLUTION RESPONSIBILITIES OF FINANCIAL INSTITUTIONS BY PAYMENT RAIL

PAYMENT RAILS	LIABILITIES AND ERROR RESOLUTION RESPONSIBILITIES OF FINANCIAL INSTITUTIONS BY PAYMENT RAIL
FedNow	Under Regulation E financial institutions must require written confirmation of an error from the consumer no later than 10 days after receiving an oral notice and provide an address where said confirmation has to be sent. After this, the FI must perform the following steps: • Promptly investigate the allegation of the error • Complete the investigation within 10 business days • Report the findings within 3 business days of completing the investigation • Correct the error within one business day after determining that an error was made The FI is allowed to extend the length of the investigation to 45 calendar days if it provides provisional funds (including interest, where applicable) to the consumer within the original 10 business days if it advises the consumer within 2 business days of the provisional credit and gives full use of the funds to the consumer during the extended investigation. The FI is not required to provide provisional funds if the consumer fails to give a written confirmation of the original error notice. After completing the investigation, if no error has been found, the FI is required to give a written explanation to the consumer, including the documents based on which it made the determination. Under UCC4A, receiving financial institutions are required to correct errors they made when executing payment orders, whether the error was related to the amount or the identity of the beneficiary. This includes collecting an additional amount from the sender if the amount debited to the beneficiary is lower than the original payment order, recovering and returning the excess funds if the amount collected is larger than the original payment order, and recovering and returning the full amount of a payment order that was debited to a beneficiary different to the one specified in the payment order. Note: these obligations apply only when UCC 4A governs the transaction, which typically excludes consumer accounts unless otherwise agreed.
TCH RTP	
Same Day ACH	
Mastercard Send	
Visa Direct	

Role of client/customer agreements and disclosures with respect to Faster Payments products and services

Client/customer agreements serve as the legal cornerstone of the relationships between Faster Payment stakeholders. These agreements outline the terms and conditions governing the use of faster payment products and services, establishing the rights, responsibilities, and liabilities of the various parties involved. These agreements allow providers to communicate policies concerning transaction limits, fees, dispute resolution mechanisms, and data privacy policies and serve as mechanisms for regulatory compliance and enforcement because regulators mandate the inclusion of certain provisions and disclosures to safeguard the interests of consumers.

Disclosures serve as measures to promote transparency and mitigate risk associated with these Faster Payment systems. Financial institutions and operators are obligated to furnish comprehensive disclosures that elucidate the features, functionalities, and associated potential risks of Faster Payment products and services. These disclosures must be provided to customers and clients in a clear, concise, and accessible manner, ensuring that customers comprehend the implications of using these services.

Required agreements for each payment rail

TABLE 63: REQUIRED AGREEMENTS FOR FASTER PAYMENT RAILS

	SAME DAY ACH	RTP NETWORK	FEDNOW	MASTERCARD	VISA
Required agreements	• Origination agreements between relevant parties • Authorization agreements between relevant parties	• RTP participant Agreement and Indemnity • Designation of Third Party Service Provider Agreement	• FedNow Service Security Procedure Agreement • Service Provider Agreement for the FedNow Service	• Digital Activity Agreement • Program service agreement • Payment transfer agreement	Visa Direct terms and conditions

Required disclosures for each payment rail

TABLE 64: REQUIRED DISCLOSURES FOR FASTER PAYMENT RAILS

	SAME DAY ACH	RTP NETWORK	FEDNOW	MASTERCARD	VISA
Required disclosures	• Designated data disclosure • Disclosure of nested relationships	• Daily reconciliation reports • Exceptions handling • Pricing for participants	• Transaction records • Fraud data disclosures	• Fee schedules • Disclosure of standards • Privacy policies	• Fee schedules • Privacy agreements

Roles of agreements among Faster Payments stakeholders

Same Day ACH

For Same Day ACH payments several agreements must be in place to execute transactions and ensure legal compliance. First and foremost, an origination agreement must be established between the ODFI and originator, including any third parties involved. These agreements bind the originator to the Nacha Operating Rules, authorize the ODFI to receive transactions on their behalf, specify limitations to transactions, and include rights to agreement termination and the right to audit originators for compliance with legal standards. Additionally, authorization agreements are necessary between originators and receivers to permit the receipt of ACH entries, with the specifications of these agreements depending on whether the parties involved are consumers or corporate entities and the specific dynamics of their relationship. ODFIs and RDFIs must also have authorization agreements in place with their ACH Operator, whether this be EPN or

Fed ACH, setting into place the name and routing number of these institutions and the settlement arrangements for transactions.

RTP Network

The Clearing House requires two primary agreements be in place for the RTP Network. The first is the RTP Participant Agreement and Indemnity, which ensures that participant institutions agree to the terms and conditions governing participation in the network. This agreement requires participants to adhere to RTP rules and regulations, including security protocols and transaction processing guidelines. Additionally, participants agree to indemnify The Clearing House against any liabilities arising from their access to the RTP Network. The second major agreement for the RTP Network applies to participants that utilize a third- party service provider to access the network. This agreement designates the third-party service provider to act on behalf of the participant, outlining the scope of services provided and ensuring compliance with RTP Network standards.

FedNow

For FedNow, the first agreement that participant financial institutions must agree to is the FedNow Service Security Procedure Agreement. FIs agree to the provisions outlined in the Federal Reserve's Operating Circular 8, which governs access to the FedNow service. As the name suggests, this agreement relates to the security procedures involved with FedNow and the Reserve banks acting upon authenticated instructions coming from service participants.

The second major agreement for FedNow is the Service Provider Agreement and is necessary for participants that wish to engage another entity to act on their behalf as a service provider. This agreement authorizes the designated entity to perform specific services on the FI's behalf within the FedNow ecosystem. These services may include sending and receiving credit transfers, managing settlement processes, and handling liquidity management transfers. Signing this agreement means that FIs ensure that their chosen service providers adhere to FedNow's standards and regulations. It should be noted that these FedNow-specific agreements come on top of the agreements necessary for the FedLine interface solution that allows access to electronic Federal Reserve services.

Mastercard Send

Mastercard Send requires several different agreements to legally ground and bind the relationship between participants and the Mastercard scheme. The first of these agreements is the Digital Activity Agreement, which commits users to a series of rules concerning the customer's digital participation, sponsorship rights, obligations, responsibilities, and usage abilities, among other concerns. Mastercard also requires compliance with its Program Service Agreement, which reflects the customer's responsibilities for establishing all management and operating policies described in Mastercard standards. The third major agreement for Mastercard customers is the Payment Transfer Agreement. This agreement applies to all customers engaging in payment transfer activity and establishes rule compliance, responsibility for transactions, transaction and authorization requirements, obligations to provide information to Mastercard, privacy and data security requirements, and various other subject areas.

Visa Direct

FIs must comply with a series of agreements to use the Visa Direct services. Among the various agreements required by stakeholders for the usage of Visa services generally, the list of which is extensive, Visa Direct users must also agree to the terms and conditions. These terms outline the rights and responsibilities of developers, financial institutions, and merchants/consumers that access Visa Direct services. The agreement clearly defines

the parameters of usage for the Visa Direct web developer portal and the usage of the service for push and pull transfers. This agreement grants users a limited license to use the Visa Direct API, while imposing restrictions on distribution, sublicensing, and prohibited activities. Visa also reserves the right to regulate API calls and may modify its methods periodically according to this agreement. In summation, this agreement governs the utilization of Visa Direct services, API usage via the developer website, approval processes, usage limitations, and Visa's discretion over API access and support, to further codify the relationship between the scheme and the users.

Roles of disclosures among Faster Payments stakeholders

Same Day ACH

Required disclosures for Same Day ACH payments are sourced from the Nacha Operating Rules. Same Day ACH payments necessitate several disclosures among stakeholders. Firstly, transparency regarding any nested third-party relationships between third parties and their originating depository financial institutions must be reported to the ODFI. Secondly, all participants must disclose their transaction data retention policies to ensure accountability and facilitate effective communication among parties. ACH Operators must disclose designated data related to entries transmitted throughout the system to Nacha. Comprehensive disclosure of pricing structures, the process of exception handling, and other pertinent information must be provided to participants to foster transparency and facilitate smooth operations within the system.

RTP Network

The Clearing House's RTP Network mandates several disclosures for system operation. The Clearing House must disclose typical information concerning pricing, exceptions handling, policies, practices, and standards to maintain clarity within the network. Participants must disclose pricing, standards and rules for RTP payments, and exceptions processing handling to their customers using the service. Funding participants are required to disclose their relationships with non-funding participants to TCH, enhancing transparency and oversight.

Participants must provide their respective customers with any necessary information contained in the designated fields of Payment Messages, Payment Message Responses, or Non-Payment Messages as outlined in the RTP Technical Specifications. Furthermore, TCH provides reconciliation reports to participants at the end of each operating day, facilitating accountability and transparency. Funding participants are also obliged to furnish non-funding participants with reconciliation reports detailing their activities at day's end.

FedNow

The need for disclosures among FedNow stakeholders arises from stipulations within Operating Circular 1, Account Relationships. FedNow Participants utilizing the FedNow Service agree to allow the Reserve Banks to utilize and/or disclose all transaction records and related information within the confines of applicable law. This anticipated collection and utilization encompass various aspects, including directory or fraud services essential for executing, managing, or enforcing transactions as well as guarding against actual or potential fraud, unauthorized transactions, or liability claims. FedNow Participants affirm that they have provided all requisite disclosures and acquired necessary consents or permissions for such usage by the Reserve Banks. Furthermore, FedNow Participants are obligated to indemnify and safeguard the Reserve Banks against any claims, losses, costs, or expenses resulting from failure to fulfill their participant obligations outlined in Operating Circular 8, which governs the FedNow service.

Mastercard Send

A set of required disclosures is necessary to ensure transparency and compliance for Mastercard Send. Institutions must be provided with the necessary standards relevant to service providers in the way of required privacy policies, fees and pricing, and any other standards they will be held to before they become customers of Mastercard. Participating financial institutions are also required to regularly disclose customer reports of transaction activities to Mastercard. Participating FIs are also required to disclose all fees and appropriate information to their customers using the Mastercard Send service as well by all applicable laws.

Visa Direct

Originators are mandated to furnish customers with comprehensive Terms and Conditions, ensuring customer acknowledgment before utilizing any Visa Direct-based service. It is the originator's responsibility to regularly update these Terms and Conditions to encompass necessary disclosures in alignment with legal requirements and industry standards. Visa reserves the right to request copies of the updated Terms and Conditions for compliance verification. Visa's Core Rules are also disclosed to all participants, with compliance being required. Originators are obligated to furnish senders with transparent information concerning fees and essential terms associated with push payment services. This entails providing a detailed breakdown of all assessed fees in compliance with relevant laws and regulations. Senders must be allowed to consent to these fees or opt to cancel the transaction. In cases where third-party agents are involved, strict adherence to Visa's rules and regulations is required and the same sorts of disclosures must be provided.

Warranties and indemnities by rail and rule set

Same Day ACH

As is the case with all of these Faster Payment rails, the central infrastructure is usually only liable in cases of gross negligence or misconduct. ODFIs are obligated to indemnify RDFIs and ACH Operators for any claims, losses, liabilities, or expenses, including legal fees, stemming from breaches of warranties made by the ODFI under the Nacha Operating Rules or from the debiting or crediting of an entry to a receiver's account in line with entry terms. This indemnity extends to situations such as the return of items or entries due to insufficient funds caused by a debit entry, ensuring that RDFIs and ACH Operators are protected from financial repercussions resulting from ODFI actions. RDFIs must indemnify ODFIs and ACH Operators against all claims, losses, liabilities, or expenses, including legal fees, resulting directly or indirectly from any breach of warranty under Section 1.2 (general rule compliance) of the Nacha operating rules. Gateway providers also assume these same warranties under Nacha rules.

RTP Network

The RTP Participant indemnity agreement covers indemnity arrangements between participants and TCH itself. Participants agree to mutual indemnification to each other for transactions facilitated via the RTP Network and to their pre-funded account balance used to facilitate transactions. Regarding acts between participants related to participation in the RTP Network, TCH can only be held indemnible in cases of its own gross negligence or intentional misconduct. The amount and basis of indemnity will correspond to the gross negligence or misconduct that is directly attributable to TCH. TCH's rules also stipulate that it shall not be held liable for any fraud that occurs except in cases where it is deemed to be negligent. The recipient bank of fraudulently sent payments is expected to cooperate with the sending bank in recovering said funds, typically within 10 business days. The Zelle service, which in some cases utilizes the RTP Network to facilitate its transactions, has its own set of liability and indemnity arrangements.

FedNow

The liabilities of the Federal Reserve for the FedNow service are delineated within the framework of Operating Circular 8. In the provision of other services outlined in the circular, the liability of a Reserve Bank is limited as explicitly stated within the circular or, in the absence of such explicit limitation, to losses directly resulting from willful misconduct or failure to exercise ordinary care. Under no circumstances is a Reserve Bank liable for special or consequential damages, regardless of foreseeability or prior notification. Moreover, the Reserve Banks' liability for damages resulting from the failure to exercise ordinary care or good faith in processing a message is restricted to the amount of any fee paid for that message. Furthermore, the Reserve Banks disclaim liability for any loss or damage arising from a FedNow Participant's or Service Provider's use of third-party products or services not supplied, owned, or operated by the Reserve Banks, even if these products or services facilitate access to the FedNow Service. This disclaimer extends to any warranty, express or implied, concerning the accuracy, timeliness, completeness, merchantability, fitness for a particular purpose, title, quality, or non-infringement of such third-party products or services. These liability provisions are designed to delineate the boundaries of responsibility and manage risks associated with the FedNow service.

Mastercard Send

Each customer is required to indemnify and hold harmless Mastercard, its parent and subsidiaries, directors, officers, employees, and agents from any claims, losses, liabilities, or expenses resulting from acts or omissions of the customer or any associated persons. This includes compliance with standards, access or use of the Interchange System, failure to perform as required, or any actions involving Mastercard assets. Mastercard disclaims all warranties regarding its systems and limits liability for damages, with any total liability capped at the compensation received by Mastercard from the customer or USD 250,000, whichever is less, unless prohibited by law. This indemnity and limitation of liability clause is a fundamental aspect of Mastercard Send's standards and conduct of activity.

Visa Direct

Visa assumes liability solely for the loss or misdirection of funds in push-to-account Original Credit Transactions if such issues are directly attributed to Visa, within the limitations outlined in the Visa Core Rules and Visa Product and Service Rules. However, Visa disclaims responsibility for any indirect, incidental, consequential, or special damages, including loss of revenue, profits, or business opportunities, irrespective of whether Visa was aware of the possibility of such losses. Furthermore, Visa holds no liability for actions or omissions of downstream entities, including partners, banks, and processing schemes, emphasizing that Visa bears no responsibility for their conduct in the transaction

Appendix A

Glossary

Account Funding Transaction (AFT)

A Visa network transaction that enables funds to be pulled from an eligible Visa account.

ACH Network / ACH

In the context of the US, an electronic funds transfer system governed by the Nacha Operating Rules that provides for the interbank clearing of electronic entries for participating financial institutions.

ACH Operator

An ACH Operator is an entity that acts as a central facility for the clearing, delivery and settlement of entries between or among participating DFIs. An ACH Operator may be a Federal Reserve Bank or an entity that executes an annual agreement with Nacha.

Beneficiary

A person designated as the recipient of funds (i.e.,, a person who has applied for and is entitled to Federal benefits payable by law.)

Correspondent Bank

A correspondent bank provides services on behalf of another bank, serving the role of a middleman between the initiating bank and the receiving bank.

Direct Access

Direct access is a situation in which an Originator or third-party (either a DFI's Third-Party Service Provider or an Originator's Third-Party Sender) transmits ACH files (transactions) directly to an ACH Operator using a financial institution's routing and transit number and settlement account.

Faster Payment(s)

Electronic payment services that provide funds to the payee from seconds or up to a few hours of initiation by the payer. In the context of this Handbook, the term Faster Payments refers to Same Day ACH, RTP, FedNow, Mastercard Send, and Visa Direct.

Funding Agent

In the context of RTP, a depository institution that has become a party to the Prefunded Balance Account Agreement and is either a Funding Manager or a Funding Provider.

Funding Manager

A depository institution that has been approved by TCH to serve as an agent to one or more Non-funding Participants to (A) provide the Prefunded Requirement and supplemental funding on behalf of each such Non-funding Participant (to the extent it is a Sending Participant) and (B) request disbursements as permitted by RTP Operating Rules for each of its Non-funding Participants.

Funding Participant

In the context of RTP, a Participant that has become a party to the RTP Prefunded Balance Account Agreement and that (i) requests and receives disbursements from the Prefunded Balance Account, as permitted by the RTP Operating Rules, to its Federal Reserve account and (ii) if the Participant is a Sending Participant, prefunds for itself in accordance with these RTP Participation Rules and the RTP Operating Rules.

Funding Provider

A depository institution that has been approved by TCH to have a funding arrangement with a group of Non-Funding Participants under which the depository institution provides funding for use by the Non-funding Participants consistent with these Rules.

Non-funding Participant

In the context of RTP, a Participant that is not a Funding Participant and that has designated a Funding Agent to act on its behalf to prefund in accordance with these RTP Participation Rules and the RTP Operating Rules.

Original Credit Transfer (OCT)

A Visa network transaction that enables eligible Visa cards to receive push payment credits.

Originating Depositary Financial Institution / Sending Participant / Sender FI

The financial institution that holds the sender's account and initiates a payment.

Originator

An organization or natural person that has authorized an ODFI (directly or through a Third-Party Sender) to transmit, for the account of that organization or natural person, a credit entry, debit entry, or non-monetary entry to the Receiver's account at the RDFI.

Push-to-card

A process whereby money is sent to a debit or prepaid card belonging to an individual or business.

Receiver

An individual, corporation or other entity who has authorized an Originator to initiate a credit entry, debit entry, or non-monetary entry to the Receiver's account held at the RDFI.

Receiving Depository Financial Institution (RDFI)

A Participating Depository Financial Institution that receives ACH entries from the ACH Operator to the accounts of Receivers. An RDFI must agree to abide by the Nacha Operating Rules.

Request for Payment (RfP)/Request To Pay (R2P – EU equiv to RfP)

A process (usually built around API/messages) that allows a payee and payer to interact and request a funds transfer, providing details of the reason for the request, alterations to the payment, allowing automated account payable/receivable processing.

Respondent

A financial institution that sends ACH entries to a correspondent bank.

RTP Prefunded Balance Account

The special deposit account established for the joint benefit of all Funding Participants and Funding Agents by the Prefunded Balance Account Bank for the purpose of supporting the operations of the RTP System

RTP Prefunded Balance Account Bank

The bank that holds the Prefunded Balance Account

Sending Point

A processing site from which entries are transmitted to the network. It may be the ODFI on its own behalf or a financial institution or a Third-Party Service Provider on behalf of the ODFI.

Settlement

The final stage of a payment transaction to complete the funds transfer after the clearing/exchange has already been validated and completed.

The Clearing House (TCH)

The Clearing House Payments Company L.L.C.

The Clearing House's Real-time Payments Platform (RTP)

The Faster Payments system owned and operated by The Clearing House

The Federal Reserve's FedNow Service (FedNow)

The Faster Payments system owned and operated by The Federal Reserve

Third-Party Service Provider / Service Provider / Third Party

A commercial data processing service organization, a person operating a data transmission facility, or any other participant that acts on behalf of a participant, as such participant's agent, to transmit and receive payment messages, payment message responses and non-payment messages through a Faster Payment System.

Value-Added Services

An extra or special features added by a company or producer to increase the value of a product or service.

List of acronyms

Accept Without Post (ACWP)

Account Funding Transaction (AFT)

Account Verification Service (AVS)

account-to-account (A2A)

Anti-Money Laundering (AML)

Authorized Push Payment (APP)

Confirmation-of-payee (CoP)

Federal Financial Institution Examination Council (FFIEC)

Liquidity management transfer (LMT)

National Settlement Service (NSS)

Original Credit Transaction (OCT)

Originating Depositary Financial Institution (ODFI)

Originating Institution (OI)

Real Time Payments (RTP)

Receiving Depositary Financial Institution (RDFI)

requests-for-payment (RfP)

Service Level Agreements (SLAs)

Strong Customer Authentication (SCA)

The Clearing House (TCH)

Transaction Initiator (TI)

Value-added services (VAS)

Appendix B

Typical parties on a Faster Payments transactions and the terminology usedby each network

Parties involved in the Faster Payments value chain include an Originator, the Originating Depositary Financial Institution, the Faster Payments operator, the Receiving Depositary Financial Institution, and the Receiver. Optionally, Third-Party Providers / Third-Party Service Providers may be an additional party to the transaction. They may be used to connect to the network and/or to submit payments to the Fast Payment Network on the Originator's or the Originating Depositary Financial Institution's behalf.

The terminology that each Faster Payments network uses for each party may vary slightly but it generally refers to the same actor. The table below aligns some of the terms used. For simplicity, a unified term will be used throughout this handbook.

TABLE 65. TYPICAL PARTIES ON A FASTER PAYMENTS TRANSACTIONS & THE TERMINOLOGY USED BY EACH NETWORK

TYPICAL PARTIES TO A FASTER PAYMENTS TRANSACTION					
NETWORK	**SENDER**	**SENDING FINANCIAL INSTITUTION**	**RECEIVER FINANCIAL INSTITUTION**	**RECEIVER**	**THIRD-PARTY SERVICE PROVIDER**
FedNow	Sender	Sender FI	Receiver FI	Receiver	Service Provider
RTP	Sender	Sending Participant	Receiving Participant	Receiver	Third-party Service Provider
Same Day ACH	Originator	Originating Depositary Financial Institution (ODFI)	Receiving Depositary Financial Institution (RDFI)	Receiver	Third-Party Service Provider
Mastercard Send	Sender	Originating Institution (OI) / Transaction Initiator (TI) (issuers, acquirers, fintechs)	Receiving Institution	Recipient	Third Parties (registered Third-Party Processors and Service Providers)

Source: Nacha, based on operational guidelines and rules from the Faster Payments networks

Appendix C

Nacha – Standard Entry Class (SEC)

STANDARD ENTRY CLASS (SEC) CODES				
SEC CODE	**TITLE**	**TRANSACTION TYPE**	**ACCOUNT TYPE**	**USER AGREEMENT OR AUTHORIZATION REQUIRED**
ACK	ACH Payment Acknowledge	Non-Monetary (Acknowledges CCD Credit Entry)	N/A	N/A
ADV	Automated Accounting Advice	Non-Monetary	N/A	Agreement — ODFI/RDFI and ACH Operator
ARC	Accounts Receivable Entry	Debit Single Entry	Consumer or Non-Consumer	Written Notice = Authorization
ATX	Financial EDI Acknowledgement	Non-Monetary (Acknowledges CTX Credit Entry)	N/A	N/A
BOC	Back Office Conversion Entry	Debit Single Entry	Consumer or Non-Consumer	Posted Notice and Written Notice = Authorization
CCD	Corporate Credit or Debit Entry	Credit or Debit Single Entry or Recurring Entry	Non-Consumer	Agreement — Originator and Receiver
CIE	Customer Initiated Entry	Credit	Consumer Originated	Agreement — Originator and Receiver
COR	Notification of Change or Refused Notification of Change	Non-Monetary	Consumer or Non-Consumer	N/A
CTX	Corporate Trade Exchange	Credit or Debit Single Entry or Recurring Entry	Non-Consumer	Agreement — Originator and Receiver
DNE	Death Notification Entry	Non-Monetary	Consumer	N/A
ENR	Automated Enrollment Entry	Non-Monetary	Non-Consumer (Federl Government Agency)	Request by Receiver / RDFI use only
IAT	International ACH Transaction	Credit or Debit Single Entry Recurring Entry	Consumer Non-Consurner	Consumer Debit — in writing and signed or similarly authenticated = authorization Consumer Credit — orally other non-written means authorization Corporate — Agreement between Originator and Receiver.
MTE	Machine Transfer Entry	Credit or Debit Single Entry	Consumer	In writing and signed similarly authenticated = authorization
POP	Point-of-Purchase Entry	Debit Single Entry	Consumer or Non consumer	Posted Notice and in writing and signed or similarly authenticated

STANDARD ENTRY CLASS (SEC) CODES *(continued)*				
SEC CODE	**TITLE**	**TRANSACTION TYPE**	**ACCOUNT TYPE**	**USER AGREEMENT OR AUTHORIZATION REQUIRED**
POS	Point-of-Sale Entry	Credit or Debit Single Entry	Consumer	In writing and signed similarly authenticated
PPD	Prearranged Payment and Deposit	Credit or Debit Single Entry or Recurring Entry	Consumer	Consumer Debit — in writing and signed or similarly authenticated. Consumer Credit — orally or other non-written means.
RCK	Re-presented Check Entry	Debit Single Entry	Consumer	Posted Notice = authorization
SHR	Shared Network Transaction	Credit or Debit Single Entry	Consumer	Agreement — ODFI and RDFI
TEL	Telephone-initiated Entry	Debit Single Entry or Recurring Entry	Consumer	Orally authorized over telephone = authorization
TRC	Truncated Entry	Debit Single Entry	Consumer or Non-Consumer	Agreement — ODFI and RDFI
TRX	Truncated Entries Exchange	Debit Single Entry	Consumer or Non-Consumer	Agreement — ODFI and RDFI
WEB	Internet-Initiated/ Mobile Entry	Debit or Consumer to Consumer Credit Single Entry or Recurring Entry	Consumer	Authorization — via any form (written or oral) that is communicated from the Receiver to the Originator via the Internet or a Wireless Network (excluding an Oral Authorization via a telephone call)
XCK	Destroy Check Entry	Debit	Consumer or Non-Consumer	N/A

Source: 2023 Nacha Operating Rules & Guidelines

Appendix D

FedNow validation items prior to message submission

TABLE 66. VALIDATIONS FOR CUSTOMER CREDIT TRANSFER MESSAGES – FEDNOW
VALIDATIONS FOR CUSTOMER CREDIT TRANSFER
• Valid message signature
• Valid message size
• XML syntax
• Authorized sender
• Confirm unique message ID
• FedNow Service ISO 20022 schema rules followed
• Sending and Receiving FIs are FedNow participants, are active and signed on, and have profiles to send/receive payments, or receive only. Sender must have a profile for sending and receiving payments
• Payment timeout clock has not been exceeded
• Message is not future dated
• Message is within Receiver FI reserved response time
• Message is within FedNow Service network transaction limit or Sender FI transaction limit, if configured to be less than the FedNow Service limit
• Sender and Receiver FIs have valid settlement relationships
• Optional: Negative list check, if enabled

TABLE 67. VALIDATIONS FOR RETURN REQUEST MESSAGES – FEDNOW
VALIDATIONS FOR RETURN REQUEST
• Valid message signature
• Valid message size
• XML syntax
• Authorized sender
• Confirm unique message ID
• FedNow Service ISO 20022 schema rules followed

TABLE 68. VALIDATIONS FOR LIQUIDITY MANAGEMENT TRANSFER MESSAGES – FEDNOW
VALIDATIONS FOR LIQUIDITY MANAGEMENT TRANSFER
• Valid message signature
• Valid message size
• XML syntax
• Authorized sender
• Confirm unique message ID
• FedNow Service ISO 20022 schema rules followed
• Sending and Receiving FIs are FedNow participants
• Message is not future dated
• Within the FedNow Service transaction limit for LMT or the Sender FI transaction limit for LMT if configured for less than the FedNow Service limit
• Receiver FI is active and profile is set to send and receive or receive liquidity management transfers
• Sender FI is active and profile is set to Send and Receive Liquidity Management Transfers
• LMT is sent within the valid LMT window

TABLE 69. VALIDATIONS FOR REQUEST FOR PAYMENT MESSAGES – FEDNOW
VALIDATIONS FOR REQUEST FOR PAYMENT MESSAGE
• Valid message signature
• Valid message size
• XML syntax
• Authorized sender
• Confirm unique message ID
• FedNow Service ISO 20022 schema rules followed

TABLE 70. VALIDATIONS FOR REQUEST FOR PAYMENT CANCELLATION REQUEST MESSAGES – FEDNOW
VALIDATIONS FOR REQUEST FOR PAYMENT MESSAGE
• Valid message signature
• Valid message size
• XML syntax
• Authorized sender
• Confirm unique message ID
• FedNow Service ISO 20022 schema rules followed

TABLE 71. VALIDATIONS FOR INFORMATION REQUEST MESSAGES – FEDNOW
VALIDATIONS FOR REQUEST FOR PAYMENT MESSAGE
• Valid message signature
• Valid message size
• XML syntax
• Authorized sender
• Confirm unique message ID
• FedNow Service ISO 20022 schema rules followed

AFPP Handbook Review

The 2026 AFPP examination will consist of up to 120 multiple choice questions covering all aspects of payments risk management as described within this Handbook. Candidates will be unable to distinguish pilot test questions from those questions that will be scored. *Please note, these questions are not intended to be indicative of those that may appear on the examination.* **The questions contained in this chapter should be used to test your knowledge of the material contained in the Handbook or other resources. A passing or perfect score on the practice test does not guarantee passage of the AFPP examination.**

1. **What must participants ensure before originating or receiving messages in Faster Payments systems?**
   A) Sufficient liquidity
   B) Continuous availability
   C) Steady network connection

2. **What must RTP participants do before submitting a payment message?**
   A) Validate the receiving account
   B) Ensure prefunding requirements are met
   C) Conduct an annual compliance assessment

3. **What is a requirement for Same Day ACH participants?**
   A) Conduct risk assessments
   B) Implement an OFAC compliance program
   C) Use public/private key pairs for digital message signing
4. **The current dollar limit for Same-Day ACH Entries is:**
   A) $500,000.
   B) $1,000,000.
   C) $10,000,000.

5. **What is a non-value message in the context of Faster Payments?**
   A) A message that instructs funds to be moved
   B) A message that does not generate an accounting entry
   C) A message that confirms a transaction

6. **Prior to submitting a Payment Message to the RTP System, a Sending Participant must**
   A) perform fraud checks
   B) confirm message formatting
   C) satisfy its Prefunded Requirement

7. **What is the time frame for resolving 'accept without posting' responses in RTP?**
    A) By midnight Eastern Time of the next standard business day
    B) Within 10 banking days
    C) By 11:59 p.m. local time the next business day

8. **What is a method used to mitigate message rejections and failures in push-to-card networks?**
    A) Use of Account Verification Service
    B) Validate card eligibility
    C) Conduct fraud checks

9. **What is a method used to reduce exceptions processing in FedNow?**
    A) Use of a reliable time server
    B) Continuous availability
    C) Monitoring and alerting capabilities

10. **What is a reason for using 'accept without posting' responses in FedNow?**
    A) Compliance or legal reasons
    B) Returning funds to a closed account
    C) The name of the recipient does not match the account name

11. **What is the time frame for resolving 'accept without posting' responses in FedNow?**
    A) By midnight Eastern Time of the next standard business day
    B) By 11:59 p.m. local time the next business day
    C) Within 10 banking days

12. **_____ governs wholesale credit entries, including wire transfers, non-consumer ACH credits and instant payments.**
    A) Regulation E
    B) UCC3 and 4
    C) UCC4A

13. **The primary ISO 20022 message in RTP and FedNow is:**
    A) pacs.008 – Customer Credit Transfer.
    B) pacs.002 – Payment Status Report.
    C) pacs.004 – payment return.

14. **In the RTP network, a request for a participant to return some or all of a previously settled payment or liquidity transfer is a:**
    A) camt.056 Return Request.
    B) pain.013 - Request for Payment.
    C) camt.026 – Request for Information

15. **____ message type specifies a common interface by which financial transaction card-originated messages can be interchanged between acquirers and card issuers, specifies message structure, format and content, data elements and values for data elements.**
    A) ISO 8583
    B) ISO 20022
    C) ANSI EDI 820

16. **What is a reason for allowing 'requests for returns' in RTP?**
    A) Incorrect Amount
    B) Error
    C) Unauthorized transactions

17. **What can help increase the adoption of Faster Payments among end users?**
    A) Value-added services (VAS)
    B) Familiarity with traditional payment methods
    C) Established corporate processes

18. **What is a permissible reason for reversing a Same Day ACH entry?**
    A) Unauthorized transactions
    B) Erroneous Entry
    C) Untimely Entry

19. **What is a reason for allowing 'requests for returns' in push-to-card networks?**
    A) Lack of funds
    B) Incorrect payment credentials
    C) Duplicate processing

20. **What is the time frame for processing 'requests for returns' in FedNow?**
    A) Within 10 banking days
    B) Within 5 banking days
    C) By midnight of the next standard business day

21. **What is the time frame for processing 'requests for returns' in RTP?**
    A) By midnight of the next standard business day
    B) Within 10 banking days
    C) Within 5 banking days

22. **What is the time frame for returning a non-consumer Same Day ACH consumer return)**
    A) By the opening of business 2 banking days after receipt of the entry
    B) By the opening of business 10 banking days of the entry
    C) By the opening of business 60 banking days of the entry

23. **What is the time frame for processing 'requests for returns' in push-to-card networks?**
    A) Within 10 banking days
    B) Within 30 minutes
    C) Within 5 banking days

24. **What does interoperability in the context of Faster Payments mean?**
    A) Payment messages are processed manually
    B) Payment messages are routed and settled seamlessly
    C) Payment messages are only exchanged within the same system

25. **What must Mastercard Send participants do before using the APIs?**
    A) Register for required programs and services
    B) Complete API testing and validation
    C) Establish a settlement account

26. **What is a key challenge for Faster Payments in terms of security?**
    A) Slower transaction times
    B) Increased risk of fraud
    C) Lack of user trust

27. **What is a requirement for participants in the RTP network?**
    A) Increase transaction fees
    B) Delay payment processing
    C) Use of multi-factor authentication

28. **What is the purpose of confirmation-of-payee (CoP)?**
    A) To speed up payment processing
    B) To reduce transaction fees
    C) To confirm the receiver's information before approving a payment

29. **Same-Day ACH allows:**
    A) both credit and debit entries.
    B) only debit entries.
    C) only credit entries.

30. **_____ requires that FIs provide evidence that necessary and reasonable actions were taken in order to identify an incident relating to money laundering and criminal activities.**
    A) FFIEC
    B) GLBA
    C) BSA

31. **What is a common fraud scenario in Faster Payments?**
    A) Payment initiation by a fraudster using stolen payment data
    B) Delayed payment processing
    C) High transaction fees

32. **What is required for FedNow participants to send or receive transactions?**
    A) Establish a participant profile
    B) Notify the Federal Reserve of any issues
    C) Conduct an annual assessment

33. **What is a key component of Strong Customer Authentication (SCA)?**
    A) Transaction fees
    B) Biometric verification
    C) Delayed payment processing

34. **What is an example of a value message in FedNow?**
    A) Service Rejections
    B) Customer credit transfer
    C) Accept Without Post (ACWP)

35. **What is the impact of Faster Payments on end-user cash flow management?**
    A) Delays in fund availability
    B) Quick availability of funds
    C) Increased transaction fees

36. **What is the role of directories in Faster Payments?**
    A) To increase transaction fees
    B) To simplify payment routing by enabling payment initiation using aliases
    C) To delay payment processing

37. **What is a challenge for financial institutions with the implementation of Faster Payments?**
    A) Increased liquidity management processes
    B) Reduced liquidity management processes
    C) Decreased need for automation

38. **What is a benefit of correspondent-respondent relationships in Faster Payments?**
    A) Increased liquidity costs for respondents
    B) Reduced need for liquidity management for respondents
    C) Increased operational costs for correspondents

39. **A regulatory consideration for financial institutions regarding Real-Time Payments and FedNow include:**
    A) impacts to the Nacha Operating Rules.
    B) impacts to Regulation E.
    C) impacts to Regulation CC.

40. **What is a role of The Clearing House in the RTP network?**
    A) Managing the RTP network and scheme rulebook
    B) Providing liquidity management services
    C) Offering settlement services to other FIs

41. **What is a responsibility of ACH Operators for Same Day ACH?**
    A) Providing liquidity management services
    B) Facilitating the electronic transfer of funds between financial institutions
    C) Offering settlement services to other Fis

42. **The USA PATRIOT Act established standards for identifying consumers at account opening utilizing a:**
    A) Know Your Customer Program.
    B) Beneficial Ownership Program.
    C) Customer Identification Program.

43. **What is a requirement for Visa Direct participants?**
    A) Adherence to Visa Direct Program Standards and Rules
    B) Maintaining a Federal Reserve Master Account
    C) Offering settlement services to other FIs

44. **What is the purpose of the Bank Secrecy Act (BSA)?**
    A) To increase transaction fees
    B) To prevent criminals from misusing financial institutions to hide or launder money
    C) To delay payment processing

45. **What is a common SLA for Faster Payments networks?**
    A) Maximum response times
    B) Minimum transaction fees
    C) Maximum transaction limits

46. **What is operational risk in the context of Faster Payments?**
    A) Risk of financial loss due to market fluctuations
    B) Risk of financial loss due to internal process failures or external events
    C) Risk of financial loss due to currency exchange rates

47. **What must participants do before submitting messages to the RTP rail?**
    A) Conduct fraud checks
    B) Validate the receiving account
    C) Ensure messages are consistent with system requirements

48. **What is a major challenge for operational risk in Faster Payments systems?**
    A) Limited transaction volume
    B) Continuous system availability 24/7/365
    C) High transaction fees

49. **What is liquidity risk in Faster Payments systems?**
    A) Risk of financial loss due to currency exchange rates
    B) Risk of a counterparty lacking adequate funds to settle obligations
    C) Risk of financial loss due to market fluctuations

50. **What type of settlement model does FedNow use?**
    A) Real-time gross settlement
    B) Deferred net settlement
    C) Prefunded settlement

51. **What is the purpose of device binding in Faster Payments?**
    A) To increase transaction fees
    B) To delay payment processing
    C) To link a user's identity to a specific device for authorized payments

52. **What is the main focus of Regulation E?**
    A) To regulate commercial transactions
    B) To manage liquidity risks
    C) To establish a framework for consumer electronic fund transfers

53. **What is risk-based authentication (RBA)?**
    A) Authentication based on the risk associated with a transaction
    B) Authentication based on transaction fees
    C) Authentication based on delayed payment processing

54. **What is a responsibility of financial institutions in a Customer Identification Program (CIP)?**
    A) Increase transaction fees
    B) Verify the identity of each customer through risk-based procedures
    C) Delay payment processing

55. **What is a method used to detect fraud in Faster Payments?**
    A) Increase transaction fees
    B) Trend analysis and anomaly detection
    C) Delay payment processing

56. **What is a barrier to implementing internal controls for Faster Payments?**
    A) Resource allocation and costs
    B) Increase transaction fees
    C) Delay payment processing

57. **What is the liability of the Federal Reserve Banks for the FedNow service as per Operating Circular 8?**
    A) Unlimited liability for all damages
    B) Liability limited to losses directly resulting from willful misconduct or failure to exercise ordinary care
    C) No liability under any circumstances

58. **What is the impact of service disruptions on Faster Payments?**
    A) Damage to system integrity and user trust
    B) Increased transaction fees
    C) Delayed payment processing

59. **What is an example of a non-value message in RTP?**
    A) Credit Transfer
    B) FI-to-FI Credit Transfer
    C) Request for Information

60. **What is a reason for implementing prefunding requirements in Faster Payments?**
    A) To increase transaction fees
    B) To avoid the buildup of credit risk
    C) To delay payment processing

61. **What is a key aspect of customer onboarding programs for managing risk in Faster Payments?**
    A) Collecting and verifying customer information
    B) Increasing transaction fees
    C) Delaying payment processing

62. **What is a type of authentication method used in Faster Payments?**
    A) Increasing transaction fees
    B) Delaying payment processing
    C) Biometric authentication

63. **What is a requirement for Same Day ACH participants?**
    A) Compliance with Nacha Operating Rules
    B) Maintaining a Federal Reserve Master Account
    C) Offering settlement services to other FIs

64. **What is the role of the Federal Reserve in FedNow?**
    A) Increasing transaction fees
    B) Operator and regulator of the system
    C) Delaying payment processing

65. **What is the main difference between ubiquity and interoperability?**
    A) Ubiquity refers to the ability of different systems to work together, while interoperability refers to the ability to reach all end users
    B) Ubiquity refers to the ability to reach all end users, while interoperability refers to the ability of different systems to work together
    C) Ubiquity and interoperability are the same

66. **What party must an ODFI of a Same Day ACH enter into an agreement with before submitting ACH entries?**
    A) Originator
    B) Nacha
    C) Receiver

67. **What is a common technique used to mitigate fraud in Faster Payments?**
    A) Confirmation-of-Payee (CoP)
    B) Increasing transaction fees
    C) Delaying payment processing

68. **What is a key factor in managing liquidity risk in Faster Payments?**
    A) Increasing transaction fees
    B) Prefunding requirements
    C) Delaying payment processing

69. **What is a value message in the context of Faster Payments?**
    A) A message that instructs funds to be moved
    B) A message that provides transaction status
    C) A message that requests information

70. **What is a major risk associated with technology in Faster Payments?**
    A) Cyber-attacks
    B) Increasing transaction fees
    C) Delaying payment processing

71. **What is a method used to mitigate message rejections and failures in Same Day ACH?**
    A) Use of notification of change
    B) Conduct risk assessments
    C) Validate messages prior to submission

72. **What is a benefit of using QR codes in payment transactions?**
    A) Simplicity of use and flexibility
    B) Increased transaction fees
    C) Delayed payment processing

73. **What is the main objective of data privacy laws in the context of Faster Payments?**
    A) To protect consumers from fraud, scams, and identity theft
    B) To increase transaction fees
    C) To delay payment processing

74. **What is the purpose of encryption in Faster Payments?**
    A) To increase transaction fees
    B) To secure data by encoding information into an indecipherable form
    C) To delay payment processing

75. **What is the purpose of log-in and API access management APIs?**
    A) To increase transaction fees
    B) To facilitate the administration of tasks like adding or adjusting account settings
    C) To delay payment processing

76. **What is the function of tokenization in payment security?**
    A) To increase transaction fees
    B) To delay payment processing
    C) To replace sensitive payment data with a random string of characters

77. **What is the importance of network connectivity in Faster Payments?**
    A) To ensure secure exchange of payment and non-payment messages among participants
    B) To increase transaction fees
    C) To delay payment processing

78. **What is the role of authentication in Faster Payments?**
    A) To increase transaction fees
    B) To verify the identity of a user before payment initiation
    C) To delay payment processing

79. **What is the purpose of core banking systems in Faster Payments?**
    A) To increase transaction fees
    B) To delay payment processing
    C) To manage account information and balance updates continuously

80. **_____ is responsible for developing uniform reporting systems and guidance for federally supervised financial institutions, their holding companies, and the nonfinancial institution subsidiaries of those institutions and holding companies.**
    A) BSA
    B) OFAC
    C) FFIEC

81. **What is the function of registration APIs in Faster Payments?**
    A) To increase transaction fees
    B) To delay payment processing
    C) To enable corporations to enroll in a provider's Faster Payments API service

82. **What is the role of payment initiation and processing-related APIs?**
    A) To support various Faster Payment processing functions
    B) To increase transaction fees
    C) To delay payment processing

83. **A primary operational consideration for faster payments includes:**
    A) the benefits of instant payments for your customers.
    B) how 24x7x365 operations can be supported.
    C) the use cases supported by instant payments.

84. **What is a validation requirement for FedNow participants sending customer credit transfers?**
    A) Invalid message size
    B) Valid message signature
    C) HTML syntax

85. **A company that initiates an RTP message is referred to as the:**
    A) Originator.
    B) Sender.
    C) Payer.

86. **What is the purpose of embedded and additional function APIs?**
    A) To increase transaction fees
    B) To provide supplementary services like fraud prevention controls
    C) To delay payment processing

87. **In the RTP Network a depository institution that has become a party to the Prefunded Balance Account Agreement is either a:**
    A) Funding Manager or a Funding Provider.
    B) Sending Participant or a Receiving Participant.
    C) Liquidity Management Provider.

88. **What is the purpose of PCI DSS in payment security?**
    A) To set a baseline level of protection for customers, merchants, and service providers
    B) To increase transaction fees
    C) To delay payment processing

89. **What is the function of a centralized directory in Faster Payments?**
    A) To increase transaction fees
    B) To map an alias to its PSP and the corresponding account number
    C) To delay payment processing

90. **As the Network Operator, the primary function of The Clearing House in the RTP Network is to provide:**
    A) transaction initiation and funding.
    B) settlement of each payment.
    C) account reconciliation.

91. **What is the purpose of confirmation-of-payee in Faster Payments?**
    A) To verify the identity of the payee before payment initiation
    B) To increase transaction fees
    C) To delay payment processing

92. **What is the function of tokenization services in Faster Payments?**
    A) To enhance the security of financial transactions by substituting payment information with a unique identifier
    B) To increase transaction fees
    C) To delay payment processing

93. **What is the role of fraud scoring in Faster Payments?**
    A) To increase transaction fees
    B) To assess a transaction based on various risk indicators
    C) To delay payment processing

94. **What is the primary role of the PCI Security Standards Council (PCI SSC)?**
    A) To increase transaction fees
    B) To establish technical and operational criteria to safeguard cardholder data
    C) To delay payment processing

95. **An agent of a FedNow Participant authorized to initiate, transmit, or receive messages on behalf of that FedNow Participant best defines a:**
    A) Service Provider.
    B) FedNow Sender.
    C) Correspondent.

96. **FedNow Participants must enter into the Service Security Procedure Agreement and agree to security procedures used in connection with:**
    A) data stored at rest.
    B) authenticated instructions.
    C) Requests for Payments.

97. **What is the transaction limit for FedNow as of November 2025?**
    A) $50,000
    B) $100,000
    C) $10,000,000

98. **What is the role of the Federal Financial Institutions Examination Council (FFIEC) in Faster Payments?**
    A) To develop uniform principles, standards, and reporting forms for the examination of financial institutions
    B) To increase transaction fees
    C) To delay payment processing

99. **What is a key requirement for financial institutions under the Customer Identification Program (CIP)?**
    A) Increase transaction fees
    B) Verify the identity of customers who open accounts
    C) Delay payment processing

100. **What is the role of client/customer agreements in Faster Payments?**
    A) To increase transaction fees
    B) To delay payment processing
    C) To outline the terms and conditions governing the use of Faster Payment products and services

AFPP Handbook Review Answer Key

1. C	2. B	3. A	4. B
5. B	6. C	7. C	8. B
9. A	10. A	11. A	12. C
13. A	14. A	15. A	16. C
17. A	18. B	19. A	20. C
21. B	22. A	23. B	24. C
25. C	26. B	27. C	28. C
29. A	30. C	31. A	32. A
33. B	34. B	35. B	36. B
37. A	38. B	39. B	40. A
41. B	42. C	43. A	44. B
45. A	46. B	47. A	48. B
49. B	50. A	51. C	52. C
53. A	54. B	55. B	56. A
57. B	58. A	59. C	60. B
61. A	62. C	63. A	64.B
65. B	66. A	67. A	68. B
69. A	70. A	71. C	72. A
73. A	74. B	75. B	76. C
77. A	78. B	79. C	80. C
81. C	82. A	83. B	84. B
85. B	86. B	87. A	88. A
89. B	90. B	91. A	92. A
93. B	94. B	95. A	96. B
97. B	98. A	99. B	100. C